

Merged research report for the Surreal project

Notations for Omnific Integers

Holonomic towers, shielded and rational-omega calculi,
and the complexity of equality

Built from three manuscripts dated September 23, 2026

Abstract

An omnific integer may have a transfinite Conway normal form, arbitrary real coefficients at its positive exponents and an ordinary integer constant term. Finite notation is therefore not the same as finite normal form, and decidable syntax is not the same as decidable equality. This report studies which finite representations of omnific integers have decidable validity, equality, order, membership in **Oz** and omnific floor, and measures the failure of these properties for the representations that lack them.

On the positive side three effective systems are constructed. Over any effective ordered subfield of $\mathbb{R}$, certified Euler operators with finite compatible jets name D-finite series, a leading-polynomial argument computes the exceptional integer indices even over non-Archimedean coefficient fields, and iterated fraction fields give effective finite-rank surreal fields whose omnific parts have a unique triangular form, exact truncation, a computable floor and a decidable ordinal slice; a union over hereditary Cantor labels has ordinal trace exactly the ordinals below ε_0 . A general shielding theorem embeds $D + XL[X]$ into **Oz** whenever the monomial interpreting X dominates the exponents of a Hahn field L , and transfers the equality problem of L many-one; with hereditary finite forms and $X = \varepsilon_0$ it gives a decidable omnific ring with infinite normal forms, and finite matrix data certify leading terms at arbitrary positive scales, with an exact collision-aware form. Finite expressions in rational constants, field operations and Conway's map $x \mapsto \omega^x$ compile into separated rational grids; this decides membership in **Oz** and computes the floor, every truncation at a named exponent and every coefficient. Its ordinals are exactly those below ε_0 , depth by depth, and its supports have order type below ω^ω , sharply.

On the negative side, primitive-recursive coefficient streams on d fixed positive blocks always denote omnific integers, but their equality is Π_1^0 -complete and their positivity is many-one complete for the d -c.e. sets, already on values with at most d monomials; no algorithm translates these names into any system with decidable equality, although every value has a rational name; for full-support rational switch streams the minimal realization dimension is exactly $s + 1$ and has no computable bound. With a decidable dyadic support in $[1, 2)$ and all coefficients one, support validity is Π_1^1 -complete.

This report merges three independently written manuscripts; it states which source proved each result, prints shared results once, and keeps every source's limitations. Several results were already in the collection and are cited rather than reprinted. It is an AI-assisted draft: it has not been refereed, and nothing in it has been formalized in Lean.

Sources: manuscripts 08 (the base text), 05 and 09 of the batch placed in commit [a4dcb91](#), all pinned to repository commit [a45d722](#); their titles are in [subsection 1.2](#). The provenance of every section is recorded in [Appendix A](#); non-claims are collected in [Appendix B](#). The accompanying programs check finite identities and examples only.

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1 Scope, results, and conventions

An ordinal notation is a finite object with an ordinal interpretation. Hereditary Cantor normal forms are the favourable case: validity and comparison are decidable, and canonical expressions name exactly the ordinals below ε_0 . More general recursive ordinal systems make well-foundedness part of the semantics, so recognizing valid expressions can itself be highly undecidable. For omnific integers the question has more dimensions. A name must describe real coefficients, surreal exponents and possibly transfinite supports, and must then account for cancellation; an ordinal notation for the order type of a support supplies none of these data. A system can be very rich in its values while being algorithmically tame, or very restricted in its values while having an undecidable equality relation. This report makes both possibilities explicit. The ordinal-notation overview [16] and Kleene’s original paper [12] motivate the distinction; the arguments below use the underlying mathematical facts, not that overview, as their technical source (08, 05, 09).

The positive constructions are three effective notation systems: *certified holonomic towers* (Part I), *hereditary finite forms and an ε_0 -shielded calculus* with finite-state series (Part II), and *rational-omega expressions* (Part III). They share a general shielding theorem (section 3). The negative constructions (Part IV) remove, one at a time, the certificates that make the positive systems work: first the certificate that a coefficient stream vanishes identically, then the certificate that a support is well ordered.

Representation	Semantic validity	Equality	Order / positivity
Certified holonomic towers (08)	Decidable	Decidable	Decidable
Hereditary forms; ε_0 -shielded fractions; matrix series (05)	Decidable	Decidable (finite certificates)	Decidable
Rational-omega expressions (09)	Decidable	Decidable	Decidable
d primitive-recursive positive blocks (08; $d = 1$: 05, 09)	Every well-formed name is valid	Π_1^0 -complete	d -c.e.-complete
Rational switch streams, coefficients 1, 2 (05)	Every name is valid	Π_1^0 -complete; no computable realization bound	Strict comparison c.e. (one block)
Tree-controlled dyadic supports (05, 08, 09)	Π_1^1 -complete	Co-c.e. under the validity promise; checked equality Π_1^1 -complete	Not needed for the validity lower bound

The last row distinguishes an equality question under a promise from a question that must also check the promise (08). Here “c.e.” means computably enumerable, equivalently semidecidable; a Π_1^0 predicate has the form $\forall n R(n)$ with R computable; a Π_1^1 predicate allows a universal quantifier over functions $\mathbb{N} \rightarrow \mathbb{N}$ followed by an arithmetical condition. Completeness is under computable many-one reductions, with the valid input domains specified explicitly (05).

1.1 Principal results

Bracketed numbers name the source manuscripts (subsection 1.2). Throughout, $k \subseteq \mathbb{R}$ is an effective ordered coefficient field, typically $\mathbb{Q}$ or the real algebraic numbers $\mathbb{R}_{\text{alg}}$ (subsection 1.4).

- (1) **Shielding and equality transfer** [05]. If $L \subseteq k((t^G))$ is a subfield of a set-sized Hahn field, $\mathbb{Z} \subseteq D \subseteq L \cap \mathbf{Oz}$, and $|g| < \Lambda$ for all $g \in G$, evaluation at $X = \omega^\Lambda$ embeds $D + XL[X]$ into $\mathbf{Oz}$ with unique coefficient lists and the sign of the top block; with effective arithmetic, equality in the shielded ring is many-one equivalent to equality in L (theorems 3.1 and 3.3). The triangular theorem of the towers and the grid-ring theorem of the rational-omega expressions are the characterizations of the whole omnific part in their fields (remark 3.2).
- (2) **Certified holonomic towers** [08]. Over an effective leading-data field the nonnegative integer roots of a polynomial are computable (theorem 4.3); a certified Euler operator with a jet through the last exceptional index names a unique D-finite series, with effective validity, coefficients, zero test and ring operations (theorems 5.2 and 5.4); iterating fraction fields gives countable subfields $F_r \subseteq \mathbf{No}$ with decidable equality and order (theorem 6.2), effective truncation (theorem 7.1), triangular omnific parts (theorem 7.2) and a computable floor (theorem 7.4). Supports have order type at most ω^r , and every $\eta < \omega^r$ is realized in $F_r \cap \mathbf{Oz}$ (theorem 8.1); the ordinals of F_r are $\mathbb{N}[U_1, \dots, U_r]$ (theorem 8.3); the union over finite sets of hereditary Cantor labels has decidable equality, order, recognition and floor, a computable canonicalization, and ordinal trace $\mathbf{On}_{<\varepsilon_0}$ (theorem 9.2).
- (3) **Hereditary forms and the ε_0 -shielded calculus** [05]. Finite hereditary forms over $k = \mathbb{Q}$ or $\mathbb{R}_{\text{alg}}$ have canonical trees, decidable order and omnific membership, and a computable floor; their ordinals are exactly those below ε_0 , with natural arithmetic (theorems 11.1 to 11.3). The shielded ring $\mathcal{S}_{\varepsilon_0} = \mathcal{D} + \varepsilon_0 \text{Frac}(\mathcal{H})[\varepsilon_0]$ has decidable validity, equality and order and contains infinite normal forms, including supports of type ω^r for every r (theorem 12.2 and corollary 12.4); $\text{Frac}(\mathcal{H})$ is not real closed (proposition 12.3).
- (4) **Finite fronts of matrix series** [05]. For commuting matrix data of dimension m at independent positive scales, the leading term occurs at total degree below m ; at arbitrary positive scales, including dependent and non-Archimedean ones, the collected adjugate numerator certifies zero and the leading term, with the sharp bound $(m-1) \sum \delta_i$ (theorems 13.1 and 13.2).
- (5) **Rational-omega expressions** [09]. For finite terms in rational constants, $+$, $-$, $\times$, $/$ and $\Omega(x) = \omega^x$, validity, equality and order are decidable (theorem 14.2); every term compiles into a rational function of separated monomials (theorem 15.2); membership in $\mathbf{Oz}$, the floor, truncation at every named exponent and every coefficient are computable (theorem 16.3). Supports have order type below ω^ω , sharply at depth two; an explicit computable strong sum of such values escapes the language, and so does $\sum_n \omega^{1/n}$ (theorems 18.2 to 18.4 and proposition 18.5).
- (6) **Ordinal traces** [09; 08; 05]. At omega-nesting depth d the ordinals named are exactly those below β_d , where $\beta_0 = \omega$ and $\beta_{d+1} = \omega^{\beta_d}$ (theorem 17.2). The labelled tower union, the hereditary ring and the rational-omega field all have ordinal trace exactly $\mathbf{On}_{<\varepsilon_0}$, yet the omnific rings of the three systems are pairwise incomparable, with one exception: for towers over $k = \mathbb{Q}$ it is shown only that the rational-omega ring is not contained in the tower ring (proposition 17.3, [merge]).
- (7) **Stream equality and no translation** [08; 05, 09]. For every $d \geq 1$, equality of raw primitive-recursive d -block names is Π_1^0 -complete, already on values with at most one monomial (theorem 20.2); no algorithm translates these names, or the full-support rational switch names, into any system with decidable equality, although every value has a finite rational name (theorems 20.4 and 21.1). The two reductions are those of `cas:thm-zerotest` in the computer-algebra report; what is new is their omnific placement, the explicit translator corollaries, and the exact minimal realization dimension $s+1$ with no computable bound (theorem 21.3).

- (8) **Positivity** [08]. For each $d \geq 1$ positivity of raw d -block names is many-one complete for the d -c.e. sets, with at most d monomials in the hard instances ([theorem 22.2](#)); this strengthens the two-scale sign obstruction **cas:prop-leading**.
- (9) **Support validity** [05; 08, 09]. With an explicit dyadic coding, validity of $\sum_{s \in T} \omega^{\kappa(s)}$ is Π_1^1 -complete while every support is a decidable subset of the dyadics in $[1, 2)$ ([theorem 23.3](#)); checked equality is Π_1^1 -complete and promised equality is co-c.e. ([proposition 23.5](#)). This strengthens the undecidability of **cas:prop-validation** to an exact classification.

These statements are proved here, but proving a statement here is not a claim that nobody has previously proved it. Each source names its own proposed contributions and credits its classical inputs ([Appendix B](#)); no exhaustive priority claim is made.

1.2 Provenance of this report

This report is built from three manuscripts written independently and dated September 23, 2026. They are items 08, 05 and 09 of the batch placed in commit [a4dcb91](#); their shipped file prefixes are **08-holonomic-towers-**, **05-hereditary-shielding-** and **09-rational-omega-**. All three pin repository commit [a45d722](#) (30 commits before that placement). At the pin the collection already contained the computer-algebra report, the computable-surreals report and the omnific Diophantine report with the labels cited below; the manuscripts read only parts of them ([Appendix A.4](#)).

Source	Manuscript	Principal contribution to this report
08	<i>Notations for Omnific Integers: Exact Holonomic Towers and the Complexity of Equality</i> (31 pp.)	Base text and structure; notation systems and checked equality; the integer-root algorithm; certified Euler series; holonomic towers, truncation, triangular integer parts, floor, support spectrum, ordinal slice and the ε_0 -labelled union (Part I); raw d -block names, Π_1^0 equality, the translator theorem, the d -c.e. positivity classification, checked equality of tree supports.
05	<i>Equality for Omnific Notations</i> (26 pp.)	The shielding and equality-transfer theorems (section 3); hereditary finite forms, the ε_0 -shielded calculus, finite and collision-aware fronts of matrix series (Part II); strict comparison; the rational-realization gap with exact minimal dimension; the explicit dyadic coding with decidable supports; effective boundedness.
09	<i>Effective Notations for Omnific Integers</i> (28 pp.)	Rational-omega expressions: decidable equality, separated grids, grid-ring theorem, recognition, floor, truncation at named exponents, per-depth ordinal trace, support barrier, escape and lattice obstructions, recurrence and jet certificates (Part III); the ω_1^{CK} barrier with its caveat.

Where the merge had to choose.

- *Base.* Among the proofs of the shared theorems, 08's have the weakest hypotheses: any effective ordered coefficient field, an abstract leading-data field, d blocks where 05 and 09 treat one, an exact order classification, and checked equality besides validity. Its text is the base, with three exceptions printed from the members as main statements: 05's shielding and transfer theorems as the general embedding, 05's dyadic coding because only it gives decidable supports, and 09's per-depth ordinal trace.
- *Embedding versus characterization.* 05 proves only that $D + XL[X]$ embeds in **Oz**; it does not prove that the image is the whole omnific part of any field. That converse is 08's triangular theorem and 09's grid-ring theorem, each in its own field. The three are printed as one general embedding and two characterizations ([remark 3.2](#)).
- *Printed once.* The omnific floor formula is `odg:thm:floor` of the omnific Diophantine report and `cas:eq-floor` of the computer-algebra report; it is cited ([subsection 2.2](#)), and only the effectivity statements of the three systems are proved here. 09's division with remainder and its Diophantine transfer are `odg:prop:division` and `odg:cor:H10`, stronger there; the computable part is kept ([corollary 16.4](#) and [proposition 19.3](#)). The stream reductions are those of `cas:thm-zero-test`; 08's two-block sign example is `cas:prop-leading`; 08's integer-root algorithm is the effective form of `hol:lem:integer` of the holonomic-rigidity report; 09's grid fields reduce to `cas:thm-core` after compilation. The equality theorems for one stream block, the no-translator theorems, the canonicalization observation, the size obstruction and the ω_1^{CK} barrier, each proved by two or three sources, are printed once.
- *Second routes.* 08's and 09's midpoint codings of the Kleene–Brouwer order are kept as second routes to the validity theorem; they do not give decidable supports ([remark 23.4](#)).
- *Corrections.* 08's statement of its promised-equality proposition says “a co-c.e. test for inequality”; its proof and its own summary table show that inequality is c.e., so promised equality is co-c.e.; the statement is corrected ([proposition 23.5](#)).

Results added by the merge carry the tag [merge] and have complete proofs. Appendix [A](#) records, section by section, which source proved what and every symbol renamed from a source.

1.3 What “decidable” means here

All algorithms are ordinary terminating algorithms on finite strings, with no oracle unless explicitly stated. A procedure need not have a practical complexity bound to be decidable. The coefficient field at the bottom is part of the input convention, not an oracle for real numbers: it is an effectively presented ordered subfield $k \subseteq \mathbb{R}$ whose valid codes, equality, order and field operations are decidable or computable. Typical choices are $\mathbb{Q}$ and the real algebraic numbers $\mathbb{R}_{\text{alg}}$ with exact algebraic codes, for instance a normalized irreducible primitive polynomial over $\mathbb{Z}$ with the index of the root in increasing order; isolating intervals may accompany this pair as certificates but are not part of its canonical identity, and standard polynomial and real-root algorithms provide the field interface (05). Arbitrary Cauchy names for all computable reals do not satisfy this equality contract ([proposition 20.7](#)). Some results need more: the floor needs an effective integer test and ordinary floor in k , and the separated-basis lemma of [Part III](#) needs $k = \mathbb{Q}$.

Every set and sum in the proofs is set-sized. The proper class **No** is used only as an ambient semantic domain; in a universe-relative formalization one may choose a universe containing the countable fields and supports under consideration, and no truth predicate for the entire

universe is required (08). All index sets and supports are sets; no proper-class sum is formed (09).

1.4 Notation and conventions

Sign convention. Exponents are those of Conway normal forms: supports are reverse well ordered and larger exponents correspond to larger monomials. In the valuation convention $t^\gamma = \omega^{-\gamma}$, Hahn series have increasing well-ordered supports; the realization $\sum_g a_g t^g \mapsto \sum_g a_g \omega^{-g}$ carries a minus sign, which is essential (05, 09). All three sources use this growth convention, as do the omnific reports of the collection.

One meaning per symbol. The sources use several letters for different notions. The following choices are fixed for the whole report; the complete list of renamings is [table 2](#).

- $\deg_\omega x = \max \text{supp } x$ is the leading (greatest) exponent of $x \neq 0$ and $\text{lc}(x)$ its coefficient, as in the omnific Diophantine report. 08 wrote $\ell(x)$ and 09 $\lambda(x)$ for this. The order type of $\text{supp } x$, read in decreasing order, is $\text{ord}(\text{supp } x)$; 09 wrote $\ell(x)$ for that, so the letter ℓ is not used for either notion. In the t -convention the least exponent of a Hahn series Φ is its valuation $v(\Phi)$ (05's λ).
- ε_0 is always written out. 05 abbreviated it E ; 08's exponential series $E(t)$ is $\text{Ex}(t)$ here.
- k is the effective ordered coefficient field inside $\mathbb{R}$ (08's C , 05's K); K is 08's effective leading-data field, a subfield of $\mathbf{No}$. $\mathbb{R}_{\text{alg}}$ is the field of real algebraic numbers (05's $\mathbb{A}$).
- The code set of a notation system is N (08 and 09 wrote D). In the shielding theorem D is the constant ring and L the coefficient field (05 wrote F); $\mathcal{D} = \mathcal{H} \cap \mathbf{Oz}$ is the hereditary omnific core.
- The three fields: $F_r, \mathcal{F}_A, \mathcal{F}_\varepsilon$ (08, with omnific parts $R_r, \mathcal{R}_A, \mathcal{R}_\varepsilon$); $\mathcal{H}$, $\text{Frac}(\mathcal{H})$ and the ring $\mathcal{S}_{\varepsilon_0}$ (05; 05 wrote $\mathcal{F}$ for $\text{Frac}(\mathcal{H})$ and $\mathcal{C}$ for $\mathcal{S}_{\varepsilon_0}$); the rational-omega field $\mathcal{Q}$ with depth levels $\mathcal{Q}_d$ and omnific part $\mathcal{I} = \mathcal{Q} \cap \mathbf{Oz}$ (09's $\mathcal{F}, \mathcal{F}_d, \mathcal{A}$).
- $\tau_{\geq g}$ and $\tau_{>g}$ are the truncations keeping the exponents $\geq g$, respectively $> g$ (08's Tr , 09's τ).
- $h_e(n) = 1$ exactly when machine e halts *for the first time* at step n (08, 09); $\bar{h}_e(n) = 1$ exactly when e has halted *within* n steps (05, who wrote h_e). These are Reductions B and A of `cas:thm-zerotest`.
- $B = \omega^\omega / (1 - \omega^{-1}) = \sum_{n \geq 0} \omega^{\omega - n}$ is the same element in all three sources (05's B , 09's B , 08's $U \text{Geo}(t)$).

Promises. Eq_ν is equality on valid names (a promise problem), $\text{Eq}_\nu^{\text{chk}}$ is *checked* equality, which must also verify validity ([definition 2.2](#)). A promise algorithm may diverge or answer wrongly on inputs violating the promise (08).

1.5 Relationship with the repository and the literature

Conway normal forms and the characterization of omnific integers are classical [6, 10]; L'Innocente and Mantova develop a factorization theory for generalized power series and

omnific integers and settle a specific primality conjecture [13]. The subject here is not factorization but finite representations and their decision problems; no claim about the decidability of factorization, of elementary theories or of Diophantine solvability is inferred from the equality algorithms (05). Lurie studied the effective content of surreal algebra [14]; Hamkins’s account of joint work with Turetsky describes a much larger notion of computable surreal and records its rediscovery of Lurie’s work [11]. The fields here are deliberately smaller: the real trace of the rational–omega field is only $\mathbb{Q}$, and “computable” always refers to a stated representation (09). D-finite power-series algebra is classical [18], and Chen, Fang and van der Hoeven give a zero test for D-algebraic transseries [5]; the towers of Part I are not presented as the first zero test beyond rational functions (08).

Within the collection, the computable-surreals report (`docs/foundations-and-computation/computable-surreals`) separates structural cut names from numerical series names and stresses that equality and normalization must not be inferred from abstract algebraic closure; its no-go theorem says a successful alternative representation must, among other options, use a more informative real representation, which is the route all three systems here take. The computer-algebra report (`docs/foundations-and-computation/computer-algebra`, prefix `cas:`) already contains the effective rational monomial core, both halting reductions for coefficient streams, a two-scale sign obstruction and the $\Pi \oplus \mathbb{Z}$ floor. The omnific Diophantine report (`docs/surreal/omnific-diophantine-geometry`, prefix `odg:`) contains the floor, division with remainder and the Hilbert’s-tenth-problem transfer. The precise relations are in section 25.

2 Normal forms, notation systems, and the floor

2.1 The semantic domain

We use Conway’s monomial map $x \mapsto \omega^x$, an order-preserving homomorphism from $(\mathbf{No}, +)$ to the positive monomials with $\omega^{x+y} = \omega^x \omega^y$ and $\omega^0 = 1$; it is not an unspecified extension of ordinary exponentiation and is not defined as $\exp(x \log \omega)$ (05, 08, 09). Every surreal has a unique normal form

$$x = \sum_{\gamma \in S} c_\gamma \omega^\gamma, \quad c_\gamma \in \mathbb{R} \setminus \{0\}, \quad (2.1)$$

where $S = \text{supp } x \subseteq \mathbf{No}$ is a set and is reverse well ordered: each nonempty subset has a greatest element. For $x \neq 0$ put $\deg_\omega x = \max S$ and $\text{lc}(x) = c_{\deg_\omega x}$; the sign of x is the sign of $\text{lc}(x)$, and

$$\deg_\omega(xy) = \deg_\omega x + \deg_\omega y, \quad \text{lc}(xy) = \text{lc}(x) \text{lc}(y). \quad (2.2)$$

No leading exponent is assigned to zero. Addition collects equal exponents; multiplication uses $\omega^\gamma \omega^\eta = \omega^{\gamma+\eta}$ and a finite convolution at each exponent. These operations are justified by the standard support lemmas for generalized power series [6, 10, 13]. For a set-sized ordered additive subgroup $G \subseteq \mathbf{No}$ we write $k((t^G))$ for the Hahn field of series with well-ordered support in the t -convention; we use the Hahn support and product theorem: sums of well-ordered supports are well ordered, and the coefficient fibres of a product are finite. For a finite set of strictly positive g_i the nonnegative integer monoid they generate is well ordered and each of its elements has finitely many representations $\sum n_i g_i$; this is the positive-support form of Neumann’s lemma (05; see also [3, Proposition 3.7]).

Strong sums are not limits. An indexed family of normal forms is strongly summable if the union of the supports is reverse well ordered and only finitely many summands contribute at each exponent; its sum is then defined coefficientwise. Every infinite sum in this report is

meant in this sense; its existence is a support assertion, not convergence of partial sums in the order topology of $\mathbf{No}$, and no such topological convergence is needed anywhere (05, 08, 09).

Definition 2.1 (Omnific integers). The class of omnific integers is

$$\mathbf{Oz} = \{x \in \mathbf{No} : \text{supp } x \subseteq \mathbf{No}_{\geq 0} \text{ and } \text{ct}(x) \in \mathbb{Z}\}, \quad (2.3)$$

where $\text{ct}(x)$ is the coefficient of ω^0 , read as zero when $0 \notin \text{supp } x$. Equivalently $z = m + \sum_{\gamma \in S} c_\gamma \omega^\gamma$ with $m \in \mathbb{Z}$ and $S \subseteq \mathbf{No}_{>0}$ reverse well ordered. The positive-exponent coefficients need not be ordinary integers.

This is the classical criterion [6, 10, 13], and it agrees with the collection's $\mathbf{Oz} = \Pi \oplus \mathbb{Z}$ (`found:eq:omnific`). For example $\sqrt{2}\omega - \frac{1}{3}\omega^{1/2} + 7$, $\omega - 1$ and $\omega^{1/2}$ are omnific integers, whereas $\frac{1}{2}$, $\omega + \frac{1}{2}$ and $\omega \pm \omega^{-1}$ are not (08, 05). The support criterion shows that $\mathbf{Oz}$ is a subring of $\mathbf{No}$: a sum of nonnegative exponents is zero only if both are zero, so $\text{ct} : \mathbf{Oz} \rightarrow \mathbb{Z}$ is a ring homomorphism which is the identity on $\mathbb{Z}$ (09). $\mathbf{Oz}$ is discretely ordered: a positive element with nonempty positive support exceeds every real number, and otherwise it is a positive ordinary integer, so 1 is its least positive element. These are elementary classical consequences, not new theorems (09).

It is useful to write

$$x = x_{>0} + \text{ct}(x) + x_{<0}, \quad (2.4)$$

where $x_{>0}$ collects the terms with positive exponents and $x_{<0}$ those with negative exponents. The constant term is an ordinary real number, $x_{<0}$ is zero or infinitesimal ($|x_{<0}| < 1/n$ for every positive integer n), and $x_{>0}$ is zero or purely infinite. This is a normal-form decomposition; $x_{>0}$ is not generally the positive part in the order-theoretic sense (08).

2.2 The omnific floor

For every surreal x there is exactly one $z \in \mathbf{Oz}$ with $z \leq x < z + 1$, namely

$$\lfloor x \rfloor_{\mathbf{Oz}} = x_{>0} + \lfloor \text{ct}(x) \rfloor - \begin{cases} 1, & \text{ct}(x) \in \mathbb{Z} \text{ and } x_{<0} < 0, \\ 0, & \text{otherwise.} \end{cases} \quad (2.5)$$

This is the integer-part formula `odg:thm:floor` of the omnific Diophantine report, credited there to Ehrlich and Kaplan, and the $\Pi \oplus \mathbb{Z}$ floor `cas:eq-floor` of the computer-algebra report (whose ring $\Pi \oplus \mathbb{Z}$ is $\mathbf{Oz}$); both reports prove it, and all three sources prove it again (05's Theorem 3.2, 08's Theorem 7.4, 09's equation (8)). It is printed once, here. The omnific continued-fraction report proves it again as `ocf:lem:floor` (batch 33), where it is the digit map of the continued-fraction algorithm. If $\text{ct}(x)$ is not an integer, its distances from the adjacent integers are positive reals which an infinitesimal cannot cross; if it is an integer, the sign of $x_{<0}$ selects the side. The correction is not optional:

$$\begin{aligned} \left\lfloor -\omega + 3 - \omega^{-1} \right\rfloor_{\mathbf{Oz}} &= -\omega + 2, & \left\lfloor \omega - \omega^{-1} \right\rfloor_{\mathbf{Oz}} &= \omega - 1, \\ \left\lfloor \omega + \frac{3}{2} - \omega^{-1} \right\rfloor_{\mathbf{Oz}} &= \omega + 1, & \left\lfloor \omega + \frac{1}{2} - \omega^{-1} \right\rfloor_{\mathbf{Oz}} &= \omega, \end{aligned}$$

and deleting the negative-exponent terms and flooring the constant gives the wrong answer in the first two cases (08, 05, 09). What each system of this report adds is *effectivity*: its names allow $x_{>0}$, $\text{ct}(x)$ and the sign of $x_{<0}$ to be computed as names of the same system ([theorems 7.4, 11.2 and 16.3](#)). The computer-algebra report records that the operation is effective only when the backend can split the normal form, decide the ordinary integer predicate and determine the infinitesimal sign; the three systems supply exactly these.

2.3 Notation systems, validity, and canonical codes

Definition 2.2 (Notation system). A notation system consists of a decidable set N of finite syntactic codes, a subset $V \subseteq N$ of semantically valid codes, and a denotation map $\nu : V \rightarrow \mathbf{Oz}$ (or into $\mathbf{No}$ for an ambient field). Its equality problem is the promise problem

$$\text{Eq}_\nu = \{(a, b) \in V^2 : \nu(a) = \nu(b)\}$$

on valid inputs. Its *checked equality* relation on all syntactic inputs is

$$\text{Eq}_\nu^{\text{chk}} = \{(a, b) \in N^2 : a, b \in V \text{ and } \nu(a) = \nu(b)\}.$$

The system is total if $V = N$. A canonicalization is a computable map $\text{can} : V \rightarrow V$ with $\nu(\text{can}(a)) = \nu(a)$, and $\text{can}(a) = \text{can}(b)$ exactly when $\nu(a) = \nu(b)$.

Four problems are logically distinct: recognizing valid codes, comparing values, producing a preferred code for a value, and deciding whether a value belongs to a specified subring; a promise such as “this program is total” is not a validity algorithm (09). A parser may decide membership in a syntactic language without deciding semantic validity; a comparison procedure may be required to work only under a validity promise; a code for an exact coefficient is different from a code for approximations to it. These alternatives change the decision problem and are never left implicit (05). A coefficient-access algorithm computes a specified coefficient or a specified finite collection of coefficients; it does not by itself certify that all remaining coefficients vanish, and a support generator is not automatically a support-validity certificate (08).

Proposition 2.3 (Canonicalization and decidable quotients [08; 05, 09]). *For a total notation system with effectively coded finite strings, equality is decidable if and only if it has a computable canonicalization.*

Proof. A canonicalizer decides equality by comparing its outputs. Conversely, number all finite strings by natural numbers. For an input code a , test the finitely many codes $b \leq a$ for syntactic validity and for semantic equality to a , and return the least successful one. The tested set contains a , so it is nonempty. The result is the least code in the whole equivalence class, since no code larger than a can be the least; equal inputs therefore give the same result. $\square$

This supplies a canonical form in the computability sense, not a useful algebraic expression or an efficient normalizer (08). A canonical denotation in the mathematical sense does not supply an algorithm to find its canonical code; the least-code search may be useless computationally, but it separates existence from efficient normal-form design (05). A reduced rational function is canonical only relative to a fixed exponent coordinate system (09).

Proposition 2.4 (Size obstruction [08, 05, 09]). *No notation system over a finite or countable alphabet names every omnific integer, or even every $c\omega$ with $c \in \mathbb{R}$. More generally, the image of any set of names is set-sized.*

Proof. Finite strings over a countable alphabet form a countable set, whereas $c \mapsto c\omega$ is injective on $\mathbb{R}$. Moreover $\mathbf{Oz}$ contains every ordinal and is a proper class; replacement gives the general assertion. $\square$

This obstruction is not an undecidability theorem, and it does not prevent large useful countable subrings. A language containing a symbol for a very large specified ordinal may have trivial equality on that symbol; restrictions involving ω_1^{CK} concern computably presented well-orders, not the largest ordinal one may mention by a finite semantic convention (05).

2.4 A sparse baseline

Let $G \subseteq (\mathbf{No}, +, <)$ be an effectively presented ordered abelian group with fixed embedding and group operations. Over k , use finite records

$$(m; (\gamma_1, c_1), \dots, (\gamma_s, c_s)), \quad m \in \mathbb{Z}, \gamma_i > 0, \quad (2.6)$$

allowing zero coefficients and repeated exponents in raw records; sort the exponents decreasingly, collect equal exponents and delete zero coefficients.

Proposition 2.5 (Sparse normalization [08]). *The records (2.6) have decidable validity, equality and order, and computable ring operations. After collection and sorting, their list of exponent-coefficient values and their integer constant are unique.*

Proof. Only finitely many comparisons and coefficient operations occur. Normal-form uniqueness gives equality and uniqueness. The sign of the first nonzero term decides order; if no positive term remains, the integer constant decides it. Products have exponents in the finite sumset of the input supports, and positive exponents stay positive. $\square$

If coefficient and exponent codes are not themselves canonical, this is uniqueness of values rather than of literal strings. The sparse core is familiar generalized-polynomial algebra and already occurs in the computer-algebra report (`cas:thm-core`) (08). Adding infinite series is not adding longer lists:

$$\sum_{n \geq 0} a_n \omega^{\omega^{-n}} = \omega^\omega \sum_{n \geq 0} a_n (\omega^{-1})^n \quad (2.7)$$

always has positive exponents and a valid support, whatever the real a_n , but an algorithm for each a_n need not decide whether every a_n vanishes. A finite certificate can supply the missing information (Part I), including the singular initial indices, which cannot simply be ignored (08). The holonomic-rigidity report reads the series (2.7) as a recurrence observable (batch 33). Starting from y_0 equal to that series, the iterates $y_n = \omega^n y_0$ under multiplication by ω stay purely infinite, and the coefficient of ω^ω in y_n is a_n (`hol:rc:thm:coding`). So the fixed-coefficient zero sets of first-order omnific recurrences realize every subset of the natural numbers, and no coefficientwise Skolem–Mahler–Lech theorem holds for them (`hol:rc:cor:nosml`). That report credits the series to this equation.

3 Shielding and the transfer of equality

This section separates the source of equality difficulty from the fact that the output is an omnific integer. The ring form $D + XL[X]$ is a familiar polynomial pullback construction; the concern here is its surreal realization and its effective equality (05).

Let $k \subseteq \mathbb{R}$ be an ordered field, $G \subseteq \mathbf{No}$ a set-sized ordered additive subgroup, and $L \subseteq k((t^G))$ a subfield, realized in $\mathbf{No}$ by $\sum a_g t^g \mapsto \sum a_g \omega^{-g}$. Let $D \subseteq L \cap \mathbf{Oz}$ be a subring containing $\mathbb{Z}$. Choose a positive surreal Λ with

$$|g| < \Lambda \quad (g \in G). \quad (3.1)$$

Such a bound exists because G is a set; its existence is not a uniform algorithm for extracting a bound from arbitrary programs, and in an effective application Λ and a justification of the bound are part of the chosen presentation. If $G \subseteq \mathbb{R}$, the fixed choice $\Lambda = \omega$ works. Put $X = \omega^\Lambda$ and

$$S_\Lambda(D, L) = \left\{ d + \sum_{j=1}^m X^j f_j : d \in D, f_j \in L, m \in \mathbb{N} \right\}. \quad (3.2)$$

Here X is a surreal; in the abstract polynomial ring below, X is an indeterminate.

Theorem 3.1 (Shielding and unique degree separation [05]). *Evaluation at $X = \omega^\Lambda$ induces an injective ring homomorphism*

$$D + XL[X] \hookrightarrow \mathbf{Oz}$$

with image (3.2). Every element has a unique finite coefficient list, up to trailing zeros. If its largest nonzero positive degree is m , its sign is the sign of f_m ; if no positive degree survives, its sign is that of d .

Proof. Expand $f_j = \sum_g a_{j,g} \omega^{-g}$. Then

$$X^j f_j = \sum_g a_{j,g} \omega^{j\Lambda - g}. \quad (3.3)$$

For $j \geq 1$ every exponent is strictly positive by (3.1), and the support is reverse well ordered because $g \mapsto j\Lambda - g$ reverses the well-ordered Hahn support; a finite union of such supports is reverse well ordered. The degree-zero element d is omnific by assumption, so the evaluated expression is omnific.

If $j > i$ and $g, h \in G$, then

$$(j\Lambda - g) - (i\Lambda - h) = (j - i)\Lambda - (g - h) > 0,$$

since $g - h \in G$. Thus every exponent in a higher degree exceeds every exponent in a lower degree, including degree zero, whose exponents lie in $-G$. Different degree blocks cannot cancel, and normal-form uniqueness gives injectivity and uniqueness of the coefficient list. The leading coefficient of the highest nonzero block is that of f_m , which gives the sign. The finite polynomial identities together with Hahn multiplication show that evaluation is a ring homomorphism. $\square$

For $D = \mathbb{Z}$ the omnific constant term is simply d ; in general it is $\text{ct}(d)$. The construction does *not* embed L as a subfield of $\mathbf{Oz}$, since a discretely ordered ring cannot contain $\frac{1}{2}$; rather $f \mapsto Xf$ is an injective additive encoding with $(Xf)(Xg) = X^2fg$ (05).

Remark 3.2 (Embedding and characterization [merge]). **Theorem 3.1** proves one inclusion: the image of $D + XL[X]$ lies in $\mathbf{Oz}$. It does not say that this image is the whole omnific part of a field containing L and X . Two such characterizations are proved in this report, each in its own field and each by an argument not contained in **theorem 3.1**.

- In the holonomic tower (**theorem 7.2**) take $L = F_{r-1}$, $D = R_{r-1}$, $G = G_{r-1}$ and $\Lambda = \delta_r$: every element of G_{r-1} is dominated by δ_r , so (3.1) holds and $\mathcal{S}_{\delta_r}(R_{r-1}, F_{r-1}) = R_{r-1} + U_r F_{r-1}[U_r]$. The theorem there adds $F_r \cap \mathbf{Oz} = R_r$ equal to this ring.
- In a separated rational grid (**theorem 16.2**) take L the tail field $\mathbb{Q}(X_2, \dots, X_r)$, D its omnific part and $\Lambda = \gamma_1$; since $\gamma_1 \gg \gamma_2 \gg \dots$, every element of the tail lattice is dominated by γ_1 . The theorem there adds that the omnific part of $\mathbb{Q}(X_1, \dots, X_r)$ equals this ring.

The converse inclusion is where both proofs use the field: an element of the field whose top Laurent block has positive index, or whose grid expansion has a negative first coordinate, has a negative exponent and so is not omnific.

3.1 Exact transfer of equality

Assume L has a system of valid finite names on which $+$, $-$ and multiplication are uniformly computable, and that D -names translate computably to L -names. No equality algorithm for L is assumed. Polynomial-list names for the shielded ring allow zero coefficients; they need not be normalized.

Theorem 3.3 (Many-one equality transfer [05]). *Under these representation contracts the valid-name promise problems satisfy*

$$\text{Eq}_{\mathcal{S}_\Lambda(D,L)} \equiv_m \text{Eq}_L,$$

by computable maps on promised valid inputs. If both naming domains are decidable, these are ordinary total many-one reductions, invalid strings being rejected, and decidability and every many-one completeness classification of Eq_L transfer to the shielded omnific notation system. Without decidable validity the assertion concerns promised equality, not the complexity of recognizing valid names.

Proof. For the forward reduction map (f, g) to (Xf, Xg) ; injectivity in [theorem 3.1](#) proves equivalence. Conversely, pad two coefficient lists with zeros to the same length, put $h_0 = d - e$ and $h_j = f_j - g_j$ for $j \geq 1$, and compute the single coefficient-field expression

$$H = h_0^2 + h_1^2 + \cdots + h_m^2. \quad (3.4)$$

In an ordered field $H = 0$ exactly when every $h_j = 0$, and degree separation identifies this with equality of the two omnific values. The output pair $(H, 0)$ is a single instance of Eq_L , not a list of oracle queries; only the stipulated arithmetic and translation operations were used. If the naming domains are decidable, first reject invalid inputs and map them to a fixed unequal pair of valid names; this extends the promised reductions to total reductions of the ordinary equality sets. $\square$

The field L is ordered, being a subfield of the ordered Hahn field $k((t^G))$ with $k \subseteq \mathbb{R}$, so the sum-of-squares step needs no hypothesis beyond those stated. Over an arbitrary coefficient field, individual coefficient tests still give a finite truth-table reduction, but this many-one argument may fail (05). If equality and order in L are decidable, comparison in the shielded ring follows by finding the highest nonzero degree and applying the sign rule; the abstract existence of L does not by itself make its raw-name equality decidable.

Part I

Certified holonomic towers

This part is source 08's. It builds effective fields inside **No** by iterating fraction fields of certified D-finite series, and describes their omnific parts, floors, supports and ordinals exactly. The base field is an effective ordered subfield $k \subseteq \mathbb{R}$ ([subsection 1.3](#)).

4 Computing integer singularities over non-Archimedean fields

The recursive construction needs to solve one modest problem: given $P(X) \in K[X] \setminus \{0\}$, compute all nonnegative ordinary integers n with $P(n) = 0$, although K is non-Archimedean.

Bounding roots by the absolute values of the coefficients does not solve this: the bound may be an infinite surreal with no ordinary integer above it.

Definition 4.1 (Effective leading-data field). An effective leading-data field over k is an effectively presented ordered subfield $K \subseteq \mathbf{No}$ with support in an effective ordered exponent group G , together with algorithms that compute $\deg_\omega a \in G$ and $\text{lc}(a) \in k$ for $a \neq 0$.

At rank zero, $K = k$: every nonzero element has exponent zero and is its own leading coefficient.

Lemma 4.2 (Archimedean integer search [08]). *Given $c \in k$, an ordinary integer greater than $|c|$ can be found effectively, and $\lfloor c \rfloor \in \mathbb{Z}$ can be computed.*

Proof. Test $2^j > |c|$ for $j = 0, 1, \dots$; Archimedeaness guarantees termination. Within the resulting finite integer interval, exact comparisons determine the greatest integer at most c ; binary search is optional. $\square$

Theorem 4.3 (Leading-polynomial integer-root algorithm [08]). *Let K be an effective leading-data field over k . There is an algorithm that, given $0 \neq P \in K[X]$, returns the finite set*

$$I(P) = \{n \in \mathbb{N} : P(n) = 0\},$$

and in particular computes $B(P) = \max(I(P) \cup \{-1\})$.

Proof. Write $P(X) = \sum_{i=0}^m a_i X^i$, omitting zero coefficients when taking leading exponents, and put

$$g = \max_{a_i \neq 0} \deg_\omega a_i, \quad p(X) = \sum_{\deg_\omega a_i = g} \text{lc}(a_i) X^i \in k[X].$$

The polynomial p is nonzero, since its contributions sit at different powers of X . For each ordinary $n \geq 0$ the coefficient of ω^g in $P(n)$ is exactly $p(n)$, so

$$P(n) = 0 \implies p(n) = 0, \tag{4.1}$$

including $n = 0$. If p is a nonzero constant, return the empty set. Otherwise write $p(X) = b_d X^d + \dots + b_0$ with $b_d \neq 0$ and choose an integer $M > 1 + \max_{i < d} |b_i/b_d|$ by [lemma 4.2](#). By the elementary Cauchy bound p has no real root of absolute value at least M : for $|x| > 1 + \max_{i < d} |b_i/b_d|$ the lower-degree terms have total absolute value below $|b_d x^d|$ by the geometric-series estimate. Thus (4.1) reduces the problem to $n = 0, \dots, M$; evaluate $P(n)$ in K and keep exactly its zeros. $\square$

The theorem does not compute all roots in K and does not require K to be real closed. The leading polynomial is only a filter: cancellation at one of its roots can leave a nonzero lower-scale term, which is why the final exact tests are needed.

Remark 4.4 (Relation to the holonomic-rigidity report [merge]). The same leading-polynomial reduction is `hol:lem:integer` (“integer evaluation”) of the holonomic-rigidity report (`docs/surcomplex/holonomic-rigidity-for-entire-hahn-functions`), which shows that $P(n) \neq 0$, with valuation equal to the least coefficient valuation, for all but finitely many n . That lemma is not effective: it does not bound the exceptional n . [Theorem 4.3](#) adds the Cauchy bound on the reduced polynomial and the exact tests, which make it an algorithm over any effective leading-data field. 08 did not cite the holonomic-rigidity report; the credit is the merge’s.

Example 4.5. In a field containing $t = \omega^{-1}$ let $P(X) = (X - 3)\omega + (X - 2)$. Its leading polynomial is $X - 3$; the only candidate is 3, but $P(3) = 1$, so $I(P) = \emptyset$. By contrast $(X - 3)(\omega + 1)$ has the same leading polynomial and the integer root 3.

5 Certified differential-series names

Fix an effective leading-data field K and an independent formal variable t . Differentiation with respect to t kills K ; put $\theta = t d/dt$.

5.1 Euler-normalized operators

An Euler-normalized operator is

$$\mathbf{L} = \sum_{j=0}^J t^j P_j(\theta), \quad P_j \in K[X], \quad P_0 \neq 0. \quad (5.1)$$

Every nonzero linear differential operator over $K(t)$ can be put into this form without changing its power-series solutions: clear rational-function denominators, use

$$t^a \left(\frac{d}{dt} \right)^b = t^{a-b} \theta(\theta-1) \cdots (\theta-b+1), \quad (5.2)$$

collect equal powers of t , and multiply by a power of t to make the least occurring power zero. Multiplying the equation by a nonzero rational function does not change its solution set in $K((t))$, and the triangular change of basis between powers and falling factorials keeps a nonzero operator nonzero. For $f = \sum_{n \geq 0} a_n t^n$ the equation $\mathbf{L}f = 0$ is equivalent to

$$P_0(n)a_n + \sum_{j=1}^J P_j(n-j)a_{n-j} = 0 \quad (n \geq 0), \quad a_m = 0 \quad (m < 0). \quad (5.3)$$

The argument of P_j is $n-j$, not n .

Definition 5.1 (Certified holonomic name). A certified holonomic name over K is an operator (5.1) together with the finite jet $(a_0, \dots, a_B)$, $B = B(P_0)$, empty if $B = -1$, such that every equation (5.3) with $0 \leq n \leq B$ holds. A well-formed but incompatible jet is invalid and is rejected.

Theorem 5.2 (Finite certification, zero testing, and leading data [08]). *A certified holonomic name has a unique denotation in $K[[t]]$. Validity, coefficient extraction and the zero test are effective; explicitly,*

$$f = 0 \iff a_0 = \cdots = a_B = 0. \quad (5.4)$$

For a nonzero denotation the least nonzero index is the least nonzero index in the jet, hence at most B . Conversely, every D-finite power series over K has such a name.

Proof. Compute B by [theorem 4.3](#); there are finitely many compatibility equations to check. For $n > B$, $P_0(n) \neq 0$, so

$$a_n = -P_0(n)^{-1} \sum_{j=1}^J P_j(n-j)a_{n-j} \quad (5.5)$$

is a uniquely determined finite computation from earlier coefficients. It gives a formal series satisfying every coefficient equation, hence $\mathbf{L}f = 0$, and any solution with that jet satisfies the same recursion, which proves uniqueness. If the jet is zero, induction in (5.5) gives the zero series; the converse of (5.4) is immediate. If $B = -1$ the induction starts at $n = 0$ and the only solution is zero. A D-finite series satisfies a nonzero linear differential equation over $K(t)$; normalize it as above and supply the jet of that series through the computed bound. $\square$

“Every D-finite series” refers to series with coefficients in K , whose finitely many initial coefficients have K -names. No algorithm is asserted for recognizing that an arbitrary coefficient program is D-finite, or for finding its differential equation (08).

Example 5.3 (The order is not a jet bound [08]). For an ordinary integer $M > 0$, $L = \theta(\theta - M)$ has the solutions 1 and t^M . Its order is two, but t^M agrees with zero through index $M - 1$; the bound from $P_0(X) = X(X - M)$ is $B = M$, and a name for t^M must include its exceptional coefficient at M . No valid argument replaces the computed singular-index bound by the operator order.

5.2 Closure algorithms, not just closure assertions

Let $\text{DF}_K[[t]]$ be the ring of D-finite series over K . Its familiar closure theorem [18] has the following effective form.

Theorem 5.4 (Effective addition and multiplication [08]). *From certified names for $f, g \in \text{DF}_K[[t]]$ one can compute certified names for $f + g$, $-f$ and fg . No oracle for algebraic independence or for minimal differential equations is needed.*

Proof. A nonzero series satisfying an operator of order $r \geq 1$ has a derivative vector $v_f = (f, f', \dots, f^{(r-1)})^\top$ with $v'_f = A_f v_f$, where A_f is the companion matrix over $K(t)$; the zero case is decidable and treated separately. For a sum use the direct sum $A_f \oplus A_g$ with an output row selecting $f + g$; for a product use $A = A_f \otimes I + I \otimes A_g$ on $v_f \otimes v_g$. These systems have dimensions $r + s$ and rs . If $h = c_0 v$ in a system $v' = Av$, define $c_{i+1} = c'_i + c_i A$, so $h^{(i)} = c_i v$. In dimension q the rows $c_0, \dots, c_q$ are linearly dependent over $K(t)$, and Gaussian elimination finds a nontrivial relation $\sum_i b_i(t) c_i = 0$, giving the nonzero annihilator $\sum_i b_i(t) (d/dt)^i$ of h . All rational-function operations and zero tests reduce to operations in K . Normalize the operator, compute its bound B , and compute the jet by finite coefficient addition or Cauchy convolution; the genuine sum or product is a solution, so the jet is compatible, and [theorem 5.2](#) verifies the name. Negation keeps the operator and negates the jet. $\square$

Repeated operator construction can be expensive and can produce nonminimal orders; that affects performance, not termination. Two different equations for the same series are compared by computing an annihilator and the finite jet of their difference, not by comparing the operators literally (08).

5.3 Passing to a field

D-finite series do not in general form a field, so we take

$$H(K, t) = \text{Frac}(\text{DF}_K[[t]]) \subseteq K((t)). \quad (5.6)$$

A name is a pair (p, q) of certified power-series names with $q \neq 0$, a decidable condition by [theorem 5.2](#).

Proposition 5.5 (Effective fraction field [08]). *$H(K, t)$ has computable arithmetic and equality; its Laurent valuation, leading coefficient in K and individual Laurent coefficients are computable.*

Proof. Use the fraction formulas and [theorem 5.4](#); equality is $p/q = r/s \iff ps - rq = 0$. For a nonzero fraction write $p = t^a(p_a + p_{a+1}t + \dots)$ and $q = t^b(q_b + q_{b+1}t + \dots)$ with $p_a q_b \neq 0$, reading a, b from the certified jets; the valuation is $a - b$ and the leading coefficient p_a/q_b . A coefficient at any specified Laurent index follows by triangular series division after removing t^b , using finitely many earlier coefficients and division by $q_b \neq 0$. $\square$

A reciprocal in $H(K, t)$ is represented as a fraction; it is not asserted to be D-finite. Rational functions are included, and every D-finite Laurent series belongs to the field after multiplication by a power of t .

6 Finite-rank holonomic towers inside the surreal field

6.1 Scales and their order

Choose a finite increasing list of ordinal labels $\alpha_1 < \dots < \alpha_r$ and set

$$\delta_j = \omega^{\alpha_j}, \quad t_j = \omega^{-\delta_j}, \quad U_j = t_j^{-1} = \omega^{\delta_j}, \quad G_j = \mathbb{Z}\delta_1 + \dots + \mathbb{Z}\delta_j. \quad (6.1)$$

The coefficients in G_j are ordinary integers and addition is surreal addition. The δ_j are linearly independent over $\mathbb{Q}$ by normal-form uniqueness, and their integer span is ordered lexicographically from the *largest* index: the sign of a nonzero vector is the sign of its last nonzero coordinate, so every positive multiple of δ_j dominates G_{j-1} . The labels are not exponents of a formal analytic exponential; they specify actual Conway monomials and so a definite embedding of the exponent lattice. Different label choices can have the same finite-rank algorithmic structure but very different magnitudes. Define

$$F_0 = k, \quad F_j = H(F_{j-1}, t_j) \quad (1 \leq j \leq r). \quad (6.2)$$

At level j the field F_{j-1} is constant for differentiation in t_j . The recursion on levels is essential: an operator may not use its own unknown denotation among its coefficients.

Lemma 6.1 (Iterated Laurent embedding [08]). *There is an injective ordered-field embedding $k((t_1)) \cdots ((t_r)) \rightarrow \mathbf{No}$ with the substitutions (6.1). For $f = \sum_{n \geq N} a_n t_r^n \neq 0$ with $a_N \neq 0$,*

$$\deg_\omega f = \deg_\omega a_N - N\delta_r, \quad \text{lc}(f) = \text{lc}(a_N). \quad (6.3)$$

Proof. Induct on r . Every exponent of $a_n t_r^n$ has the form $g - n\delta_r$ with $g \in G_{r-1}$. If $n < m$, every exponent of the n th block exceeds every exponent of the m th, because $(m - n)\delta_r$ dominates G_{r-1} . Each block is reverse well ordered by induction and the blocks are indexed by a set of integers bounded below, so the ordered union is reverse well ordered; distinct integer vectors give distinct exponents, so blocks do not overlap. In a product, a prescribed top coordinate receives finitely many pairs of top indices, and within each pair the lower-rank convolution is legitimate by induction. Nonzero series have the leading term (6.3), which gives injectivity and order preservation, and a field homomorphism respects inverses. Rank zero is $k \subseteq \mathbb{R} \subseteq \mathbf{No}$. $\square$

Theorem 6.2 (Effective holonomic tower [08]). *For every finite list of labels, (6.2) gives a countable subfield $F_r \subseteq \mathbf{No}$ with a decidable finite-name representation, and computable field operations, equality, order, leading exponent, real leading coefficient and coefficient extraction at each specified exponent of G_r .*

Proof. Rank zero is the contract for k . Suppose the assertions hold at rank $j - 1$. Theorem 4.3 computes the exceptional indices of every Euler operator over F_{j-1} , so certificates over that field have decidable validity and zero tests, and theorem 5.4 and proposition 5.5 give the field algorithms at rank j . Lemma 6.1 supplies the denotation and computes the leading exponent and real leading coefficient by descending one rank; the sign is that of the latter. To extract the coefficient at $g = m\delta_j + h$, take the Laurent coefficient at index $-m$ and then its coefficient at $h \in G_{j-1}$. Each algorithm calls lower-rank algorithms and finitely many same-rank operations, never an unconstructed next rank. The codes are finite trees, so there are countably many. $\square$

There is no claim of analytic convergence: a formal factorial series is as legitimate as an exponential series. Nor is real closedness asserted: the tower is a specific effective subfield, not an algebraic closure or the field of all surreals with a given support group (08).

6.2 A modular generalization

The role of differential equations can be isolated. Suppose a class $\mathfrak{C}(K, t) \subseteq K[[t]]$ has decidable finite names, computable ring operations and coefficients, and a computable finite zero-test bound for each name, and contains $K[t]$. Then the same fraction-field and finite-rank construction applies, provided the resulting fields carry the leading data needed at the next stage; certified D-finite series are one completely specified realization. This permits later replacement by stronger certified classes without changing the omnific arguments below. It does *not* justify inserting arbitrary differential-equation solvers: a zero test, a chosen-solution convention and terminating coefficient algorithms remain hypotheses (08). This is stated as an observation, not as a theorem with proof.

7 Truncation, triangular integer forms, and floor

7.1 Finite descriptions of infinite truncations

For $g \in G_r$, $\tau_{\geq g}(f)$ is the sum of the terms of f with exponent at least g , and $\tau_{>g}(f)$ the sum of those above g ; a truncation need not have finite support.

Theorem 7.1 (Effective truncation closure [08]). *F_r is effectively closed under $\tau_{\geq g}$ and $\tau_{>g}$ for every $g \in G_r$. In particular $f_{>0}$, $\text{ct}(f)$ and $f_{<0}$ are computable as finite names.*

Proof. Rank zero is immediate. At rank r take $f = \sum_{n \geq N} a_n t_r^n$ and $g = m\delta_r + h$ with $h \in G_{r-1}$; for $f = 0$ return zero. Block dominance gives

$$\tau_{\geq g}(f) = \sum_{N \leq n < -m} a_n t_r^n + t_r^{-m} \tau_{\geq h}(a_{-m}), \quad (7.1)$$

with an empty sum zero and coefficients below N zero; the same formula with $\tau_{>h}$ in the last term gives $\tau_{>g}(f)$. The first sum is finite, each coefficient has an effective F_{r-1} -name, and the last term is available by induction. Constant extraction is the iterated coefficient at zero, and subtraction gives the negative part. $\square$

An exponent cutoff and a number-of-terms cutoff are different interfaces: (7.1) returns a finite *expression* for the exact truncation, whose support can have order type ω^{r-1} or more; it does not print an infinite list as a finite one (08).

7.2 The exact omnific part

Put $R_r = F_r \cap \mathbf{Oz}$, with $R_0 = \mathbb{Z} \subseteq k$.

Theorem 7.2 (Triangular omnific normal form [08]). *For each $r \geq 1$,*

$$R_r = R_{r-1} + U_r F_{r-1}[U_r]. \quad (7.2)$$

Every element of R_r has a unique finite decomposition by coefficient values

$$z = m + \sum_{j=1}^r \sum_{i=1}^{d_j} U_j^i a_{j,i}, \quad m \in \mathbb{Z}, \quad a_{j,i} \in F_{j-1}, \quad (7.3)$$

after deleting trailing zero coefficients at each level. Recognition of R_r inside F_r and computation of this decomposition are decidable, and arithmetic, equality and order in this notation are computable.

Proof. Write $f = \sum_{n \geq N} a_n t_r^n$. All exponents of a nonzero block with $n < 0$ are positive, since $-n\delta_r$ dominates every lower-rank exponent; all exponents of a nonzero block with $n > 0$ are negative; the block $n = 0$ has exactly its lower-rank normal form. So f is omnific if and only if the positive- t_r tail is zero and $a_0 \in R_{r-1}$. There are finitely many negative Laurent indices, with arbitrary coefficients in F_{r-1} . This proves (7.2) inductively.

For the algorithm, extract the finite nonpositive Laurent prefix and compare it with f by the field equality test; if they differ, the positive tail is nonzero and $f \notin R_r$. Otherwise apply the lower-rank recognition to a_0 . At rank zero, lemma 4.2 computes $\lfloor c \rfloor$ and decides whether it equals c , which is membership in $\mathbb{Z}$. Different top indices have disjoint support blocks, so equality of two such expressions forces equality of every top coefficient and of the constant block; descending the ranks proves uniqueness. The field algorithms decide arithmetic and order, and results are normalized again. Closure also follows from (7.2): products of positive-degree polynomials keep positive top degree, and R_{r-1} is a subring of the coefficient field. $\square$

The triangular form is a finite outer normal form with exact field coefficients, not a claim that every coefficient has a finite Conway normal form. Applying proposition 2.3 to the coefficient names would give a literal canonical string, which is not needed for the equality test (08). In the language of section 3, (7.2) says that the omnific part of F_r is exactly the shielded ring $\mathcal{S}_{\delta_r}(R_{r-1}, F_{r-1})$ (remark 3.2).

Corollary 7.3 (Normal-form membership test [08]). *For $x \in F_r$ let $p = x_{>0}$, $c = \text{ct}(x)$ and $\epsilon = x_{<0}$. Then $x \in R_r$ iff $\epsilon = 0$ and $c \in \mathbb{Z}$. Both tests are effective, and $p \in R_r$ always.*

Theorem 7.4 (Computable omnific floor [08]). *For every $x \in F_r$ the omnific floor $\lfloor x \rfloor_{\mathbf{Oz}}$ lies in R_r and is computable; with the decomposition of corollary 7.3 it is given by (2.5).*

Proof. By theorem 7.1 the names of p , c and ϵ are computable; $\lfloor c \rfloor$ and the test $c \in \mathbb{Z}$ are computable at rank zero by lemma 4.2, and the sign of ϵ by theorem 6.2. The right side of (2.5) lies in R_r by corollary 7.3, and it is the floor by subsection 2.2. $\square$

8 Support order types and the ordinal part

8.1 Sharp finite-rank support bounds

$\text{ord}(\text{supp } f)$ is the order type of the support read in decreasing exponent order, zero for $f = 0$. It is a property of the normal form, not the birthday.

Theorem 8.1 (Support spectrum [08]). *For $r \geq 1$,*

$$f \in F_r \Rightarrow \text{ord}(\text{supp } f) \leq \omega^r, \quad z \in R_r \Rightarrow \text{ord}(\text{supp } z) < \omega^r.$$

The field bound is attained by a rational expression, and for each $\eta < \omega^r$ a rational expression in R_r has support order type exactly η .

Proof. At rank zero a nonzero element has one term. At rank r an iterated Laurent series has at most countably many top blocks, each of type at most ω^{r-1} , so the ordinal sum has type at

most ω^r . An omnific integer has finitely many negative- t_r blocks and then one R_{r-1} block; for $r = 1$ its support is finite, and for $r > 1$ its type is at most $\omega^{r-1}d + \beta < \omega^r$ with $d < \omega$, $\beta < \omega^{r-1}$.

For sharpness, $\text{Geo}_r = \prod_{j=1}^r (1 - t_j)^{-1}$ has one nonzero coefficient at every vector of $\mathbb{N}^r$, read in the iterated lexicographic order, so its support has type ω^r . For $\eta = \omega^{r-1}d_r + \dots + d_1$ with $d_j \in \mathbb{N}$, the element

$$z_\eta = \sum_{j=1}^r \sum_{i=1}^{d_j} U_j^i \prod_{l < j} (1 - t_l)^{-1} \quad (8.1)$$

is omnific by [theorem 7.2](#). Each (j, i) block has support type ω^{j-1} , all coefficients are positive and the blocks are disjoint, so nothing cancels; higher j blocks precede lower ones, giving exactly the Cantor normal form of η . For $\eta = 0$ take zero. $\square$

These bounds contrast with [section 20](#): arbitrarily large finite-rank support types are compatible with decidable equality, while a raw name with at most one monomial can have an undecidable zero test (08).

8.2 Recognizing actual ordinals

Three operations must not be conflated: surreal addition, ordinary ordinal addition and natural (Hessenberg) addition. On ordinals the first agrees with the third, not generally with the second: $1 + \omega = \omega + 1$ in $\mathbf{No}$, while $1 +_{\text{ord}} \omega = \omega$. The constructors here are field operations; ordinary ordinal operations can be performed on recognized Cantor normal forms (08; also 05, [theorem 11.3](#)).

Lemma 8.2 (Ordinal points of the exponent group [08]). *For the scales (6.1), $G_r \cap \mathbf{On} = \mathbb{N}\delta_1 + \dots + \mathbb{N}\delta_r$.*

Proof. A nonnegative integer combination is the Cantor normal form of an ordinal. Conversely an element of G_r has Conway normal form $\sum_j n_j \omega^{\alpha_j}$ with distinct ordinal exponents; an ordinal's Conway normal form is its Cantor normal form, whose coefficients are positive integers, so every nonzero n_j is positive. $\square$

Theorem 8.3 (Ordinal slice and its recognition [08]). *The ordinals in F_r , equivalently in R_r , are exactly*

$$F_r \cap \mathbf{On} = \mathbb{N}[U_1, \dots, U_r]. \quad (8.2)$$

Given an F_r -name one can decide whether its value is an ordinal and, if so, compute its Cantor normal form. For the labels $\alpha_j = j - 1$ the slice is exactly $\mathbf{On}_{<\omega^{\omega^r}}$.

Proof. An ordinal has a finite Cantor normal form with ordinal exponents and positive integer coefficients; every exponent of an element of F_r lies in G_r , so by [lemma 8.2](#) it is a natural-coefficient polynomial in the U_j , and every such polynomial is an ordinal. For recognition, extract the finite Laurent prefix of indices at most zero and test that the positive-index tail vanishes; if not, some exponent has negative top coordinate and is not an ordinal exponent. Otherwise recursively test each coefficient of the finite polynomial in U_r for membership in $\mathbb{N}[U_1, \dots, U_{r-1}]$; at rank zero test membership in $\mathbb{N}$. The recursion terminates and yields the Cantor normal form, since $\prod_j U_j^{n_j}$ has exponent $\sum_j n_j \omega^{\alpha_j}$ and these exponents can be sorted exactly. For consecutive labels the exponent-group ordinals are those below ω^r , and finite Cantor normal forms with such exponents give exactly the ordinals below ω^{ω^r} . $\square$

The recognition algorithm does not decide whether all terms of an arbitrary recurrence are nonnegative; it uses a zero test for the positive-index tail and finitely many lower-rank tests, so no unproved positivity procedure is hidden (08).

9 One notation system for all ordinals below epsilon-zero

9.1 Canonical labels

Use hereditary Cantor-normal-form trees: zero is a tree, and a nonzero tree is a finite list $(\beta_1, n_1), \dots, (\beta_s, n_s)$ with trees $\beta_1 > \dots > \beta_s$ and $n_i \in \mathbb{N}_{>0}$, interpreted as $\omega^{\beta_1} n_1 + \dots + \omega^{\beta_s} n_s$. Comparison is lexicographic, recursing into the exponents; finite-tree induction proves termination and decidable validity. The classical Cantor-normal-form theorem identifies these trees with $\mathbf{On}_{<\varepsilon_0}$ (see also the formal treatment [2]), ε_0 being the least fixed point of $\beta \mapsto \omega^\beta$. For a finite set $A = \{\alpha_1 < \dots < \alpha_r\}$ of labels write $\mathcal{F}_A, \mathcal{R}_A$ for the fields and rings above; a global name is a finite label set with a valid local name, and empty label sets give k and $\mathbb{Z}$.

Lemma 9.1 (Effective insertion of scales [08]). *If $A \subseteq A'$ there are computable denotation-preserving embeddings of local names for $\mathcal{F}_A$ into those for $\mathcal{F}_{A'}$, preserving the omnific parts; the semantic embeddings commute under successive insertions.*

Proof. Insert missing labels in increasing order. At an existing variable, map all lower-level operator coefficients and jet entries through the embedding already constructed; a nonzero polynomial stays nonzero and its ordinary integer roots are unchanged under an injective field embedding, so the exceptional bound and compatibility equations are preserved, and the induced series has the mapped coefficients by the uniqueness recursion. Fractions map likewise. At an inserted variable, regard every old lower-field element c as the constant series with certificate $\theta f = 0$ and jet $f(0) = c$. Lemma 6.1 identifies both expressions with the same surreal under the fixed scales, so different insertion orders agree and omnific membership is preserved. $\square$

Theorem 9.2 (Epsilon-labelled decidable omnific notation [08]). *The directed unions*

$$\mathcal{F}_\varepsilon = \bigcup_{A \subseteq_{\text{fin}} \varepsilon_0} \mathcal{F}_A, \quad \mathcal{R}_\varepsilon = \bigcup_{A \subseteq_{\text{fin}} \varepsilon_0} \mathcal{R}_A$$

have finite-name representations with decidable validity, equality and order, computable arithmetic, computable omnific recognition and floor, and computable truncation at every finitely named exponent of their exponent groups. Moreover

$$\mathcal{F}_\varepsilon \cap \mathbf{On} = \mathcal{R}_\varepsilon \cap \mathbf{On} = \mathbf{On}_{<\varepsilon_0}. \quad (9.1)$$

Recognition and conversion of this slice are effective, and the whole omnific system has a computable canonicalization.

Proof. Given two names, embed both in the field of the sorted union of their label sets (lemma 9.1) and apply the local algorithms there; a cutoff exponent is handled by the same union. Validity is finite label comparison plus the finite certificate checks. Local floor and recognition commute with insertion, so the answers do not depend on the common field.

If an element of the union is an ordinal, theorem 8.3 writes it as a natural-coefficient polynomial in finitely many ω^{ω^α} with $\alpha < \varepsilon_0$; every exponent of its Cantor form is a finite natural sum of such ω^α and is below ε_0 , so the ordinal is below ε_0 . Conversely, for $\xi < \varepsilon_0$ collect the exponents of the Cantor normal forms of the exponents of ξ as labels A ; each ω^β is then a

monomial in the $U_\alpha = \omega^{\omega^\alpha}$, $\alpha \in A$, so $\xi \in \mathcal{R}_A$, and the local recognition converts between the forms. Finally apply [proposition 2.3](#) to the total system of all locally valid names. $\square$

“Total notation system” uses the decidable set of locally valid codes as its syntactic domain; malformed strings and incompatible certificates are rejected first, and no noncomputable choice of representative is hidden (08).

Corollary 9.3 (Cofinality and an ambient ordinal bound [08]). *For a nonempty finite label set A with greatest label α , the positive powers of $U_\alpha = \omega^{\omega^\alpha}$ are cofinal in $\mathcal{F}_A$, and its least ordinal upper bound is $\omega^{\omega^{\alpha+1}}$. For the union, $\mathcal{F}_\varepsilon \subset (-\varepsilon_0, \varepsilon_0)$, and ε_0 is its least ordinal upper bound. These are ordinal upper bounds, not suprema in **No**.*

Proof. For nonzero $x \in \mathcal{F}_A$ the leading exponent lies in the span of the scales; choose N with $\deg_\omega x < N\omega^\alpha$, so $|x| < U_\alpha^N$, an ordinal of $\mathcal{F}_A$. In the ordinal order, $\sup_N \omega^{\omega^\alpha \cdot N} = \omega^{\omega^\alpha \cdot \omega} = \omega^{\omega^{\alpha+1}}$ (ordinary ordinal products). For the empty label set the naturals are cofinal in k and the bound is ω . For $\alpha < \varepsilon_0$ the local bound is below ε_0 , so every element of the union has absolute value below ε_0 , and (9.1) shows no smaller ordinal bounds the union. Indeed $\varepsilon_0/2$ is already a smaller surreal upper bound: every element is below some ordinal $\beta < \varepsilon_0$, and $\beta + \beta$ is still an ordinal below ε_0 , so $\beta < \varepsilon_0/2$. This prevents reading “least ordinal upper bound” as “surreal supremum”. $\square$

Remark 9.4 (Epsilon-zero is a design choice [08]). The theorem does not make ε_0 an absolute ceiling for effective omnific notation. The labels are canonical hereditary Cantor trees; a different, independently justified decidable ordinal-label system can be substituted in the finite-rank construction, but its validity and comparison properties must be proved separately, and an unrestricted well-foundedness promise cannot be smuggled in as decidable syntax (08). The union names ε_0 in no form, which is compatible with 05’s use of ε_0 as a primitive symbol ([theorem 12.2](#)), because the languages differ; compare the remark after [proposition 2.4](#) ([merge]).

Example 9.5 (Large magnitudes at small rank [08]). The labels $A = \{0, \omega\}$ give the ordinal $\omega^{\omega^\omega} + \omega^3 + 2$ at rank two, far beyond the consecutive-label cutoff ω^{ω^2} , with two nested coefficient fields. Label magnitude, finite rank, support order type and birthday are four different measurements.

10 Worked infinite-support notations

Take $k = \mathbb{Q}$, consecutive labels $0, 1$, and abbreviate $t = t_1 = \omega^{-1}$ and $U = U_2 = \omega^\omega$. For every $f \in F_1$, Uf is omnific, because the exponents of f are ordinary integers bounded above and ω dominates them; so $\mathbb{Z} + UF_1[U] \subseteq R_2$. This restricted ring is what the delivered prototype implements ([section 24](#)).

A factorial tail. $\text{Fac}(t) = \sum_{n \geq 0} n! t^n$ has the certificate $L = \theta - t(\theta + 1)^2$ with $a_0 = 1$: $P_0(X) = X$ has the sole integer root 0 and (5.3) reads $na_n - n^2 a_{n-1} = 0$ for $n \geq 1$, so $a_n = n!$. Thus

$$U \text{Fac}(t) = \omega^\omega + \omega^{\omega-1} + 2\omega^{\omega-2} + 6\omega^{\omega-3} + \dots$$

is a finitely named omnific integer of infinite support. That Fac has no positive radius of convergence is irrelevant to its formal or surreal denotation.

Exponential notation without a global exponential. $\text{Ex}(t) = \sum_n t^n/n!$ and $\text{Ex}(-t)$ have the certificates $\theta - t$ and $\theta + t$ with constant coefficient one. The formal product has derivative zero and constant term one, and in characteristic zero a series with zero derivative is constant, so

$$(U \text{Ex}(t))(U \text{Ex}(-t)) = U^2. \quad (10.1)$$

Both factors have infinite normal forms and are omnific. The identity is proved in formal series and transported to **No**; it needs no choice of a global surreal exponential. The closure algorithm independently produces a certificate for the difference and checks its finite jet, without recognizing the identity from the printed expressions.

Cancellations and leading terms. $\text{Geo}(t) = (1 - t)^{-1}$ has the certificate $\theta - t(\theta + 1)$ with initial coefficient one. The first two coefficients of $\text{Geo} - \text{Ex}$ vanish and the next is $\frac{1}{2}$, so

$$\text{lt}(U(\text{Geo}(t) - \text{Ex}(t))) = \frac{1}{2}\omega^{\omega-2}, \quad (10.2)$$

found after a finite certified zero test, not a fixed truncation length. At higher rank a whole lower-rank infinite series can cancel in a single top coefficient; the recursive equality test lets the leading-term algorithm move past it, which enumerating the expanded normal form would not guarantee. Note that $U \text{Geo}(t) = \mathbf{B}$.

Ordinal versus nonordinal outputs. $U(t^{-2} + 3) + 5t^{-2} + 7 = \omega^{\omega+2} + 3\omega^\omega + 5\omega^2 + 7$ is an ordinal, and recognition outputs this Cantor normal form. $U \text{Fac}(t)$ is not an ordinal: it has infinitely many terms, and $\omega - n$ for $n > 0$ is not an ordinal although it is a positive exponent; integer coefficients do not remedy either issue. Similarly $\sqrt{2}U$ is omnific when $\sqrt{2} \in k$ but is not an ordinal; confusing arbitrary real coefficients with Cantor coefficients would exclude much of the intended domain.

Part II

Hereditary forms, the epsilon-zero-shielded calculus, and finite-state series

This part is source 05's. Its coefficient field is an exact effective ordered subfield $k \subseteq \mathbb{R}$ with canonical codes, effective field operations and decidable comparison; the implementation uses $k = \mathbb{Q}$ and the mathematics also applies to $k = \mathbb{R}_{\text{alg}}$ (subsection 1.3). No arbitrary computable-real coefficient oracle is assumed.

11 A hereditary finite normal-form calculus

11.1 Syntax and denotation

A hereditary form is a finite tree of the shape

$$a_0 + \sum_{i=1}^m a_i \omega^{e_i}, \quad (11.1)$$

where $a_0 \in k$, $a_i \in k \setminus \{0\}$, and the e_i are nonzero hereditary forms of strictly smaller tree height in strictly decreasing numerical order; at height zero there are only constants. Let

$\mathcal{H} = \mathcal{H}_k$ be the class of values of such trees. The exponent zero is absorbed into a_0 . Negative exponents are allowed in this auxiliary calculus; omnific membership is imposed afterwards rather than making exponent arithmetic artificially one-sided. A convenient raw expression language has constants from k and constructors $+$, $-$, $\cdot$ and $\Omega(e)$ with denotation $\omega^{\nu(e)}$; its normal forms are exactly the hereditary trees. It contains, for example, $(\omega + 1)^2 = \omega^2 + 2\omega + 1$ and $\omega^{\omega^{-1}} + 3\omega^{1/2} - 7$, and over algebraic coefficients $\omega^{\sqrt{2}\omega+1/3}$. It has no division and no limits as raw constructors.

Theorem 11.1 (Hereditary normalization [05]). *There are terminating exact algorithms for normalization, equality, comparison, addition, negation, multiplication and the omega map on hereditary expressions. Two canonical trees denote the same surreal exactly when their coefficient and exponent trees are identical. In particular $\mathcal{H}$ is an ordered subring of $\mathbf{No}$.*

Proof. Proceed by tree height. At height zero use the field algorithms. For normalized exponents of lower height, comparison is available by induction: merge equal exponents, add their coefficients, remove zero coefficients, absorb exponent zero into the constant and sort the rest. For addition merge the two finite lists; for multiplication form all pairs of terms, using $(a\omega^e)(b\omega^f) = ab\omega^{e+f}$, where the exponent addition takes place at strictly smaller height, and normalize. Negation changes signs. To normalize $\Omega(e)$, normalize e and output 1 if $e = 0$, and otherwise the single monomial with coefficient one and exponent e . A raw expression is normalized along its finite syntax tree. To compare normal forms, inspect the greatest exponent at which the lists differ; the sign is that of the coefficient difference, and comparing candidate exponents uses only smaller heights. The procedures terminate by induction on height. Normal-form uniqueness says equal values have the same exponents and coefficients, and the induction hypothesis turns equality of exponent values into identity of their canonical trees. Hahn arithmetic gives the ring identities. $\square$

The proof is a decidability argument, not a polynomial bit-complexity bound: a short product can have many monomials, and exponent trees may contain large expressions. Hash-consing and directed acyclic graphs are useful implementation choices but do not change the semantics (05).

11.2 Omnific membership and the floor

Put $\mathcal{D} = \mathcal{H} \cap \mathbf{Oz}$, and assume also that the integer test and the ordinary floor are effective in k , as they are for $\mathbb{Q}$ and $\mathbb{R}_{\text{alg}}$.

Theorem 11.2 (Decidable omnific core and floor [05]). *Membership in $\mathcal{D}$ is decidable; $\mathcal{D}$ has decidable equality and order; and the omnific floor restricts to an effective map $\lfloor \cdot \rfloor_{\mathbf{Oz}} : \mathcal{H} \rightarrow \mathcal{D}$ with $\lfloor x \rfloor_{\mathbf{Oz}} \leq x < \lfloor x \rfloor_{\mathbf{Oz}} + 1$.*

Proof. In the normal form of x , test whether every nonconstant exponent is positive and whether the constant is in $\mathbb{Z}$; criterion (2.3) gives membership, and closure under ring operations follows from positive supports or from the ring property of $\mathbf{Oz}$. The decomposition (2.4) of a hereditary form is read off its finite normal form: $x_{>0}$ is the finite sum of positive-exponent terms, $\text{ct}(x) \in k$, and $x_{<0}$ is the finite sum of negative-exponent terms, whose sign is the sign of its leading coefficient. Formula (2.5) therefore produces a hereditary form with positive exponents and integer constant, which lies in $\mathcal{D}$ and is the floor by subsection 2.2. $\square$

11.3 The ordinal fragment

Theorem 11.3 (Ordinal fragment [05]). *For $k = \mathbb{Q}$ or $k = \mathbb{R}_{\text{alg}}$, the ordinals in $\mathcal{H}$ are exactly the ordinals below ε_0 . They are recognized recursively: the normal form must have natural-number coefficients and hereditarily ordinal exponents. Arithmetic inherited from **No** restricts to natural (Hessenberg) addition and multiplication, not ordinary ordinal arithmetic.*

Proof. The normal form of an ordinal is its Cantor normal form: positive natural coefficients, ordinal exponents, natural constant; conversely such a finite form is an ordinal. Apply the criterion recursively to the exponent subtrees. Finite hereditary Cantor normal forms are exactly the ordinals below ε_0 : downward recursion on leading exponents constructs each such ordinal, and induction on tree height bounds each constructed ordinal below a finite tower of omega powers, whose supremum in the ordinal order is ε_0 . The coefficientwise sum and distributive product of Cantor forms are the natural sum and product, which are the operations the normalizer uses. $\square$

A program supporting both surreal and ordinary ordinal operations must give them different names or types (05, 08).

Lemma 11.4 (Hereditary values have smaller ordinal bounds [05]). *For every $h \in \mathcal{H}_k$, $k \subseteq \mathbb{R}$, there is an ordinal $\alpha < \varepsilon_0$ with $|h| < \alpha$.*

Proof. For a constant take a large finite ordinal. For a normal form, induction gives a common positive ordinal $\beta < \varepsilon_0$ exceeding the absolute values of all exponent subtrees. Each monomial is smaller than ω^β up to a finite real factor, so a finite sum is bounded by $\omega^{\beta+1}$; increasing once more gives a strict bound, still below ε_0 . $\square$

Although ε_0 is the least *ordinal* bounding all hereditary ordinals, it is not their least surreal upper bound: $\varepsilon_0 - 1$ is also an upper bound of every ordinal below ε_0 . No surreal supremum assertion is made (05; compare [corollary 9.3](#), where 08 exhibits $\varepsilon_0/2$).

12 The epsilon-zero-shielded calculus

Let $k = \mathbb{Q}$ or $\mathbb{R}_{\text{alg}}$, and consider $\text{Frac}(\mathcal{H})$ and $\mathcal{D} = \mathcal{H} \cap \mathbf{Oz}$. A valid $\text{Frac}(\mathcal{H})$ -name is a pair (p, q) of hereditary forms with $q \neq 0$, denoting p/q ; equality is decided by

$$\frac{p}{q} = \frac{r}{s} \iff ps - rq = 0 \quad (12.1)$$

using [theorem 11.1](#), and signs come from the signs of numerator and denominator. Division introduces infinite normal forms without destroying exact equality.

Lemma 12.1 (Support of hereditary fractions [05]). *Every member of $\text{Frac}(\mathcal{H})$ has a Hahn normal form with coefficients in k and exponents in the additive group $\mathcal{H}$.*

Proof. This is immediate for hereditary forms. For $q \neq 0$ factor its leading term, $q = c\omega^a(1+u)$ with $a \in \mathcal{H}$, $c \in k^\times$ and u a finite sum with strictly negative exponents in $\mathcal{H}$. The geometric inverse $q^{-1} = c^{-1}\omega^{-a} \sum_{n \geq 0} (-u)^n$ is Hahn summable by the positive-support lemma, and its exponents are finite sums of elements of $\mathcal{H}$ shifted by $-a$. Multiplication by p keeps both properties. $\square$

Theorem 12.2 (The ε_0 -shielded omnific calculus [05]). *Let ε_0 be embedded as a surreal, so that $\varepsilon_0 = \omega^{\varepsilon_0}$. Then*

$$\mathcal{S}_{\varepsilon_0} = \left\{ d + \sum_{j=1}^m \varepsilon_0^j f_j : d \in \mathcal{D}, f_j \in \text{Frac}(\mathcal{H}) \right\} \quad (12.2)$$

is a subring of $\mathbf{Oz}$ with decidable validity, equality and order on finite polynomial-fraction names. It contains $\mathcal{D}$, ε_0 is transcendental over $\text{Frac}(\mathcal{H})$, and $\mathcal{S}_{\varepsilon_0}$ contains elements with infinite normal forms.

Proof. Take $G = \mathcal{H}$. Lemma 11.4 gives $|g| < \varepsilon_0$ for all $g \in G$, and lemma 12.1 puts $\text{Frac}(\mathcal{H})$ in the corresponding Hahn field. Apply theorem 3.1 with $\Lambda = \varepsilon_0$ and $X = \omega^{\varepsilon_0} = \varepsilon_0$; thus $\mathcal{S}_{\varepsilon_0} = \mathcal{S}_{\varepsilon_0}(\mathcal{D}, \text{Frac}(\mathcal{H}))$. The denominator and omnific-base tests are decidable, as are coefficient arithmetic and equality, so polynomial normalization and the sign rule give the algorithms. Injectivity of evaluation is transcendence of ε_0 over $\text{Frac}(\mathcal{H})$. For an infinite support, $\varepsilon_0/(1 - \omega^{-1}) = \sum_{n \geq 0} \omega^{\varepsilon_0 - n}$ has positive exponents and all coefficients one. $\square$

The polynomial representation is canonical only after coefficient values have been identified; fractions need not be reduced to a unique quotient, since cross multiplication already decides equality. An inefficient least-code canonicalizer is available (proposition 2.3), but no multivariate gcd algorithm is hidden in the procedure (05).

Proposition 12.3 (An algebraic limit of the fraction core [05]). *$1 + \omega$ has no square root in $\text{Frac}(\mathcal{H})$. In particular $\text{Frac}(\mathcal{H})$ is not real closed, even for $k = \mathbb{R}_{\text{alg}}$.*

Proof. Suppose p/q were a square root with $p, q \in \mathcal{H}$, $q \neq 0$. Take the finite-dimensional $\mathbb{Q}$ -vector space spanned by 1 and the outer exponents of p and q , with a basis $\beta_1 = 1, \beta_2, \dots, \beta_r$; clearing the rational coordinates gives $N > 0$ such that p, q are Laurent polynomials over k in $x_i = \omega^{\beta_i/N}$. The x_i are algebraically independent over k : distinct integer exponent vectors give distinct exponents by $\mathbb{Q}$ -independence, and normal-form uniqueness prevents a relation. So $1 + x_1^N$ would be a square in $k(x_1, \dots, x_r)$. But $1 + x_1^N$ is nonconstant and squarefree in characteristic zero (it is coprime to Nx_1^{N-1}), and in the factorial ring $k[x_1, \dots, x_r]$ a square in the fraction field has even order at every irreducible factor. $\square$

Corollary 12.4 (One-state supports of type ω^r [05]). *For every positive integer r ,*

$$\zeta_r = \varepsilon_0 \prod_{j=0}^{r-1} (1 - \omega^{-\omega^j})^{-1} \quad (12.3)$$

lies in $\mathcal{S}_{\varepsilon_0}$. Its normal form has all coefficients one and support $\{\varepsilon_0 - \sum_{j < r} n_j \omega^j : (n_0, \dots, n_{r-1}) \in \mathbb{N}^r\}$, of decreasing order type ω^r .

Proof. Each denominator is a nonzero hereditary form, so the product of reciprocals lies in $\text{Frac}(\mathcal{H})$, and the expansion of the geometric factors is jointly Hahn summable. The monomials $1, \omega, \dots, \omega^{r-1}$ are $\mathbb{Q}$ -independent, so distinct tuples give distinct exponents; their nonnegative integer combinations in increasing order are the ordinals below ω^r , and subtraction from ε_0 reverses the order. $\square$

“One-state” refers to the matrix language of section 13: every matrix is the 1×1 matrix (1). Already for $r = 2$ an interval of exponents below a fixed positive Hahn bound contains infinitely many terms, so a proof by scanning all terms below a numerical cutoff would not work. This agrees with the bound $< \omega^{r+1}$ for omnific supports in a rank- $(r + 1)$ lattice, since

ζ_r uses $r + 1$ scales including ε_0 (theorems 8.1 and 18.2). The ring $\mathcal{S}_{\varepsilon_0}$ is deliberately not all of $\mathbf{Oz}$: it is not asserted to be closed under the omega map on arbitrary inputs, under root extraction, or under ordinal-indexed sums, and these restrictions keep its interface tractable (05).

13 Finite-state series and finite leading-term certificates

Let k be an effective ordered field and $\delta_1, \dots, \delta_r$ strictly positive elements of an effective ordered additive group $G \subseteq \mathbf{No}$. For dimension $m \geq 1$ choose $u \in k^{1 \times m}$, $v \in k^{m \times 1}$ and pairwise commuting $M_1, \dots, M_r \in k^{m \times m}$, and put

$$\Phi = \sum_{\mathbf{n} \in \mathbb{N}^r} a_{\mathbf{n}} t^{w(\mathbf{n})}, \quad a_{\mathbf{n}} = u M_1^{n_1} \cdots M_r^{n_r} v, \quad w(\mathbf{n}) = \sum_i n_i \delta_i, \quad (13.1)$$

adding contributions of coinciding weights. The positive-monoid support lemma makes this a legitimate Hahn series even when G is non-Archimedean, and a shielding parameter turns $X\Phi$ into an omnific integer (theorem 3.1). In commuting formal variables z_i the associated series is

$$\hat{\Phi}(\mathbf{z}) = u \prod_{i=1}^r (I - z_i M_i)^{-1} v. \quad (13.2)$$

Finite linear representations and their zero tests are standard in rational series and weighted-automata theory [9, 1]; the underlying theory goes back to Schützenberger’s rational series, a reference 05 does not give. The question here is what the finite data certify after evaluation at positive Hahn scales, particularly when different tuples collide.

13.1 Independent scales: a small simplex controls the front

If the scales are $\mathbb{Q}$ -linearly independent, $w : \mathbb{N}^r \rightarrow G$ is injective and the coefficient at $w(\mathbf{n})$ is $a_{\mathbf{n}}$.

Theorem 13.1 (Independent-scale finite front [05]). *If $\Phi \neq 0$ in (13.1) and the scales are $\mathbb{Q}$ -independent, its least Hahn exponent occurs at a multi-index with*

$$|\mathbf{n}| = n_1 + \cdots + n_r < m. \quad (13.3)$$

Consequently zero, the leading exponent, the leading coefficient and the sign are determined by the finitely many coefficients with $|\mathbf{n}| < m$. For two representations of dimensions m and m' on the same scale frame, use $m + m'$ after forming the direct-sum difference.

Proof. If $v = 0$ the series is zero. Otherwise take $\mathbf{n}$ of least weight with nonzero coefficient, which exists by well ordering; choose a word of length $N = |\mathbf{n}|$ with n_i letters i and put $v_0 = v$, $v_j = M_{i_j} v_{j-1}$. Commutativity identifies $u v_N$ with $a_{\mathbf{n}}$. If $N \geq m$, some first $j \leq m$ has $v_j = \sum_{l < j} c_l v_l$, since the earlier vectors are independent in dimension m . Apply the remaining suffix matrices and u ; the left side is nonzero, so some summand on the right is, and by commutativity it corresponds to deleting the nonempty block of letters between positions $l + 1$ and j . The new multi-index is coordinatewise at most $\mathbf{n}$ and different, so of smaller weight, a contradiction; thus $N < m$. The rest follows because independence rules out cancellation at the least weight; a direct sum with block-diagonal commuting matrices represents the difference. $\square$

There are $\binom{m+r-1}{r}$ candidate multi-indices: a finite bound, not a polynomial one when r is part of the input. Reachable-state linear algebra can reduce the zero-test work; the theorem additionally locates the leading Hahn term without enumerating the possibly infinite set of weights below it. For $r = 1$ this is the Cayley–Hamilton bound, and it is sharp: a nilpotent shift of dimension m can give $\widehat{\Phi}(z) = z^{m-1}$ (05; the same bound and example are 09’s [proposition 19.1](#)).

13.2 Dependent scales: collect collisions in a finite numerator

Checking individual small coefficients is unsound when different multi-indices have the same weight. Define

$$Q(\mathbf{z}) = \prod_{i=1}^r \det(I - z_i M_i), \quad (13.4)$$

$$P(\mathbf{z}) = u \prod_{i=1}^r \operatorname{adj}(I - z_i M_i)v, \quad (13.5)$$

so $\widehat{\Phi} = P/Q$, $Q(0) = 1$, and each variable has degree at most $m - 1$ in P . Let ev_δ map $\mathbf{z}^{\mathbf{n}}$ to $t^{w(\mathbf{n})}$ and collect identical weights.

Theorem 13.2 (Collision-aware finite front [05]). *For arbitrary strictly positive scales, including dependent and non-Archimedean ones, let $\bar{P} = \operatorname{ev}_\delta(P)$. Then*

$$\Phi = 0 \iff \bar{P} = 0, \quad (13.6)$$

and when nonzero, Φ and $\bar{P}$ have the same leading Hahn exponent and coefficient. In particular the valuation $v(\Phi)$ satisfies

$$0 \leq v(\Phi) \leq (m - 1)(\delta_1 + \cdots + \delta_r). \quad (13.7)$$

The polynomial P , its at most m^r monomial positions and the exact collection of equal weights provide an equality and sign certificate.

Proof. The adjugate identity gives $\widehat{\Phi} = P/Q$ in formal power series. Positive-scale evaluation is defined on these series and commutes with products, since supports lie in the positive finitely generated monoid and fibres are finite. $\bar{Q} = \operatorname{ev}_\delta(Q)$ has constant coefficient one and all other exponents strictly positive, with no collision at zero because every $\delta_i > 0$; so it is invertible with leading term 1, and $\Phi = \bar{P}\bar{Q}^{-1}$ is zero iff $\bar{P}$ is, with the same leading term otherwise. Every monomial of P has $0 \leq n_i \leq m - 1$, so its weight lies in the stated range; collecting collisions removes or combines terms but creates no new weights. $\square$

The proof uses only the ordered product in the adjugate formula; commutativity is part of the chosen series language and of the shift-state interpretation, not a hidden positivity assumption, and entries and coefficients may be negative (05).

Example 13.3 (The independent-scale bound cannot be reused [05]). Let $N_0 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, $M_1 = N_0$, $M_2 = I - N_0$, $u = (0, 1)$, $v = (1, 0)^T$; the matrices commute and $m = 2$. Then

$$\widehat{\Phi}(z_1, z_2) = \frac{z_1 - z_2 - z_1 z_2}{(1 - z_2)^2},$$

with coefficients 1 and -1 at $(1, 0)$ and $(0, 1)$. If $\delta_1 = \delta_2 = \delta > 0$ these cancel and $\Phi = -t^{2\delta}/(1 - t^\delta)^2$: the leading exponent is 2δ , not the weight of a surviving coefficient of total degree below 2, and the collision-aware bound $(m - 1)(\delta_1 + \delta_2) = 2\delta$ is attained.

13.3 Closure and transfinite support

Direct sums give addition of descriptions; tensor products give coefficientwise (Hadamard) products of the multivariate arrays before evaluation, which with colliding scales need not be the Hadamard product of the collected Hahn coefficients. Hahn multiplication is effective within a fixed frame.

Proposition 13.4 (Cauchy-product closure [05]). *Two commuting matrix-series descriptions of dimensions m and m' on r fixed scales have a Cauchy-product description by commuting matrices of dimension at most $(m + m')^r$, and positive-scale evaluation of the multivariate product is the Hahn product, even when scales collide.*

Proof. Before evaluation write the series as $P / \prod_i q_i(z_i)$ and $R / \prod_i s_i(z_i)$ by the adjugate formulas. The product numerator has degree at most $m + m' - 2$ in each variable, and $q_i s_i$ has degree at most $b = m + m'$ and constant coefficient one. Multiplying by the denominator shows that the coefficients satisfy a homogeneous constant-coefficient recurrence in each coordinate once that coordinate is at least b , the others held fixed (missing high recurrence coefficients are zero). Store the initial array on $\{0, \dots, b - 1\}^r$ as a vector of dimension b^r ; in coordinate i the companion shift advances that coordinate using its recurrence at the last position. The shifts act on separate tensor factors and commute, and selecting the all-zero coordinate after the shifts recovers every coefficient. Evaluation respects the Cauchy product by the support lemma, including the finite regrouping of colliding weights. $\square$

For ζ_r of (12.3) take $m = 1$, $u = v = 1$, $M_i = (1)$ and $\delta_i = \omega^{i-1}$: the state dimension controlling the front is independent of the ordinal type of the support, by finite algebraic dependence of the coefficient states, not because transfinite supports became finite. Not every rational multivariate function has this separated-resolvent form: $1/(1 - z_1 - z_2)$ is rational, but as the second coordinate varies the sequences in the first coordinate include polynomials of unbounded degree, which cannot all satisfy one fixed-order recurrence. Such functions can still be treated in the fraction-field calculus; the finite-state sublanguage is a certificate format, not an identification of all rational multivariate series (05).

Part III

Rational–omega expressions

This part is source 09's. It studies one particular language, not a universal decision claim. Write $\Omega(x) = \omega^x$ for Conway's omega map and consider finite terms

$$s ::= q \mid -s \mid s + s \mid s s \mid s/s \mid \Omega(s), \quad q \in \mathbb{Q}, \quad (13.8)$$

where division requires a nonzero denominator. Let $\mathcal{Q}$ be the set of their values and $\mathcal{I} = \mathcal{Q} \cap \mathbf{Oz}$. The integer terms are the valid terms whose *values* lie in $\mathcal{I}$; intermediate values need not be omnific, and infinitesimal intermediate expressions are essential for the compressed infinite series below. The symbol Ω is not a shorthand for the full surreal exponential: no unrestricted exp, log, trigonometric function, arbitrary computable real constant or program-controlled infinite sum is included, and the restriction is substantive (09). Rational constants are given as exact integer numerator–denominator pairs.

The three main theorems of the part are: validity, equality, order and membership in $\mathbf{Oz}$ are decidable, and from names of $x, a \in \mathcal{Q}$ one computes names of $\lfloor x \rfloor_{\mathbf{Oz}}$ and $\tau_{\geq a} x$ and the

rational coefficient of ω^a in x , all exactly (theorems 14.2 and 16.3); the ordinals of $\mathcal{Q}$ are exactly those below ε_0 , while every support has order type below ω^ω , sharply at depth two (theorems 17.2, 18.2 and 18.3); and nevertheless equality of an always-valid stream language of omnific integers is Π_1^0 -complete on a subfamily whose values lie in the depth-two field with at most one term (theorem 20.4). The ε_0 bound is about which ordinal values occur, the ω^ω bound about the order type of the support of one value, and the complexity result about a different input language; none is a birthday bound (09).

14 Finite rational–omega terms and decidable equality

14.1 The depth hierarchy

Set

$$\mathcal{Q}_0 = \mathbb{Q}, \quad \mathcal{Q}_{d+1} = \text{Frac}(\mathbb{Q}[\omega^{\mathcal{Q}_d}]), \quad \mathcal{Q} = \bigcup_{d < \omega} \mathcal{Q}_d, \quad (14.1)$$

where $\mathbb{Q}[\omega^{\mathcal{Q}_d}]$ consists of *finite* sums $\sum q_i \omega^{g_i}$ with $g_i \in \mathcal{Q}_d$, negative exponents allowed: the group algebra embedded by Conway monomials, not a Hahn completion, with fraction field taken in **No**. The hierarchy is increasing ($\mathbb{Q} \subseteq \mathcal{Q}_1$, and an expression of an element of $\mathcal{Q}_d$ as a fraction over exponents in $\mathcal{Q}_{d-1}$ is one over $\mathcal{Q}_d$). A term has depth zero if it has no Ω node; $+$, $\times$, $/$ take the maximum of the input depths and Ω adds one. Induction shows that $\mathcal{Q}_d$ is exactly the set of values of terms of depth at most d .

Lemma 14.1 (Sparse-fraction representation [09]). *Every term of depth at most $d+1$ can be computed as a pair (P, Q) , $Q \neq 0$, of finite rational-coefficient sums of monomials with exponents of depth at most d . Addition and multiplication of these sums need only rational arithmetic, addition of exponent terms and equality of exponent values.*

Proof. A rational uses the exponent zero, and $\Omega(u)$ is the singleton monomial with exponent u . Fractions are added and multiplied by the usual identities, and division swaps a nonzero numerator and denominator. Multiplying finite sums uses $\omega^g \omega^h = \omega^{g+h}$; equal exponents are collected and zero coefficients deleted. All loops are over finite lists. $\square$

Theorem 14.2 (Recursive ground decision procedure [09]). *Validity of (13.8), equality of valid terms and their order are decidable. For a nonzero term of depth at most $d+1$ its leading exponent is computable as a term in $\mathcal{Q}_d$ and its leading coefficient as a nonzero rational.*

Proof. Induct on depth; depth zero is rational arithmetic, including rejection of zero denominators. Assume comparison and equality are decidable at depth d . Build the representation of lemma 14.1, sort each exponent list with the depth- d comparator and collect equal values. By uniqueness of normal forms a finite sum is zero exactly when its list is empty, and otherwise its greatest exponent and coefficient are its leading data; so a division at the next depth is valid exactly when the numerator of its denominator is nonzero. For valid fractions

$$\frac{P}{Q} = \frac{R}{S} \iff PS - RQ = 0, \quad (14.2)$$

decided by the same collection; if the difference is nonzero, its sign is the sign of $\text{lc}(PS - RQ)$ times the signs of Q and S . Finally

$$\deg_\omega(P/Q) = \deg_\omega P - \deg_\omega Q, \quad \text{lc}(P/Q) = \text{lc}(P)/\text{lc}(Q), \quad (14.3)$$

the exponent being a subtraction at depth d . Inputs of different depths are viewed at the larger depth. Every exponent comparison descends strictly in depth, and the rest is finite polynomial arithmetic, which proves termination. $\square$

Semantic exponent equality matters: $u + v$ and $v + u$ must be merged, and treating the written Ω subterms as independent indeterminates would miss $\Omega(u + v) = \Omega(u)\Omega(v)$ and give an incorrect equality algorithm (09).

Corollary 14.3 (Canonical representatives [09]). *There is a computable canonical finite-string representative for every value of $\mathcal{Q}$, and likewise for every value of $\mathcal{I}$ once membership has been decided.*

Proof. Apply [proposition 2.3](#) to the decidable set of valid codes. For $\mathcal{I}$ the same least representative works, since equality to a member of $\mathcal{I}$ implies membership. $\square$

This is an existence proof for a canonicalizer, not a practical implementation; the prototype uses semantic comparison instead. No polynomial-time or elementary-time claim is made for unrestricted nesting: expression expansion, coefficient growth and repeated comparison can be expensive (09).

14.2 Rational coefficients throughout the infinite normal form

Lemma 14.4 (Rational supports [09]). *Every $x \in \mathcal{Q}$ has only rational normal-form coefficients. More precisely, if $x = P/Q$ is a finite monomial fraction and Γ is the additive group generated by the supports of P and Q , then $\text{supp } x \subseteq \Gamma$.*

Proof. Write $Q = c\omega^g(1 - h)$ with $c \in \mathbb{Q}^\times$ and h of finite support strictly below exponent zero. The Hahn inverse $Q^{-1} = c^{-1}\omega^{-g} \sum_{n \geq 0} h^n$ is strongly summable by the Hahn–Neumann support lemma [3, Proposition 3.7]; every exponent is an integer combination of the original ones and every coefficient a finite sum of rational products. Multiplying by P preserves both assertions. The argument uses a semantic inverse to prove a support statement; the equality algorithm does not enumerate this expansion. $\square$

Corollary 14.5 (Real trace [09]). $\mathcal{Q} \cap \mathbb{R} = \mathbb{Q}$. *In particular $\mathcal{Q}$ is not real closed.*

Proof. A nonzero real r has the one-term normal form $r\omega^0$, so $r \in \mathbb{Q}$ by the lemma; and $X^2 - 2$ has no root in $\mathcal{Q}$. $\square$

15 Effective separated grids

Equality alone does not decide omnific membership: a rational function may hide infinitely many negative exponents. The next reduction makes the sign of every possible exponent visible in finitely many coordinates.

15.1 Elimination by leading scale

For positive surreals write $u \gg v$ when $u > nv$ for every positive integer n . For positive elements with rational leading coefficients, $\deg_\omega u > \deg_\omega v$ implies $u \gg v$.

Lemma 15.1 (Effective separated basis [09]). *Given $h_1, \dots, h_m \in \mathcal{Q}$ one can compute a basis $\gamma_1, \dots, \gamma_r$ of their rational span, with all coordinates, such that*

$$\gamma_1 \gg \gamma_2 \gg \dots \gg \gamma_r > 0; \tag{15.1}$$

the basis is empty if all inputs are zero, and it can be chosen in the same $\mathcal{Q}_d$ as the inputs.

Proof. Discard zero inputs; take a vector of greatest leading exponent, make it positive, and call it γ_1 . Replace every other vector v with the same leading exponent by $v - (\text{lc}(v)/\text{lc}(\gamma_1))\gamma_1$; the multiplier is rational by lemma 14.4 and the leading term cancels exactly, while vectors of smaller leading exponent are unchanged. Discard zero residuals and repeat. Each stage removes one pivot, so there are at most m stages, and each pivot has strictly smaller leading exponent than the previous one; after sign normalization (15.1) holds. Distinct leading exponents make the pivots rationally independent, the reversible row operations preserve the span, and tracking them gives the coordinates. All leading data and comparisons are computable by theorem 14.2, and all operations stay in the original level. $\square$

The restriction to rational coefficients is essential: the rational span of 1 and $\sqrt{2}$ is two-dimensional but has only one Archimedean scale, so it has no separated rational basis. Substituting an arbitrary computable ordered coefficient field into this lemma would be false (09).

15.2 From a finite fraction to a lexicographic lattice

Theorem 15.2 (Finite separated-grid compilation [09]). *Given $x \in \mathcal{Q}$ one can compute positive separated $\gamma_1, \dots, \gamma_r \in \mathcal{Q}$ and $R \in \mathbb{Q}(X_1, \dots, X_r)$ with $x = R(\omega^{\gamma_1}, \dots, \omega^{\gamma_r})$. The map*

$$\mathbb{Z}_{\text{lex}}^r \rightarrow \mathbf{No}, \quad (n_1, \dots, n_r) \mapsto \sum_i n_i \gamma_i \quad (15.2)$$

is an injective ordered-group map, the monomials ω^{γ_i} are algebraically independent over $\mathbb{Q}$, and $\text{supp } x$ lies in their exponent lattice.

Proof. Apply lemma 15.1 to the union of the exponent lists of a representation P/Q , and clear denominators in the coordinate matrix by replacing γ_i with γ_i/N ; every original exponent then has integer coordinates, and replacing $\omega^{\sum n_i \gamma_i}$ by $\prod X_i^{n_i}$ gives a Laurent rational function, an element of $\mathbb{Q}(X_1, \dots, X_r)$. For a nonzero integer vector with first nonzero coordinate n_i , separation makes $n_i \gamma_i$ dominate the rest, so the sign of (15.2) is that of n_i ; this gives order preservation and injectivity. A nonzero Laurent polynomial then evaluates to a finite normal form with distinct exponents, which cannot vanish; this is algebraic independence. Support containment is lemma 14.4. $\square$

The basis depends on the input and its members may have infinite normal forms: leading-scale elimination may need to distinguish $\omega + 1/(\omega - 1)$ from ω , for which the leading data of $1/(\omega - 1)$ suffice. For a fixed grid the relevant Hahn field is the iterated Laurent field

$$\mathbb{Q}((X_r^{-1}))((X_{r-1}^{-1})) \cdots ((X_1^{-1})), \quad (15.3)$$

with X_1^{-1} the *outermost* variable: a positive power of X_1 dominates every monomial in the later variables, and a negative power is smaller than every positive real multiple of such a monomial. This is the lexicographic order of (15.2), not a numerical substitution. After compilation the grid fields are the effective rational monomial core `cas:thm-core` of the computer-algebra report, with $\mathbb{Z}$ -independent exponents; what 09 adds is the compilation of arbitrary nested terms into such a frame, and the omnific recognition, floor and truncation below ([merge]).

Corollary 15.3 (Finite-grid transplantation [09]). *Replacing one positive separated r -tuple of scales by another induces an ordered-field isomorphism of the corresponding rational grid fields which preserves zero-exponent coefficients, the omnific condition and truncation at zero.*

Proof. Both evaluations identify their fields with the same rational function field inside (15.3), and their lattices have the same lexicographic order, so signs and formal coefficients agree, including the test that all exponents are nonnegative. $\square$

This concerns rational functions in the selected monomials; the isomorphism is not asserted to commute with Ω on arbitrary elements (09).

16 Computable truncation and omnific recognition

16.1 A finite polynomial-part recursion

Fix a positive separated grid and put $\mathcal{K}_r = \mathbb{Q}(X_1, \dots, X_r)$, with Hahn interpretation (15.3); $\mathcal{K}_0 = \mathbb{Q}$. For $R \in \mathcal{K}_r$ let $\tau_{\geq 0}R$ be its nonnegative-exponent part and $\text{ct}(R)$ its coefficient at exponent zero in this interpretation.

Theorem 16.1 (Recursive polynomial-part formula [09]). *For $R \in \mathcal{K}_r$, $r \geq 1$, divide in X_1 over $\mathcal{K}_{r-1} = \mathbb{Q}(X_2, \dots, X_r)$:*

$$R = \sum_{j=0}^m c_j X_1^j + \frac{U}{V}, \quad \deg_{X_1} U < \deg_{X_1} V, \quad (16.1)$$

a zero polynomial part being allowed, with then $c_0 = 0$. Then

$$\tau_{\geq 0}R = \sum_{j=1}^m c_j X_1^j + \tau_{\geq 0}(c_0), \quad (16.2)$$

$$\text{ct}(R) = \text{ct}(c_0), \quad (16.3)$$

where c_0 is read in the tail grid, and at rank zero $\tau_{\geq 0}q = \text{ct}(q) = q$. Both outputs are computable, $\tau_{\geq 0}R \in \mathcal{K}_r$ and $\text{ct}(R) \in \mathbb{Q}$.

Proof. Expand at $X_1 = \infty$. The proper fraction U/V has only strictly negative powers of X_1 : factoring the leading powers of X_1 from U and V gives a negative initial power times a power series in X_1^{-1} over $\mathcal{K}_{r-1}$, including the zero remainder. After expanding coefficients in the later variables, every exponent of such a block has negative first coordinate and so is negative, whatever the other coordinates. Every term of $c_j X_1^j$, $j > 0$, has positive first coordinate and positive exponent, and these whole blocks are kept even when c_j has an infinite normal form. Only the block $j = 0$ needs a lower-rank decision. Polynomial division over a rational function field is finite, and the recursion follows only the constant block, so there are at most r univariate polynomial-part computations; their algebraic cost can grow, but the recursion cannot become an infinite expansion. $\square$

Listing 1: Truncation at zero on a separated grid (09).

```

TRUNCATE_ZERO(R, [X1, ..., Xr]):
  if r = 0:
    return (R, R)
  P := polynomial part of R in X1 over Q(X2, ..., Xr)
  c := constant coefficient of P as a polynomial in X1
  (T, C) := TRUNCATE_ZERO(c, [X2, ..., Xr])
  return (P - c + T, C)

```

The first return value is the nonnegative part and the second the rational zero-exponent coefficient.

16.2 The omnific part of a grid

Let $\mathcal{I}_r = \mathcal{K}_r \cap \mathbf{Oz}$ in the separated grid, and $\mathcal{I}_{r-1}^{\text{tail}} = \mathbb{Q}(X_2, \dots, X_r) \cap \mathbf{Oz}$; $\mathcal{I}_0 = \mathbb{Z}$.

Theorem 16.2 (Recursive grid integer ring [09]). *For every $r \geq 1$,*

$$\mathcal{I}_r = \mathcal{I}_{r-1}^{\text{tail}} + X_1 \mathbb{Q}(X_2, \dots, X_r)[X_1], \quad (16.4)$$

with a unique decomposition. Equivalently, a rational function is omnific exactly when it is a polynomial in the largest grid variable whose positive-degree coefficients are arbitrary elements of the tail field and whose constant coefficient is omnific in the tail grid.

Proof. If R is omnific its negative part vanishes, so $R = \tau_{\geq 0}R$, which by [theorem 16.1](#) is a polynomial in X_1 with positive-degree coefficients c_j and constant coefficient $\tau_{\geq 0}(c_0)$, whose zero coefficient must be an integer; so the constant lies in $\mathcal{I}_{r-1}^{\text{tail}}$. Conversely, on the right side every positive-degree block has strictly positive exponents and the constant block has nonnegative exponents and integer zero coefficient. Uniqueness is uniqueness of polynomial expressions in the algebraically independent X_1 . $\square$

This is a pullback-shaped description, not a claim to invent the ring construction: in the notation of [section 3](#) it says that $\mathcal{I}_r$ is exactly the shielded ring $\mathcal{S}_{\gamma_1}(\mathcal{I}_{r-1}^{\text{tail}}, \mathcal{K}_{r-1})$ ([remark 3.2](#)). Its use is algorithmic: it identifies which rational denominators are harmless at a smaller scale. In rank one $\mathcal{I}_1 = \mathbb{Z} + X_1 \mathbb{Q}[X_1]$; in rank two $\mathcal{I}_2 = (\mathbb{Z} + X_2 \mathbb{Q}[X_2]) + X_1 \mathbb{Q}(X_2)[X_1]$, so a denominator in X_2 is allowed in a positive X_1 block though it may make a constant block non-omnific (09).

16.3 The effective integer-part theorem

Theorem 16.3 (Omnific recognition, floor, and internal truncations [09]). *There are terminating exact algorithms on finite rational-omega terms that, for $x, a \in \mathbb{Q}$,*

- (i) *decide $x \in \mathbf{Oz}$;*
- (ii) *return a term for $\lfloor x \rfloor_{\mathbf{Oz}} \in \mathcal{I}$;*
- (iii) *return a term for $\tau_{\geq a}x \in \mathbb{Q}$ and compute $[\omega^a]x \in \mathbb{Q}$.*

Consequently the valid omnific terms form a decidable finite-string notation system with decidable equality and order.

Proof. Compile x into a separated grid ([theorem 15.2](#)) and apply [theorem 16.1](#) to compute a rational function for $\tau = \tau_{\geq 0}x$ and the rational $c = \text{ct}(x)$; evaluating back in the grid gives a finite term of $\mathbb{Q}$, since every grid monomial is the omega map of a represented exponent. Then

$$x \in \mathbf{Oz} \iff x = \tau \text{ and } c \in \mathbb{Z}, \quad (16.5)$$

the first test by [theorem 14.2](#) and the second by reducing a rational. For the floor, compute $x_{<0} = x - \tau$ and compare it with zero, and substitute $x_{>0} = \tau - c$, c and $x_{<0}$ in (2.5); the rational floor and the test $c \in \mathbb{Z}$ are computable, also for negative rationals. For an arbitrary represented cutoff a use

$$\tau_{\geq a}x = \Omega(a) \tau_{\geq 0}(x/\Omega(a)), \quad (16.6)$$

$$[\omega^a]x = \text{ct}(x/\Omega(a)), \quad (16.7)$$

which hold because multiplication by ω^{-a} subtracts a from every exponent and keeps coefficients. The shifted input is again a finite term; a need not lie in the original lattice, and $x = 0$ is covered. $\square$

The theorem supplies *effective* truncation at a named cutoff. Abstract truncation closure of a field generated by a truncation-closed set is already known (Camacho's Proposition 4.1(i) [3]); the novelty assessment is not a claim to abstract closure. What the proof adds is a computable route from this syntax to a finite output syntax, an omnific recognizer, and a floor with the infinitesimal boundary handled explicitly (09).

Corollary 16.4 (Effective division with remainder [09]). *$\mathcal{I}$ is a discretely ordered subring and an integer part of $\mathcal{Q}$. Given $a, b \in \mathcal{I}$ with $b > 0$, the unique $q, r \in \mathbf{Oz}$ with $a = qb + r$ and $0 \leq r < b$ lie in $\mathcal{I}$ and are computable; $\text{ct} : \mathcal{I} \rightarrow \mathbb{Z}$ is a computable ring retraction.*

Proof. The ring and discrete-order properties come from (2.3), and $\lfloor x \rfloor_{\mathbf{Oz}} \in \mathcal{I}$ for $x \in \mathcal{Q}$ by theorem 16.3. Existence and uniqueness of $q = \lfloor a/b \rfloor_{\mathbf{Oz}}$ and $r = a - qb$ in $\mathbf{Oz}$ are `odg:prop:division` of the omnific Diophantine report; since $a/b \in \mathcal{Q}$, the theorem computes q in $\mathcal{I}$, and then $r \in \mathcal{I}$. The constant term is computable by (16.7) with $a = 0$. $\square$

This is not a claim that a Euclidean gcd algorithm terminates. The positive omnific integers are not well ordered ($\omega, \omega - 1, \omega - 2, \dots$ strictly decreases), and successively smaller remainders need a separate termination argument, which is not supplied (09); the omnific Diophantine report exhibits a Euclidean algorithm in $\mathbf{Oz}$ that never stops (`odg:ex:euclid`).

16.4 Worked examples

Put $X = \omega^\omega$ and $Y = \omega$, so the scales ω and 1 are separated. The rational expression

$$B = \frac{X}{1 - Y^{-1}} = \sum_{n \geq 0} \omega^{\omega - n} \quad (16.8)$$

has infinitely many terms, all with positive exponents; it is omnific by theorem 16.2, its coefficient $1/(1 - Y^{-1})$ being allowed in a positive X block. Geometric examples of this kind have classical precedents [13]; here they are tests for the decision procedure. On the other hand $1/(Y - 1) = \sum_{n \geq 1} \omega^{-n}$ has zero nonnegative part, so $B + Y^{-1}$ is not omnific, and neither is $B + \frac{1}{2}$: one fails by a negative exponent, the other by a nonintegral zero coefficient. Polynomial division gives

$$\frac{X^2 + Y}{X - 1} = X + 1 + \frac{Y + 1}{X - 1}, \quad \tau_{\geq 0} \left(\frac{X^2 + Y}{X - 1} \right) = X + 1 : \quad (16.9)$$

every term of the remainder has negative X coordinate, so the remainder is infinitesimal although its numerator contains Y ; similarly $\tau_{\geq 0}(Y/(X - Y)) = 0$. The floor correction appears without series expansion: $\lfloor B - Y^{-1} \rfloor_{\mathbf{Oz}} = B - 1$, $\lfloor B + Y^{-1} \rfloor_{\mathbf{Oz}} = B$, $\lfloor B - \frac{1}{2} \rfloor_{\mathbf{Oz}} = B - 1$. For a cutoff off the lattice, $\tau_{\geq \omega - 3/2} B = \omega^\omega + \omega^{\omega - 1}$ and $\lfloor \omega^{\omega - 3/2} \rfloor B = 0$; the shifted-grid algorithm proves these by finite algebra, without searching for a last term in an infinite stream.

17 Ordinal traces

17.1 Depth gives a tower bound, not a birthday bound

Define $\beta_0 = \omega$ and $\beta_{d+1} = \omega^{\beta_d}$; their supremum is ε_0 , the least positive fixed point of $\alpha \mapsto \omega^\alpha$.

Lemma 17.1 (Strong ordinal majorants [09]). *For every d and $x \in \mathcal{Q}_d$ there is an ordinal $\alpha < \beta_d$ with $|x| < \alpha$.*

Proof. At $d = 0$ a rational is bounded by a natural number. For nonzero $x \in \mathcal{Q}_{d+1}$, [theorem 14.2](#) gives $e = \deg_\omega x \in \mathcal{Q}_d$ and a nonzero rational leading coefficient; by induction choose $\delta < \beta_d$ with $|e| < \delta$, and $\delta + 1 < \beta_d$ since β_d is an infinite limit. Then $|x| < \omega^{e+1} < \omega^{\delta+1}$, because $|x|/\omega^e$ is a positive finite surreal with rational standard part, below ω ; and $\omega^{\delta+1} < \beta_{d+1}$. Zero is bounded by 1. $\square$

This supplies an ordinal strictly below the boundary that dominates the value, which matters because a surreal below a limit ordinal need not be below an ordinal strictly below that limit (09).

Theorem 17.2 (Exact ordinal trace at every depth [09]). *For every $d < \omega$,*

$$\mathcal{Q}_d \cap \mathbf{On} = [0, \beta_d), \quad \mathcal{Q} \cap \mathbf{On} = [0, \varepsilon_0). \quad (17.1)$$

In particular ε_0 has no rational-omega notation.

Proof. The inclusion $\subseteq$ is [lemma 17.1](#). Conversely induct on d : the ordinals below ω are rational. If $\alpha < \omega^{\beta_d}$, its Cantor normal form $\omega^{\alpha_1}n_1 + \dots + \omega^{\alpha_k}n_k$ has $\beta_d > \alpha_1 > \dots > \alpha_k$, each $\alpha_i \in \mathcal{Q}_d$ by induction, so it is a term of depth at most $d + 1$. The decreasing Cantor form agrees with the Conway normal form, so although surreal operations on ordinals are the natural ones, this decreasing sum has the required value. The union over d gives the second identity. $\square$

Division and negative or fractional intermediate values do not enlarge the ordinal trace beyond the ε_0 range of hereditary finite Cantor descriptions, but they enlarge the nonordinal omnific values very substantially. The smaller hereditary hierarchy $H_0 = \mathbb{Q}$, $H_{d+1} = \mathbb{Q}[\omega^{H_d}]$, with finite outer support and no fractions, has the same ordinal trace by the same argument but is strictly smaller than $\mathcal{Q}$: $1/(\omega - 1)$ has infinite outer support. The difference is invisible in the representable ordinals (09). The union of the H_d is 05's ring $\mathcal{H}_{\mathbb{Q}}$ ([theorem 11.3](#)), since both consist of the finite $\mathbb{Q}$ -linear combinations of ω -powers of earlier elements, starting from $\mathbb{Q}$ ([merge]).

17.2 Three systems, one trace

The three systems of this report have the same ordinal trace, $\mathbf{On}_{<\varepsilon_0}$: 08's labelled union by [theorem 9.2](#), 05's hereditary ring by [theorem 11.3](#), and $\mathcal{Q}$ by [theorem 17.2](#), which alone refines the trace depth by depth (08 refines it by rank, [theorem 8.3](#)). The trace does not determine the system.

Proposition 17.3 (The omnific rings are incomparable [merge]). *Let $\mathcal{R}_\varepsilon$ be the omnific part of the labelled holonomic union over a coefficient field k , let $\mathcal{S}_{\varepsilon_0}$ be the ε_0 -shielded ring over $\mathbb{Q}$ or $\mathbb{R}_{\text{alg}}$, and let $\mathcal{I} = \mathcal{Q} \cap \mathbf{Oz}$.*

- (i) $\mathcal{S}_{\varepsilon_0}$ and $\mathcal{I}$ are incomparable, and so are $\mathcal{S}_{\varepsilon_0}$ and $\mathcal{R}_\varepsilon$, for every choice of coefficient fields.
- (ii) $\mathcal{I} \not\subseteq \mathcal{R}_\varepsilon$ for every k , and $\mathcal{R}_\varepsilon \not\subseteq \mathcal{I}$ whenever $k \neq \mathbb{Q}$.

All three contain every ordinal below ε_0 and no other ordinal, except that $\mathcal{S}_{\varepsilon_0}$ also contains ε_0 and other ordinals $\geq \varepsilon_0$.

Proof. First, an element $x \in \mathcal{S}_{\varepsilon_0}$ which is bounded in absolute value by an ordinal $\beta < \varepsilon_0$ has finite support. Indeed write $x = d + \sum_{j=1}^m \varepsilon_0^j f_j$ with $f_m \neq 0$, $m \geq 1$. By [theorem 3.1](#) the leading exponent of x is $m\varepsilon_0 + \deg_\omega f_m$, and $\deg_\omega f_m$ lies in the group $\mathcal{H}$ ([lemma 12.1](#)), so $|\deg_\omega f_m| < \alpha$ for some ordinal $\alpha < \varepsilon_0$ ([lemma 11.4](#)). Hence $\deg_\omega x > \varepsilon_0 - \alpha$. For $\beta < \varepsilon_0$ write $\beta < \omega^\gamma$ with $\gamma < \varepsilon_0$; the natural sum $\alpha \oplus \gamma$ is below ε_0 , so $\varepsilon_0 - \alpha > \gamma$ and $|x| > \omega^\gamma > \beta$. So a bounded x has all $f_j = 0$ and $x = d \in \mathcal{D}$, a finite hereditary form.

(i) $\varepsilon_0 = \varepsilon_0^1 \cdot 1 \in \mathcal{S}_{\varepsilon_0}$, while $\varepsilon_0 \notin \mathcal{I}$ by [theorem 17.2](#) and $\varepsilon_0 \notin \mathcal{R}_\varepsilon$ because $\mathcal{F}_\varepsilon \subset (-\varepsilon_0, \varepsilon_0)$ ([corollary 9.3](#)). Conversely $B = \sum_{n \geq 0} \omega^{\omega^{-n}}$ lies in $\mathcal{I}$ ([subsection 16.4](#)) and in $\mathcal{R}_\varepsilon$ (it is $U \text{Geo}(t)$ with labels $\{0, 1\}$), is bounded by the ordinal $\omega^{\omega+1}$, and has infinite support, so $B \notin \mathcal{S}_{\varepsilon_0}$.

(ii) $\omega^{1/2} \in \mathcal{I}$. Every exponent of an element of $\mathcal{F}_A$ lies in $G_A = \sum_{\alpha \in A} \mathbb{Z}\omega^\alpha$ ([lemma 6.1](#)), and the Conway normal form of an element of G_A has integer coefficients, whereas $\frac{1}{2} = \frac{1}{2}\omega^0$; so $\omega^{1/2} \notin \mathcal{R}_\varepsilon$. If $c \in k \setminus \mathbb{Q}$, then $c\omega = cU_0 \in \mathcal{R}_{\{0\}}$ by [theorem 7.2](#), while every element of $\mathcal{Q}$ has rational coefficients ([lemma 14.4](#)).

The trace statements are [theorems 9.2, 11.3](#) and [17.2](#); $\varepsilon_0 \cdot 2$ and ε_0^2 lie in $\mathcal{S}_{\varepsilon_0}$. $\square$

For $k = \mathbb{Q}$ it is not established here whether $\mathcal{R}_\varepsilon \subseteq \mathcal{I}$; for instance, whether $U \text{Fac}(t) = \sum_n n! \omega^{\omega^{-n}}$ or $U \text{Ex}(t)$ has a rational-omega term is not decided by any source, and neither is whether they lie in the field $\text{Frac}(\mathcal{H})$.

17.3 What can and cannot be inferred from the Church–Kleene bound

Proposition 17.4 (An effective ordinal-order barrier [09; 05]). *Let $N \subseteq \mathbb{N}$ be decidable and $\nu : N \rightarrow \mathbf{On}$, with equality and order of the values decidable. Then the ordered range of ν has order type strictly below ω_1^{CK} ; in particular it cannot contain all computable ordinals. If in addition the interpretation uniformly supplies computable presentations of the well-orders of its values, the range is not cofinal in the computable ordinals.*

Proof. Put $N^* = \{e \in N : \text{no } f < e \text{ in } N \text{ has } \nu(f) = \nu(e)\}$, a decidable set since only finitely many smaller codes are tested. The semantic order on N^* is a computable well-order, so its type is a computable ordinal, below ω_1^{CK} . For the cofinality statement, take the supplied presentation for $e \in N$ and the empty order otherwise; the ordinal sum over $e \in \mathbb{N}$ (05 uses $(L_0 + 1) + (L_1 + 1) + \dots$) is a computable well-order on a decidable set of pairs, comparing first the outer index and then the component order, and its type is a computable ordinal at least as large as every value; its successor is a computable strict upper bound. $\square$

05 states the second half for a computably enumerable family of support descriptions with uniformly computable presentations of their decreasing well-orders: their order types are bounded by a computable ordinal and so are not cofinal in ω_1^{CK} (the finite case is a finite ordinal sum plus one). 09 adds the first half and the following caveat.

Remark 17.5 (The uniform-presentation hypothesis is essential [09]). The order-type conclusion alone does *not* bound the supremum of an arbitrarily interpreted range: a noncomputable increasing sequence of computable ordinals can be cofinal in ω_1^{CK} and still have the trivially decidable index order of type ω . For completely arbitrary semantic maps only the first conclusion and the prohibition on containing every computable ordinal apply; equality alone, or equality with an abstract order, is often overinterpreted.

This is a classical computability observation rather than a new theorem, and it does not apply directly to $\mathcal{I}$ with its whole order, which is not well ordered; [theorem 17.2](#) is a stronger language-specific statement that does not use it (09). A bare finite symbol denoting an ordinal is not a computable copy of its well-order, and the proposition supplies a bound for each

sound system, not a single algorithm extracting a least bound from every description. A sound effective support calculus can be very expressive but cannot effectively cover all recursive well-order types; adding larger named ordinals or rank constructors may extend a calculus without creating a universal validator. The one-state supports of type ω^r (corollary 12.4) against the hard stream family of support type ω (section 20) show that support order type alone does not measure equality difficulty; coefficient representation and the available certificates are decisive (05).

18 Support complexity and values outside the language

The outer normal form is a reverse well-ordered sequence of monomials whose length $\text{ord}(\text{supp } x)$ need not resemble the ordinal value of x , even when x is an ordinal.

Lemma 18.1 (Lexicographic support bound [09]). *If $S \subseteq \mathbb{Z}^r$ is reverse well ordered lexicographically, its decreasing order type is at most ω^r ; if $r \geq 1$ and $S \subseteq (\mathbb{Z}^r)_{\geq 0}$, it is strictly below ω^r .*

Proof. For $r = 0$ there is at most one element. For $r \geq 1$ the first coordinates of a nonempty reverse well-ordered set are bounded above, and listed decreasingly they form a finite or ω -sequence; each fibre has type at most ω^{r-1} by induction, so the sum is at most ω^r . If S is nonnegative there are finitely many fibres: for $r = 1$ this is finiteness, and for $r > 1$ the positive fibres have type at most ω^{r-1} and the zero fibre, nonnegative in the tail group, less than ω^{r-1} , so the finite sum is below ω^r . $\square$

Theorem 18.2 (The support barrier [09]). *If a term compiles into a rank- r grid, its support has type at most ω^r , and strictly less for an omnific value in a positive-rank grid. Consequently*

$$x \in \mathcal{Q} \implies \text{ord}(\text{supp } x) < \omega^\omega, \quad (18.1)$$

independently of the omega-nesting depth.

Proof. Apply lemma 18.1 to the lattice support of theorem 15.2; omnific support lies in the nonnegative cone, the rank is finite, and rank zero gives at most one term. $\square$

The theorem does not assert that every support below the bound occurs: it is necessary, not a characterization (09). The same bounds hold in the towers (theorem 8.1), where every $\eta < \omega^r$ is realized.

Theorem 18.3 (Sharpness at depth two [09]). *For every $d \geq 1$ there is $B_d^\sharp \in \mathcal{Q}_2 \cap \mathbf{Oz}$ with $\text{ord}(\text{supp } B_d^\sharp) = \omega^d$; so the supremum of the support types of $\mathcal{Q}_2 \cap \mathbf{Oz}$ is already ω^ω .*

Proof. Put $\gamma_0 = \omega^d$ and $\gamma_i = \omega^{d-i}$ for $1 \leq i \leq d$; they lie in $\mathcal{Q}_1$ and $\gamma_0 \gg \dots \gg \gamma_d > 0$. Let

$$B_d^\sharp = \omega^{\gamma_0} \prod_{i=1}^d (1 - \omega^{-\gamma_i})^{-1} = \sum_{\mathbf{n} \in \mathbb{N}^d} \omega^{\gamma_0 - n_1 \gamma_1 - \dots - n_d \gamma_d}. \quad (18.2)$$

Every exponent is positive since γ_0 dominates the later scales, distinct multi-indices give distinct exponents, and the decreasing order corresponds to lexicographically increasing $(n_1, \dots, n_d)$, a d -fold ordinal sum of type ω^d . The sum is strong, all coefficients are one and no exponent is zero. $\square$

The grid here has $d + 1$ scales, consistent with the nonnegative-cone bound $< \omega^{d+1}$ (09; compare [corollary 12.4](#)).

Theorem 18.4 (Explicit summation escape [09]). *There is an explicitly computable sequence of terms $E_d \in \mathcal{Q}_2 \cap \mathbf{Oz}$, $d \geq 1$, whose strong sum is an omnific integer $E \notin \mathcal{Q}$ with support type exactly ω^ω .*

Proof. For $d \geq 1$ put $b_d = \omega^{-d}$, $\delta_{d,i} = \omega^{-(d+i)}$ for $1 \leq i \leq d$, and

$$E_d = \omega^{b_d} \prod_{i=1}^d (1 - \omega^{-\delta_{d,i}})^{-1}, \quad (18.3)$$

so $E_d \in \mathcal{Q}_2$ with support $S_d = \{b_d - \sum_i n_i \delta_{d,i} : \mathbf{n} \in \mathbb{N}^d\}$ of type ω^d , all coefficients one and all exponents positive. For every tuple, $b_d - \sum_i n_i \delta_{d,i} > b_{d+1}$, since the difference has leading term ω^{-d} with coefficient one; so S_d lies above S_{d+1} , whose greatest member is b_{d+1} . The union in decreasing order is the concatenation $S_1, S_2, \dots$, of type $\omega + \omega^2 + \dots = \omega^\omega$, reverse well ordered with each exponent in one block; $E = \sum_d E_d$ is a strong sum with coefficient one everywhere and positive support, so $E \in \mathbf{Oz}$, and $E \notin \mathcal{Q}$ by [theorem 18.2](#). The map from d to the syntax (18.3) is computable. $\square$

This exhibits a specific limit of the language, not a cardinality obstruction: finite rational-omega arithmetic is not closed under explicitly generated strong sums of its own omnific values (09).

Proposition 18.5 (The finite-lattice obstruction [09]). *The omnific integer $H = \sum_{n \geq 1} \omega^{1/n}$ has support type ω but is not in $\mathcal{Q}$.*

Proof. The exponents $1, \frac{1}{2}, \frac{1}{3}, \dots$ decrease and are positive, so this is a valid omnific normal form of type ω . If it were in $\mathcal{Q}$, [lemma 14.4](#) would put its support in a finitely generated group Γ ; then $\Gamma \cap \mathbb{Q}$ is finitely generated, hence cyclic, inside $(1/N)\mathbb{Z}$ for some N , which does not contain all $1/n$. $\square$

Three restrictions should therefore be remembered: rational coefficients, a finitely generated exponent lattice for each input, and a support-length bound; the last alone is far from sufficient. The example is a diagnostic consequence of finite-grid support, not a new series (09).

19 Certified infinite descriptions and Diophantine scope

The contrast between finite expressions and arbitrary programs is not a finite-versus-infinite dichotomy: a finite certificate can control an infinite coefficient sequence. The two intermediate languages below use elementary linear and polynomial algebra and are not new general theorems about automata or recurrences (09).

19.1 Linear-recurrence stream certificates

Let $M \in \mathbb{Q}^{m \times m}$ and $u, v \in \mathbb{Q}^m$, and put $a_n = u^\top M^n v$. The finite triple (u, M, v) has the omnific value

$$\sum_{n \geq 0} a_n \omega^{\omega^{-n}} = \omega^\omega u^\top (I - \omega^{-1} M)^{-1} v, \quad (19.1)$$

the inverse existing because its determinant is a polynomial in ω^{-1} with constant coefficient one; the right side lies in $\mathcal{Q}_2 \cap \mathbf{Oz}$, and the formal geometric identity proves (19.1) coefficientwise.

Proposition 19.1 (Finite recurrence witness bound [09]). *For an m -dimensional triple the value is zero iff $a_0 = \dots = a_{m-1} = 0$. Two triples of dimensions m and m' are equal iff their first $m + m'$ coefficients agree, and their order is decidable. The zero-test bound m is sharp.*

Proof. By Cayley–Hamilton, M^m is a rational combination of $I, \dots, M^{m-1}$, so the a_n satisfy a recurrence of order at most m and vanish if the first m do. The block-diagonal matrix with output row $(u^\top, -u'^\top)$ represents the difference in dimension $m + m'$; its first nonzero coefficient occurs before the bound and gives the sign. For sharpness take the nilpotent shift $Me_i = e_{i+1}$, $Me_m = 0$, with $v = e_1$ and $u = e_m$: $a_n = 0$ for $n < m - 1$ and $a_{m-1} = 1$. $\square$

This is the case $r = 1$ of 05’s finite-front theorem (theorem 13.1), shifted by ω^ω . It asks whether *all* coefficients vanish; it does not solve the Skolem problem of whether *some* coefficient of a general recurrence vanishes (09).

19.2 Rational jets in several separated directions

Let $P, Q, R, S \in \mathbb{Q}[t_1, \dots, t_d]$ with $Q(0)S(0) \neq 0$, so P/Q and R/S have formal power-series expansions over $\mathbb{N}^d$. Choose $b_i \geq 0$ bounding the degree of $PS - RQ$ in each variable, for example $b_i = \max\{0, \deg_i P + \deg_i S, \deg_i R + \deg_i Q\}$, omitting the degree terms of zero polynomials.

Theorem 19.2 (Finite rectangular jet certificate [09]). *$P/Q = R/S$ iff their formal coefficients agree on the box*

$$0 \leq n_i \leq b_i \quad (1 \leq i \leq d). \quad (19.2)$$

After substituting $t_i = \omega^{-\gamma_i}$ for positive separated $\gamma_1 \gg \dots \gg \gamma_d$ the same criterion decides equality, and multiplication by ω^{γ_0} with $\gamma_0 \gg \gamma_1$ makes both series omnific without changing the criterion.

Proof. Let $F = P/Q - R/S$ and $N_F = PS - RQ = QSF$. Equality implies agreement. Conversely, if the coefficients of F vanish on the box, the coefficient of QSF at a box index is a finite sum of products with coefficients of F at indices $\mathbf{n} - \mathbf{j}$ that stay in the box, which is closed downward; so N_F vanishes on the box, and outside it by the degree bounds, hence $N_F = 0$ and $F = 0$. Separated scales make $\mathbf{n} \mapsto -\sum n_i \gamma_i$ injective and order-compatible, so no coefficients merge; after the shift every exponent is positive, and multiplication by a nonzero monomial preserves equality. $\square$

The certificate depends on the finite algebraic input; there is no universal prefix length for all rational series, since the first nonzero coefficient can be postponed by a high monomial, and the certificate cannot be transferred to arbitrary primitive-recursive streams (theorem 20.4) (09). The theorem is a special case of the fraction-field setting of Part II.

19.3 Ground equality is not Diophantine decidability

Proposition 19.3 (Classical transfer consequence [09]). *For every $P \in \mathbb{Z}[Y_1, \dots, Y_m]$, P has a zero in $\mathcal{I}^m$ iff it has one in $\mathbb{Z}^m$. Thus solvability of integer polynomial equations over $\mathcal{I}$ is undecidable, although equality of named elements is decidable.*

Proof. Apply the ring retraction $\text{ct} : \mathcal{I} \rightarrow \mathbb{Z}$ to a solution, and use $\mathbb{Z} \subseteq \mathcal{I}$ for the converse; the undecidability is the negative solution of Hilbert’s tenth problem [15]. This is the argument of `odg:eq:existencetransfer` and `odg:cor:H10` of the omnific Diophantine report, which prove

the same equivalence over all of **Oz**, for positive existential formulas, and the c.e.-completeness of the yes-instances. $\square$

This is included to prevent a scope error, not as a new omnific-Diophantine result: ground equality tests given tuples, while an existential question ranges over all tuples and an equality algorithm only semidecides it by search. 09 wrote that “similar constant-term arguments already belong to the repository’s omnific arithmetic program”, citing the catalogue; the precise reference is `odg:cor:H10` ([subsection A.4](#)).

Part IV

The complexity of equality, positivity, and validity

The positive systems carry certificates telling when an entire coefficient sequence vanishes. This part removes that certificate without introducing any uncertainty about totality or support validity ([sections 20 to 22](#)), and then releases the support geometry as well, exposing an independent well-foundedness obstruction ([section 23](#)). The base is 08’s §§11–13, with 05’s §§7–9 and 09’s §8. Throughout $U = \omega^\omega$ and $t = \omega^{-1}$ (05 wrote X for ω^ω).

20 Primitive-recursive stream names

20.1 An everywhere-valid syntax

Fix a standard decidable grammar of primitive-recursive functions $\mathbb{N} \rightarrow \mathbb{N}$. A rational-valued program is a triple (p, q, b) of such programs with value $(p(n) - q(n))/(1 + b(n))$, so its denominator is positive by syntax, not by an undecidable promise; integer-valued programs are the case $b = 0$. Every code is total, and universal evaluation is computable although not itself primitive recursive. Primitive-recursive syntax is used so that totality is guaranteed syntactically; running arbitrary purported programs for total functions would introduce a separate validity problem (05, 08). Boolean output can be enforced syntactically by a final reduction modulo two (09).

For fixed $d \geq 1$, a raw d -block name is a tuple of rational programs $a_1, \dots, a_d$ with denotation

$$Z(a_1, \dots, a_d) = \sum_{j=1}^d \sum_{n \geq 0} a_j(n) \omega^{j\omega - n} = \sum_{j=1}^d U^j \sum_{n \geq 0} a_j(n) t^n. \quad (20.1)$$

Here $j\omega$ is the surreal integer multiple of ω , the ordinal $\omega \cdot j$, not the ordinal product $j \cdot \omega$; this avoids a collapse of the blocks (08). The case $d = 1$, $Z(a) = \sum_n a(n) \omega^{\omega - n}$, is 05’s stream notation with rational a and 09’s Boolean stream notation with $a : \mathbb{N} \rightarrow \{0, 1\}$.

Proposition 20.1 (Total validity and effective ring operations [08]). *Every raw d -block name denotes an omnific integer, and distinct pairs (j, n) have distinct exponents. Allowing finitely many blocks and an integer constant gives a notation ring with computable addition, negation and multiplication of codes.*

Proof. All exponents $j\omega - n$ are positive and decrease with n within a block, and every exponent of block $j + 1$ exceeds every exponent of block j ; a finite union of such blocks is

reverse well ordered, also after zero coefficients are omitted. Distinctness is the independence of 1 and ω . Addition and negation are coefficientwise; in a product the coefficient in block j at index n is a finite sum of products with block indices adding to j and ordinary indices adding to n , and bounded sums and products of primitive-recursive rational functions are primitive recursive. The number of blocks stays finite and all exponents except the constant stay positive. $\square$

For the positivity classification of [section 22](#) the zero-constant syntax (20.1) is used; adding a positive ordinary constant changes the initial-default bookkeeping of the mind-change bound and is not silently included (08). Zero coefficients disappear from the semantic support; every name is valid, its support lies in a fixed reverse well-ordered set, and for $d = 1$ its support type is at most ω (09).

20.2 The exact equality complexity

Theorem 20.2 (Raw equality and zero testing [08; 05, 09]). *For every fixed $d \geq 1$, equality of raw d -block names is Π_1^0 -complete, and so is the zero test. Inequality is c.e., and equality is not c.e. Hardness holds on a uniformly generated family whose values have at most one nonzero monomial, with coefficients in $\{0, 1\}$. For $d = 1$, each strict comparison relation is Σ_1^0 -complete.*

Proof. Normal-form uniqueness and the fixed grid of distinct exponents give

$$Z(a_1, \dots, a_d) = Z(b_1, \dots, b_d) \iff \forall n \forall j \leq d [a_j(n) = b_j(n)],$$

a computable matrix under one universal quantifier; a mismatch is found by evaluating the programs at $n = 0, 1, \dots$, so inequality is c.e. For hardness let $h_e(n) = 1$ if machine e on empty input halts for the first time at step n and 0 otherwise; bounded simulation makes h_e primitive recursive uniformly in e . With $a_1 = h_e$ and the other blocks zero,

$$Z_e = \sum_{n \geq 0} h_e(n) \omega^{\omega^{-n}} = \begin{cases} 0, & e \text{ never halts,} \\ \omega^{\omega^{-N}}, & e \text{ first halts at step } N, \end{cases} \quad (20.2)$$

so $Z_e = 0$ is exactly nonhalting, a Π_1^0 -complete property, and every value has at most one monomial; comparison with a fixed zero name gives the pair version. If equality were c.e., halting and nonhalting would both be c.e.

For $d = 1$ and strict comparison, search for the first index where the coefficients differ; its sign decides, so $Z(a) < Z(b)$ has the c.e. witness $\exists n (a(n) < b(n) \wedge \forall i < n a(i) = b(i))$, and the search need not stop on equal inputs. Finally $0 < Z(h_e)$ exactly when e halts, which gives Σ_1^0 -hardness; reversing the pair handles the other relation. $\square$

Second reduction. 05 reduces with the cumulative predicate $\bar{h}_e(n) = 1$ iff e has halted within n steps: $Z(\bar{h}_e) = 0$ iff e never halts, and $Z(\bar{h}_e) = Ut^s/(1-t)$ if e first halts at step s , so the hard values have full tails rather than one term ([section 21](#)).

Credit. Both reductions are those of `cas:thm-zerotest` of the computer-algebra report, which proves that total coefficient access does not decide zero for Hahn series at $t = \omega^{-1}$: its Reduction A is $\bar{h}_e$, with the moral that values in a decidable rational field become undecidable when presented by unrestricted programs (`cas:rem-moralA`); its Reduction B is the first-halt spike h_e , with the moral that the obstruction survives a promise of at most

one nonzero coefficient (`cas:rem-moralB`). The report was present at the sources' pin; 08 read only its README and counted the one-monomial localization as its own, and 05 and 09 did not cite it ([subsection A.4](#)). What is new here is the placement inside always-valid omnific names on positive supports, the d -block form, the translator corollaries and the exact realization dimension of [theorem 21.3](#). The method is a standard halting reduction, not a new undecidability principle (08, 05, 09).

The coefficient field here is $\mathbb{Q}$, the support is prescribed, and every candidate exponent is an elementary hereditary expression; none of the hardness comes from exotic real coefficients, uncertain well-foundedness or large ordinal notations, but from the universal demand that no later nonzero coefficient exists (05). It cannot be evaded by requiring a support certificate, restricting coefficients to $\{0, 1\}$ or assuming that every program is total (09).

Corollary 20.3 (Known leading data do not decide equality [08]). *There is no equality algorithm for a uniformly named family of nonzero omnific integers with the same leading exponent, leading coefficient and constant term, even if each support has at most two terms.*

Proof. Compare $Y_e = U^2 + Z_e$ with U^2 : the leading exponent 2ω , the leading coefficient 1 and the constant 0 do not depend on e , and equality holds exactly when e never halts. $\square$

20.3 The same values in incompatible representations

Theorem 20.4 (Same values, no translator [08, 09]). *Every value Z_e of (20.2) has support of size at most one, lies in the tower ring R_2 (labels 0, 1) as the rational monomial Ut^N or 0, and lies in $\mathcal{Q}_2 \cap \mathbf{Oz}$ as*

$$\omega^{\omega-N} = \frac{\Omega(\Omega(1))}{\Omega(1)^N}.$$

Nevertheless no total computable map translates all these raw names into value- equivalent names of any notation system with decidable equality and an available name for zero. In particular there is no uniform compiler from these stream names into the holonomic towers or into rational-omega terms, although a suitable finite term exists for each value.

Proof. A machine has at most one first halting step, which gives the values. If a translator existed, translate Z_e and compare with the translated zero name; this would decide nonhalting, contrary to [theorem 20.2](#). $\square$

The obstruction is not that the values lack exact finite descriptions but the absence of a uniform algorithm extracting them from this input representation (08). Here the *semantic normal form is finite*, with at most one term; what is missing is an effective certificate of whether a term occurs and where. A promise of finite support is not an explicit support list or a computable bound on its positions (09). The existence of canonical Conway normal forms does not give a computable canonicalizer: on the decidable term language one exists ([corollary 14.3](#)), but on these stream indices a canonicalizer into that language would be the forbidden compiler; the input representations differ (09).

20.4 What finite-support information suffices

Proposition 20.5 (An exact support count is enough [08]). *For a raw finite-block name promised to have exactly $s < \omega$ nonzero coefficients, with s supplied, its finite normal form is computable. Replacing the exact count by the upper bound “at most one” does not give an equality algorithm.*

Proof. If $s = 0$ return zero. Otherwise evaluate all blocks on successive complete prefixes until s nonzero coefficients are found; the promise makes the search terminate and guarantees no other terms. Sort and return. The failure for an upper bound is (20.2). $\square$

This is a promise algorithm and does not certify the count; an erroneous count can make it diverge or return an incorrect truncation, so the count must come from a trusted certificate or be an explicit precondition (08).

Corollary 20.6 (No uniform finite observation bound [08]). *No total computable function of a raw name gives a prefix length whose vanishing in every block guarantees that the value is zero. There is no sound, c.e., complete proof system for equality of these names.*

Proof. A prefix bound would reduce the zero test to finitely many coefficient tests. A sound and complete enumerable proof system would make equality c.e.; inequality is c.e., so dovetailing would decide equality. Both contradict theorem 20.2. $\square$

20.5 A single computable real coefficient

Proposition 20.7 (One-term obstruction for Cauchy names [05, 09]). *A finite-list language allowing arbitrary computable real coefficients, given by fast rational Cauchy names, need not have decidable equality; the failure occurs already for one-term names $r\omega$.*

Proof. Let $r_e = 0$ if machine e never halts and $r_e = 2^{-s}$ if it first halts at step s (09; 05 uses $2^{-(s+1)}$). At precision n , simulate n steps and output the corresponding value if e has halted and 0 otherwise; a later halt changes the value by at most $2^{-(n+1)}$, so this is a uniform fast Cauchy name. Every $r_e\omega$ is omnific and is zero exactly when e never halts. $\square$

Every r_e here is a dyadic rational; the obstruction lies in extracting an exact finite answer from an intensional name (09). This is why exact algebraic codes, not arbitrary Cauchy names, were specified for the positive systems (05). The constants are those of `thm:inversehard(iv)` of the computable-surreals report, which proves Π_1^0 -complete equality on raw numerical names with the same family (section 25).

21 The rational-realization gap

This section is 05's. The obstruction can be strengthened to a family whose values all have simple rational generating functions and identical supports. Let

$$a_e(n) = 1 + \bar{h}_e(n), \quad A_e^{\text{sw}} = U \sum_{n \geq 0} a_e(n)t^n, \quad B = \frac{U}{1-t}. \quad (21.1)$$

All coefficients of A_e^{sw} are 1 or 2, and every support is exactly $\{\omega - n : n \in \mathbb{N}\}$.

Theorem 21.1 (Rational full-support hardness [05]). *The uniformly primitive-recursive family (21.1) satisfies:*

- (i) *every value is rationally generated: it is B if e never halts, and $A_e^{\text{sw}} = U(1+t^s)/(1-t)$ if e first halts at step s ;*
- (ii) *the equality $A_e^{\text{sw}} = B$ is Π_1^0 -complete, and equality among the A_e^{sw} is Π_1^0 -complete after choosing a known nonhalting baseline code;*

- (iii) *no total computable translator takes these stream codes to explicit rational-function codes denoting the same values, nor to any notation system with decidable equality and an effective notation for $\mathbf{B}$.*

Proof. If halting first occurs at s , the added coefficients are 0 before s and 1 from s on, a geometric tail $t^s/(1-t)$; without halting the added stream is zero. This gives (i), and equality to $\mathbf{B}$ is exactly nonhalting; the coefficientwise upper bound gives (ii). A translator as in (iii), followed by cross multiplication of rational functions or by the target system's equality test against its name for $\mathbf{B}$, would decide nonhalting. $\square$

The difficulty is not that arbitrary streams denote nonrational functions: every member of this syntactically valid family is rational, but the rational description cannot be extracted uniformly from the stream program (05).

Corollary 21.2 (The same decidable ring admits hard names [05]). *There is a uniformly primitive-recursive family of stream names whose values all lie in $\mathcal{S}_{\varepsilon_0}$ (theorem 12.2), but whose equality to one fixed member is Π_1^0 -complete; the same values with explicit polynomial-fraction names in $\mathcal{S}_{\varepsilon_0}$ have decidable equality.*

Proof. Replace U by ε_0 in (21.1). Each stream then denotes $\varepsilon_0/(1-t)$ or $\varepsilon_0(1+t^s)/(1-t)$, whose rational coefficient lies in $\text{Frac}(\mathcal{H})$, so the value lies in $\mathcal{S}_{\varepsilon_0}$; equality to $\varepsilon_0/(1-t)$ still means nonhalting. The explicit-name algorithm is theorem 12.2. $\square$

Restricting attention to a fixed ring with a decidable presentation does not remove representation dependence: it is the uniform availability of the finite certificate, not its existence somewhere, that supplies the algorithm (05).

Credit. Parts (i)–(iii) are anticipated in substance by `cas:rem-moralA`, whose values $t^N/(1-t)$ lie in a decidable rational field. New in 05 are the omnific placement with full support and coefficients 1 and 2, the same-ring corollary, and the exact dimension count that follows.

21.1 Exact minimal dimension and absence of an effective bound

For a rational power series $f(t)$ over k , a univariate linear realization of dimension m is $f(t) = u(I - tM)^{-1}v$ with $M \in k^{m \times m}$; no minimality promise is imposed on the input.

Theorem 21.3 (Unbounded realization complexity [05]). *Let $f_e(t) = \sum_n a_e(n)t^n$. If e never halts its minimal nonzero realization dimension is 1; if e first halts at step $s \geq 0$ it is exactly $s + 1$. Consequently no total computable function of the stream program code bounds the minimal realization dimension on this family.*

Proof. The nonhalting series $(1-t)^{-1}$ and the series $2/(1-t)$ for $s = 0$ have one-dimensional realizations. For $s \geq 1$, $f_e = (1+t^s)/(1-t)$, and numerator and denominator are coprime since the numerator is 2 at $t = 1$. In dimension m the adjugate formula writes f_e with numerator degree at most $m - 1$, and reduction cannot raise that degree; the reduced numerator has degree s , so $m \geq s + 1$. Conversely the sequence is s ones followed by twos and satisfies $a(n + s + 1) = a(n + s)$, so a companion shift on windows of $s + 1$ coefficients realizes it in dimension $s + 1$. If a computable bound $b(e)$ existed, simulating e for $b(e)$ steps (one more if the convention requires) would decide halting, since a later first halt at s would give $s + 1 > b(e)$. $\square$

This is more precise than saying normalization is difficult: a finite-state certificate exists for every input, but even its size cannot be bounded computably from this kind of name. The finite-front theorem ([theorem 13.1](#)) is not contradicted: it needs the realization itself, not a promise that one exists (05).

Corollary 21.4 (No complete effective equality certificates [05]). *No computably enumerable sound proof system proves every true equation $A_e^{\text{sw}} = B$, and there is no total computable stopping bound for the number of coefficients to check.*

Proof. Enumerating proofs would enumerate the nonhalting codes; a stopping bound would decide the same set by bounded checking. $\square$

An incomplete proof-producing simplifier is compatible with this: a sound implementation distinguishes “equal”, “different” and “not established” and never turns a timeout into a false equality (05).

22 The finite mind-change spectrum of positivity

This section is 08’s. Equality of raw names is controlled by one universal quantifier; order has a different obstruction, since a later discovery in a higher block overrides every conclusion based on a lower one. The result is an exact classification in the finite Ershov hierarchy rather than another undecidability proof.

Definition 22.1 (*d*-c.e. set). For $d \geq 1$, $A \subseteq \mathbb{N}$ is *d*-c.e. if there is a total computable approximation $a(e, s) \in \{0, 1\}$ with $a(e, 0) = 0$, $\lim_s a(e, s) = \mathbf{1}_A(e)$, and at most d changes of value for each e . The 1-c.e. sets are the c.e. sets, the 2-c.e. sets the d.c.e. sets; the finite levels and their strictness are standard [8].

Theorem 22.2 (Dominance-layer completeness [08]). *For $d \geq 1$ let Pos_d be the set of raw d -block names (20.1) denoting a strictly positive omnific integer. Pos_d is many-one complete for the d -c.e. sets. The hardness reduction can be chosen with at most one nonzero coefficient per block, each equal to ± 1 , and total support of at most d terms. Equality on the unrestricted raw d -block syntax remains Π_1^0 -complete.*

Upper bound. At stage $s \geq 1$ compute the coefficients with $0 \leq n < s$ in every block; a block is discovered if this prefix contains a nonzero coefficient, and then its first nonzero coefficient and its sign are known permanently. If no block is discovered output 0; otherwise let j_s be the largest discovered block and output 1 exactly when its first nonzero coefficient is positive; stage zero outputs 0. The output can change only when a higher block replaces the current highest, the record indices increase strictly and there are at most d of them, so there are at most d changes, including a possible initial change from 0 to 1. Every nonzero block is eventually discovered; if all blocks vanish the output stays 0, and otherwise the final highest block is the highest nonzero block, whose first nonzero coefficient is the leading coefficient of the number. $\square$

Lower bound. Let A be d -c.e. with a fixed approximation starting at 0. For input e run a machine computing the successive approximation values and emitting a change event whenever the value flips, at most d times; computing a stage may take unbounded finite time, which is harmless because the predicate below uses machine steps, not stages. Let $\chi_{i,e}(n)$ say that

machine step n is exactly the i th change event; bounded simulation makes it primitive recursive uniformly in (i, e, n) , and it is 1 for at most one n . Put

$$a_{i,e}(n) = (-1)^{i+1} \chi_{i,e}(n) \quad (1 \leq i \leq d), \quad (22.1)$$

with codes computable from e . With no changes the value is 0 and $e \notin A$. With $j \geq 1$ changes the highest occupied block is j , with sign $(-1)^{j+1}$, so the value is positive exactly when j is odd, which is exactly when the final approximation value is 1. Hence $e \in A \iff Z(a_{1,e}, \dots, a_{d,e}) > 0$. The equality statement is [theorem 20.2](#). $\square$

Corollary 22.3 (Strict order hierarchy with fixed equality complexity [08]). *One raw block has c.e.-complete positivity, two blocks d.c.e.-complete positivity, and d blocks the d th finite mind-change level; these levels are strict, although all the equality problems are Π_1^0 -complete.*

Proof. Apply [theorem 22.2](#) and the strictness of the hierarchy [8]: a complete set for level $d+1$ at level d would collapse the two levels, since the d -c.e. sets are closed under computable preimages. $\square$

The case $d = 1$ is 05's Σ_1^0 -complete strict comparison ([theorem 20.2](#)). The theorem is insensitive to the times of the change events: they only fix the offsets $-n$ in the exponents, while any term in a higher block dominates every term in a lower one. This is the mechanism converting logical mind changes into surreal dominance changes (08).

Example 22.4 (A sign obstruction under a nonzero promise [08]). For the spike Z_e put

$$W_e = U - UZ_e = U - \sum_{n \geq 0} h_e(n) \omega^{2\omega - n}.$$

Every W_e is nonzero with at most two monomials. If e does not halt, $W_e = U > 0$; if it halts, the negative term in block two dominates and $W_e < 0$. So a nonzero promise does not make the two-block sign problem decidable. For one block under a nonzero promise, scanning to the first nonzero coefficient terminates and decides the sign.

This example is `cas:prop-leading` of the computer-algebra report (the series $t^{(1,0)} - \sum_n a_n t^{(0,n)}$ with first-halt coefficients, always nonzero with one or two terms), and the one-block contrast is its `cas:rem-rankone`. [Theorem 22.2](#) strengthens that proposition from undecidability of the sign to an exact classification of positivity at every finite number of blocks.

Proposition 22.5 (Known highest occupied block [08]). *For a raw finite-block name supplied with the index of its highest nonzero block, and the promise that all higher blocks vanish, sign and leading monomial are computable. Neither this information nor the full leading monomial decides equality in general.*

Proof. Scan the specified block to its first nonzero coefficient, which gives the leading monomial and the sign; the promise makes this correct and terminating. The equality counterexample is [corollary 20.3](#). $\square$

An ordinary halting oracle $0'$ decides whether each of the finitely many blocks is identically zero, finds the highest nonzero block and then applies [proposition 22.5](#); this gives a uniform $0'$ upper bound for order over the union of all finite-block syntaxes. The one-block equality lower bound shows that this oracle strength is necessary for a general equality decision, and no stronger oracle is concealed in the finite-level results (08).

23 Support validity is a separate coanalytic problem

The raw block construction made validity automatic. To expose the separate cost of recognizing a valid support it suffices to use dyadic exponents in a fixed bounded interval and coefficient one everywhere, with no large ordinal labels, exotic real constants or varying exponent algebra (08, 05).

23.1 Trees and the Kleene–Brouwer order

A tree $T \subseteq \mathbb{N}^{<\mathbb{N}}$ is closed under prefixes (and contains the empty sequence when nonempty). Its Kleene–Brouwer order puts $s <_{\text{KB}} u$ when u is a proper prefix of s , or, at the first differing coordinate, s has the smaller entry; it is a computable linear order on $\mathbb{N}^{<\mathbb{N}}$.

Lemma 23.1 (Tree well-foundedness and its linearization [08; 05, 09]). *A tree $T \subseteq \mathbb{N}^{<\mathbb{N}}$ has no infinite branch iff $(T, <_{\text{KB}})$ is a well-order.*

Proof. An infinite branch gives the strictly $<_{\text{KB}}$ -descending chain of its longer and longer prefixes. Conversely, if T has no infinite branch, proper extension is well founded on T ; induct on it. The subtree at s is, in the Kleene–Brouwer order, the ordinal sum in increasing n of the subtrees at the children $s \frown n$, followed by the singleton s ; a countable ordered sum of well-orders followed by one point is a well-order, which applies at the root. No finite-branching assumption is used. (09 argues the converse directly: a descending chain yields a path by successively stabilizing its natural coordinates, since there is no infinite strict decrease of a natural coordinate.) $\square$

Well-foundedness of recursive, and of primitive-recursive, trees is Π_1^1 -complete: this standard fact is recorded in [4, Theorem 2.6(g)] (08), in [17, Theorems 5.2.10 and 7.1.4] (05) and in the tree normal forms discussed in [7, Section 2] (09). Tree predicates can be made primitive recursive by including bounded computation witnesses in the nodes; this is the usual effective tree normal form (09). Treehood is enforced syntactically by taking as T the set of nodes all of whose prefixes pass a primitive-recursive test (08, 09). Π_1^1 is not Π_1^0 .

23.2 An explicit dyadic coding with decidable image

Associate an open interval (a_s, b_s) to each $s \in \mathbb{N}^{<\mathbb{N}}$: start with $(a_\emptyset, b_\emptyset) = (0, 1)$ and, with $L = b_s - a_s$, put

$$a_{s \frown n} = a_s + L(1 - 2^{-n}), \quad b_{s \frown n} = a_s + L(1 - 2^{-(n+1)}). \quad (23.1)$$

Let $\iota(s) = b_s$ and

$$\kappa(s) = 2 - \iota(s). \quad (23.2)$$

For example $\iota(\emptyset) = 1$, $\iota(0) = \frac{1}{2}$, $\iota(1) = \frac{3}{4}$, $\iota(0, 0) = \frac{1}{4}$, $\iota(0, 1) = \frac{3}{8}$.

Lemma 23.2 (Dyadic coding [05]). *ι is a computable bijection from $\mathbb{N}^{<\mathbb{N}}$ onto the dyadic rationals in $(0, 1]$ which preserves the Kleene–Brouwer order and has a computable inverse. Hence κ is an order-reversing computable bijection onto the dyadic rationals in $[1, 2)$ with computable inverse.*

Proof. Every child interval lies strictly left of its parent's right endpoint; consecutive children occur from left to right with disjoint interiors, and all descendants stay in their child interval. So extensions precede prefixes and the first differing child index decides the order, which gives

order preservation and injectivity. For the inverse, write a dyadic $q \in (0, 1)$ in lowest terms $q = c/2^\ell$ with c odd, write $c - 1$ as a binary word of exactly ℓ bits with leading zeros (it ends in zero), and parse it uniquely into blocks $1^{n_0}01^{n_1}0 \cdots 1^{n_j}0$; the sequence is $(n_0, \dots, n_j)$, because each child choice appends exactly the indicated block to the binary address of the interval. The endpoint $q = 1$ is the empty sequence. This gives surjectivity and a computable inverse, and subtraction from 2 reverses the order. $\square$

For a decidable tree T put

$$\kappa(T) = \{\kappa(s) : s \in T\}, \quad \Sigma(T) = \sum_{s \in T} \omega^{\kappa(s)}. \quad (23.3)$$

This is a *candidate* notation: it is assigned a surreal value only when the displayed support is admissible. An invalid candidate is not given a default value such as zero, which would silently change the representation and its decision problem (05).

Theorem 23.3 (Bounded dyadic support validity [05]). *There is a uniformly effective family of candidates (23.3) in which every support is a decidable subset of the dyadic rationals in $[1, 2)$, every listed coefficient is one, and*

$$\Sigma(T) \text{ is an admissible omnific normal form} \iff T \text{ is well founded.}$$

Consequently admissibility of these candidates is Π_1^1 -complete; every admissible candidate denotes an omnific integer.

Proof. Given a rational, test whether it is dyadic and in $[1, 2)$; if so, apply the inverse of lemma 23.2 to 2 minus it and test the resulting sequence for membership in T . This decides the support, uniformly in a total tree predicate. The decreasing order on $\kappa(T)$ is isomorphic to $(T, <_{\text{KB}})$, so admissibility is equivalent to well-foundedness by lemma 23.1. Admissible sums have positive exponents and constant zero, so they are omnific. This reduces well-founded recursive trees to admissibility; conversely the absence of an infinite increasing sequence of support exponents is a Π_1^1 statement, and the tree completeness theorem gives the classification. $\square$

No numerical estimate repairs this validity problem: all exponents lie between one and two and all coefficients are one. The compactness of a real interval has no bearing on whether a subset is well ordered; the relevant order is that of the support, not a metric estimate for a real series (05).

Remark 23.4 (Second routes: midpoint codings [08, 09]). 08 and 09 prove the same completeness with a different coding. Enumerate $\mathbb{N}^{<\mathbb{N}}$ without repetition; at stage m find the position of the new sequence among the finitely many already placed in the *reverse* Kleene–Brouwer order, and assign the midpoint $\mu(s) = (l + u)/2$ of the values $l < u$ of its placed neighbours, using 1 and 2 when a neighbour is absent. Induction on stages shows that $\mu : \mathbb{N}^{<\mathbb{N}} \rightarrow \mathbb{Q} \cap (1, 2)$ is a computable injection into the dyadics with $s <_{\text{KB}} u \iff \mu(s) > \mu(u)$ (08’s lemma); to compute $\mu(s)$ run the construction to the stage at which s appears. 09 applies the same midpoint construction to any computable countable linear order, on all finite sequences first and then restricted to T , so that it is uniform and unaffected by the eventual well-foundedness of T . With $S_T = \sum_{s \in T} \omega^{\mu(s)}$, 08 proves that validity is Π_1^1 -complete with coefficient one and exponents in the fixed dyadic set $\mu(\mathbb{N}^{<\mathbb{N}}) \subset (1, 2)$, and 09 that the property “this is a valid Conway normal form” of a program notation for sums with coefficient one and distinct rational exponents in $(1, 2)$ is Π_1^1 -complete. Neither source shows that the image of μ , or the support of S_T , is decidable, and neither claims it; decidable supports are obtained here only with 05’s coding κ .

Proposition 23.5 (Promised versus checked equality [08]). *On admissible tree-support names,*

$$\Sigma(T) = \Sigma(T') \iff T = T',$$

so under the validity promise inequality is c.e. and equality is co-c.e. The checked equality relation, requiring validity of both inputs and equality, is Π_1^1 -complete. The same holds for 08's names S_T .

Proof. The coefficient at the distinct exponent $\kappa(s)$ (or $\mu(s)$) is the indicator of $s \in T$, so normal-form uniqueness gives the equivalence, and a finite node in exactly one of the trees witnesses inequality. Checked equality is the conjunction of two Π_1^1 validity conditions and the arithmetical condition $T = T'$, hence Π_1^1 ; on the diagonal (T, T) it is exactly validity, which gives hardness. $\square$

08's statement reads "equality under the validity promise has a co-c.e. test for inequality"; its proof, and the summary table of its introduction ("co-c.e. on valid inputs"), show that the intended and correct statement is the one printed here: inequality is c.e., so promised equality is co-c.e. (subsection A.5).

Corollary 23.6 (No complete enumerable certificate system [08]). *No sound computably enumerable family of finite certificates recognizes exactly the valid tree-support names.*

Proof. An accepted finite certificate would make validity c.e., contrary to its Π_1^1 -completeness. $\square$

23.3 What these theorems classify

The theorems classify *validity*, not equality under a validity promise: once a candidate is promised valid, its equation with itself is trivial, and Π_1^1 -hardness of promised equality cannot be inferred from the difficulty of checking the promise; the fixed-support equality theorem was proved by a separate reduction to avoid exactly this mistake (05). The statement concerns the validity of this flexible support language, not equality in the fixed-stream language, and certainly not validity of finite rational-omega terms, which is decidable; conflating these domains turns correct statements into false ones (09). Nor does it undermine the certified syntaxes of the positive parts, which certify deliberately restricted, total families of valid supports and do not purport to accept every semantically valid program-generated support (08).

The computer-algebra report proves by an arithmetical reduction that Hahn admissibility of supports output by total bounded-time tests is undecidable (**cas:prop-validation**). Theorem 23.3 and proposition 23.5 strengthen this to an exact Π_1^1 classification with bounded dyadic exponents, and for checked equality. The computable-surreals report proves Π_1^1 -completeness of *structural* validity for its cut-program coding (**prop:validity**) by the same tree reduction in a different representation.

Implementations, relations, questions, and records

24 Implementation specifications and the shipped programs

24.1 Interfaces

A practical implementation should keep separate interfaces (05). A *hereditary normal form* carries finite exact coefficient and exponent data. A *certified compressed form* carries finite fraction, holonomic or matrix data together with its denominator, scale, jet and shielding contracts. A *stream form* supplies coefficient access on a stated support but need not admit an equality decision. These are not names for one unchecked data structure: an operation converting a stream into a compressed form is partial and certificate-producing, and it cannot be a total hidden coercion by [theorems 20.4, 21.1 and 21.3](#).

For the towers (08), an exact coefficient backend supplies k with decidable validity, equality, order and arithmetic; a scale context stores a sorted finite list of canonical ordinal labels and represents each lattice exponent by its integer coordinate vector; and a recursive field name is either a k -name or a fraction of certified Euler-series names whose operator and jet coefficients are field names in the preceding context. An omnific name may use the triangular form (7.3) directly: an integer constant, positive outer powers, and arbitrary valid field names one level lower as coefficients, which makes omnific membership automatic. An importer of an unrestricted field name must instead run the recognition algorithm of [theorem 7.2](#); a nonzero field name provides an exact Laurent valuation, zero being handled first.

Listing 2: Field-to-integer conversion in the towers (08).

```
OmnificNormalize(x, level):
  if level == 0:
    m := ordinary_floor(x)
    return Integer(m) if x == m else Reject
  if x == 0:
    return Zero
  N := LaurentValuation(x, top_variable)
  prefix := sum(Coefficient(x,n) * top_variable^n
               for n from N through 0)
  # The sum is empty when N > 0.
  if x != prefix:
    return Reject
  lower := OmnificNormalize(Coefficient(x,0), level-1)
  if lower is Reject:
    return Reject
  return Triangular(lower, all nonzero coefficients at N <= n < 0)
```

Every equality sign invokes the certified field equality test; replacing it by a numerical tolerance or by comparison of a fixed number of coefficients would invalidate the proof (08).

Listing 3: Hereditary finite normalization (05).

```
normalize(expression):
  recursively normalize the operands
  for addition: merge equal exponents and add coefficients
  for multiplication:
    for each pair of terms:
      add their exponents recursively
```

```

    multiply their coefficients
    merge the resulting finite list
    discard zero coefficients
    absorb exponent zero into the constant
    sort exponents by recursive exact comparison
    return the canonical tree

```

Listing 4: Equality in the ε_0 -shielded calculus (05).

```

equal(A, B):
    validate the hereditary integer parts
    validate every fraction denominator as nonzero
    subtract A and B as finite polynomials in epsilon_0
    for each coefficient p/q - r/s:
        if normalize(p*s - r*q) is nonzero:
            return False
    return True

```

Listing 5: Collision-aware leading term of a matrix series (05).

```

front(u, M[1..r], v, delta[1..r]):
    require every delta[i] > 0
    P = u * product(adjugate(I - z[i]*M[i])) * v
    weights = empty exact map
    for each monomial c*z^n of P:
        weights[sum(n[i]*delta[i])] += c
    remove zero coefficients from weights
    if weights is empty:
        return Zero
    lambda = least key in weights
    return LeadingTerm(lambda, weights[lambda])

```

The last algorithm is finite even if the exponent interval below its answer contains infinitely many potential support points. For the omnific value $X\Phi$ it returns the omega-leading exponent $\Lambda - v(\Phi)$ and the same coefficient; for $X^j\Phi$, use $j\Lambda - v(\Phi)$ (05).

24.2 Finite certificates

At one level of a tower, a zero answer for a certified series consists of its normalized operator, the verified exceptional-index computation and the compatible zero jet; a nonzero answer identifies the first nonzero jet coefficient, whose nonzeroness is certified recursively; fractions use a certified numerator cross-difference; exact rational or algebraic arithmetic discharges the claims at the bottom. The integer-root procedure exposes a certificate—the leading polynomial p , an integer Cauchy bound M and the exact tests of $P(n)$ for $0 \leq n \leq M$ —which an independent checker can verify without trusting a black-box root solver; a faster implementation may factor p first. These are mathematical specifications: the prototype uses exact arithmetic but does not emit a proof-assistant object or a complete independent proof trace (08). For the hereditary layer, equality certificates are normalized finite trees with recursive checks of exponent order and coefficient identities; for $\text{Frac}(\mathcal{H})$, the identity $ps - rq = 0$ and the two nonzero denominators; for a shielded polynomial, the finite coefficient lists; for a matrix series, the adjugate identities, the collected numerator and the unit leading term of the denominator. For a general stream, inequality has a finite first-difference witness; equality may sometimes have a finite special-purpose proof, such as a recurrence identity, but no sound c.e. collection of such proofs covers every true equation of the switch family ([corollary 21.4](#)), which limits completeness, not usefulness (05).

24.3 Complexity and resource limits

No uniform polynomial-time bound is claimed for any of the three systems. Differential orders grow under tensor products and coefficient expressions under repeated field operations; a singularity M written in binary forces a jet of length $M + 1$ in the elementary certificate format ([example 5.3](#)). A production system could use sparse compatible jets with a proved completion procedure, shared subexpressions, cached zero tests and specialized annihilator algorithms, each with its own correctness argument (08). The hereditary normalizer and the grid compiler allow substantial expression swell (05, 09). A resource-limited implementation should return a distinct “not completed” status when a limit is exceeded; that is not a proof of inequality, invalidity or nonexistence, and failing to import a raw stream into the holonomic backend says nothing about whether its value has a holonomic or even rational name ([theorem 20.4](#)) (08).

24.4 The shipped programs

Delivered code is kept byte-identical under its placed names; the README of this directory explains how to rerun it on a copy.

Source 08. `code/08-holonomic-towers-holonomic.py` implements certified Euler series over $\mathbb{Q}$, sum and product annihilators by finite differential-system linear algebra, their fraction field, and the guarded ring $\mathbb{Z} + UF_1[U] \subset R_2$; `code/08-holonomic-towers-ordinals.py` implements hereditary Cantor-normal-form trees below ε_0 with comparison, natural and ordinary addition, natural multiplication and omega exponentiation; `code/08-holonomic-towers-verify.py` runs the tests and `code/08-holonomic-towers-demo.py` is a short example. Its scope is strictly smaller than the tower theorem: arbitrary coefficient backends, general higher-rank operators, the full R_2 recognition and floor algorithms, and all ε_0 -labelled embeddings are mathematical constructions, not claimed implementations. The recorded run (`data/08-holonomic-towers-verification.json`, Python 3.13.5, SymPy 1.14.0) passed 41 test groups, including 521 finite change-event schedules, 780 arbitrary-sign block assignments, and all 14,520 ordered pairs among 121 dyadic-coded tree nodes; the tests exercise recurrence indices, exceptional jets, series identities through certificates, Laurent leading data and finite instances of the dominance encodings. Tests of infinite-series identities use generated certificates and the finite-jet rule, not matching a fixed number of terms.

Source 05. `code/05-hereditary-shielding-verify.py` (Python 3.9 or later, standard library only, exact rational arithmetic) checks hereditary arithmetic and the floor correction, the dyadic coding and its order preservation, matrix numerator identities and collision examples, and Hankel ranks of switch sequences. The recorded run (`data/05-hereditary-shielding-verification.json`) passed 18,109 assertions in 16 categories, the largest being 14,641 Kleene–Brouwer comparisons; these are not 18,109 independent theorems.

Source 09. `code/09-rational-omega-omega_notation.py` is an exact prototype: its sparse-fraction kernel uses Python integers and `fractions.Fraction`; SymPy is used only after grid compilation, for rational-function algebra and polynomial division; floating-point input is rejected. `om(x)` is $\Omega(x)$; `equivalent` and `compare` implement [theorem 14.2](#), leading the leading data, `separated_basis` and `grid_form` [lemma 15.1](#) and [theorem 15.2](#), and `nonnegative_part`, `is_omnific`, `omnific_floor`, `truncate_at` and `coefficient_at` [theorem 16.3](#). Semantic equality is `equivalent`, not Python equality. For example, with `w = om(1)` and `B = om(w) / (1 - 1/w)`, the prototype confirms that `B` is omnific, that `B + 1/2`

is not, that $\lfloor B - \omega^{-1} \rfloor_{\mathbf{Oz}} = B - 1$, that $\lfloor \omega^{-3} \rfloor B = 1$ and that $\tau_{\geq \omega^{-3/2}} B = \omega^\omega + \omega^{\omega^{-1}}$. The recorded run (`data/09-rational-omega-verification.json`, Python 3.13.5, SymPy 1.14.0, seed 23092026) passed 197 exact assertions, covering nested monomial identities, a positive infinitesimal exponent, infinite geometric omnific values, both floor-boundary cases, absent coefficients, off-lattice cutoffs, a tail-sensitive separated basis, polynomial-part fixtures in two and three scales, and rejection of zero division and floating-point input; several expected truncations are supplied as independent fixtures.

Reruns for this report. All three suites were rerun on copies with the delivery file names restored, under Python 3.14.4 and SymPy 1.14.0, and passed with the same counts (41 groups; 18,109 assertions; 197 assertions). No test computes a halting set, verifies a Π_1^1 reduction, enumerates the support of $\mathbf{E}$, or proves a completeness, support or termination theorem.

24.5 Formalization targets

No part of this report is formalized, and `docs/FORMALIZATION.md` maps none of its labels. The sources propose the same layering. A manageable first target is the Euler recurrence theorem over an abstract effective field, then the integer-root lemma and the triangular theorem over iterated Laurent fields, with the bridge to actual surreal monomials kept as a separate semantic layer (08). 05 suggests finite hereditary trees with rational coefficients first, then the shielding theorem over an explicitly embedded set-sized Hahn field, then the adjudicate certificate theorem, which needs only finite algebra and a positive-scale evaluation interface, and finally the computability reductions in a fixed machine-index framework. 09 suggests an inductive syntax with finite sparse fractions and explicit depth measures, a denotation into the surreal carrier or a fixed Hahn workspace, and proof obligations for the collection of equal exponents, the fraction leading-data formula, leading-scale elimination, the ordered embedding of each lattice, and the polynomial-part identities, with the negative-infinitesimal floor correction as a distinct branch. A type-level assertion in Lean that equality is decidable is not by itself an executable verified normalizer; classical instances must be distinguished from the algorithms, and a classical formalized isomorphism, especially one using choice, does not implement a computable normalizer for finite strings (05, 09). The normal-form bridge note `docs/NORMAL_FORM_BRIDGE.md` describes the existing finite-evaluation and field-bridge modules.

25 Relations to neighbouring reports

- `docs/foundations-and-computation/computer-algebra (cas:)`. The two stream reductions are its `cas:thm-zerotest` with `cas:rem-moralA` and `cas:rem-moralB`; 08's two-block sign example is `cas:prop-leading` and the one-block contrast `cas:rem-rankone`; [theorem 22.2](#) strengthens the former, and [theorem 23.3](#) and [proposition 23.5](#) strengthen `cas:prop-validation`. The floor is `cas:eq-floor`. After compilation the rational-omega grids are its core `cas:thm-core`; its positive-grid extraction `cas:thm-grid` concerns coefficient extraction with collisions, to which 05's collision-aware front adds a finite zero and leading-term certificate for matrix coefficient rules. Its section on positive counterweights says that singular points require a certified recurrence start and treatment of exceptional indices; [theorems 4.3](#) and [5.2](#) supply exactly that, over non-Archimedean coefficient fields. Its remark on the 2026 zero test (`cas:rem-cfh`) applies equally to the towers.
- `docs/foundations-and-computation/computable-surreals` (bare labels). The one-coefficient obstruction uses the constants of `thm:inversehard(iv)`; its structural validity

theorem `prop:validity` is the tree reduction in another representation; its no-go theorem `thm:nogo` lists a more informative real representation among the ways out, the route taken by all three systems here. Its research directions “support certificates beyond left-finiteness” and “uniform translation complexity” are *not* answered: the transfinite supports here arise only inside finite-rank lattices or towers, which is the excluded case, with exact rather than Cauchy coefficients ([theorem 18.4](#) and the questions below only point beyond it), and the translator theorems concern stream-to-finite names, not numerical-to-structural names.

- `docs/surreal/omnific-diophantine-geometry` (`odg:`). The floor is `odg:thm:floor`, division with remainder `odg:prop:division`, the Diophantine transfer `odg:cor:H10` with `odg:eq:existencetransfer`; all are cited, with only their effective forms proved here. Its question `odg:q:roots` on root decidability for finitely represented omnific coefficients is *not* addressed: the systems here decide ground equality, order and floor, not the existence of roots; nor is `odg:q:search`. The report is being extended concurrently from the same batch; only labels present at the time of writing are cited.
- `docs/surcomplex/holonomic-rigidity-for-entire-hahn-functions` (`hol:`). [Theorem 4.3](#) is the effective form of its `hol:lem:integer` ([remark 4.4](#)). Its question `hol:nl:q:effective`, on computing candidate degree sets and optimal excluded valuation cuts for nonlinear rigidity certificates, is *not* answered; the effective leading-data fields of [Part I](#) are relevant input only.
- `docs/foundations-and-computation/foundations` (`found:`): the definition $\mathbf{Oz} = \Pi \oplus \mathbb{Z}$ (`found:eq:omnific`) agrees with [definition 2.1](#).
- `docs/foundations-and-computation/definable-surreals-and-omnific-integers`, written concurrently from the same batch, restates the same floor; its subject is set-theoretic definability, not computability, and it explicitly declines to identify the two. It is cited by directory only.

26 Questions

The sources call these directions suggested by their results, not completed results or previously published open problems (08, 05, 09).

Question 26.1 (Compact certificates [08, 05]). Can the full jet through the last exceptional index be replaced by a compressed certificate listing the free singular coefficients with a small independently checkable proof that the coefficients between them follow from the recurrence (08)? Can the adjugate numerator be computed or certified by reachability and observability reduction, structured polynomial identity testing or sparse group-algebra arithmetic with substantially smaller certificates, and what is the bit complexity when the number of scales is part of the input (05)?

Status: open. Univariate polynomial-time weighted-automata algorithms should not be quoted as a bound for every multiscale representation (05).

Question 26.2 (Larger certified coefficient classes [08, 05, 09]). Can the D-finite numerator and denominator class be replaced by richer uniquely specified differential-algebraic classes, with a precise common field, leading-data access, truncation closure and compatible parameter semantics (08)? Can algebraic extensions of $\text{Frac}(\mathcal{H})$ with explicit branch and order data serve as coefficients for the shielding theorem (05)? For algebraic real constants in rational-omega expressions, is there a block version of leading-scale elimination over effective number fields that computes the finite subgroup order and truncation cuts uniformly (09)?

Status: open. The D-algebraic zero test [5] gives tools but the obligations must be verified separately rather than inferred from the phrase “zero-test” (08); the separated rational basis fails with irrational coefficients (lemma 15.1).

Question 26.3 (Beyond finite rank [05, 09]). Is there a transfinite iteration constructor, or a certified ω -summation operation with syntactic support certificates and algebraic restrictions on block coefficients, that extends the one-state supports ω^r or the escape sum $\mathbf{E}$ while keeping equality decidable?

Status: open. Proposition 17.4 and theorem 23.3 show that no sound effective universal constructor covers all recursive support orders, and theorem 20.4 excludes all primitive-recursive coefficient programs; a specific stronger language needs its own positive theorem (05, 09). This is also the direction “support certificates beyond left-finiteness” of the computable-surreals report.

Question 26.4 (Beyond finitely many dominance priorities [08]). Can higher effective difference hierarchies be assigned to appropriately well-founded systems of dominance priorities?

Status: open. The finite proof uses a computable uniform bound on the number of record priorities; an ordinal-indexed version must specify a validity certificate and an effective descending control measure, and the finite classification is not asserted to extend automatically (08).

Question 26.5 (Intrinsic versus presentation-based invariants [08]). Which representation morphisms preserve leading-term certificates or exact support counts, and which natural subclasses of raw names admit effective translations?

Status: open. Theorem 20.4 shows that an extensional inclusion of denotation sets is much too weak to answer this (08).

Question 26.6 (Richer exponent languages and proof-theoretic strength [05]). Which additional decision assumptions are needed for exponent languages admitting transcendental real constants, global exponential identities or cut-defined exponents? What proof-theoretic strength is needed to establish termination of normalizers for richer ordinal-indexed constructors?

Status: open. Any extension must state its decision assumptions rather than subsume them under a generic “simplify” operation; the extensional question of which surreals are denoted stays distinct (05).

Question 26.7 (Size, sharing, and mechanization [09]). What complexity bounds hold for the recursive comparator and the grid compiler in terms of nesting depth, the number of distinct exponent subterms, grid rank, degrees and coefficient bit lengths? Can the ring $\mathcal{I}$, whose ground word problem is decidable and whose existential polynomial solvability is not, serve as a testbed for mechanizing omnific division?

Status: open. Finite test timings do not justify an asymptotic claim (09).

Question 26.8 (Comparison of the three systems [merge]). For $k = \mathbb{Q}$, is the tower ring $\mathcal{R}_\varepsilon$ contained in the rational-omega ring $\mathcal{I}$? In particular, does $\sum_{n \geq 0} n! \omega^{\omega^{-n}}$ have a rational-omega term, or lie in $\text{Frac}(\mathcal{H})$?

Status: open; no source addresses it (proposition 17.3).

None of these is a question named by an earlier report, and no question of an earlier report is answered here: not the research directions of the computable-surreals report, not `odg:q:roots` or `odg:q:search`, and not `hol:nl:q:effective`.

A Provenance

A.1 Section by section

Table 1: Provenance of each part of the report.

Part of this report	Sources and choices
Section 1	08 §1 with 05 §1 and 09 §1; the summary table combines 08's and 05's tables.
Section 2	08 §§2–3, 05 §2, 09 §2, printed once. The floor formula is cited from <code>odg:thm:floor</code> and <code>cas:eq-floor</code> ; 05's Theorem 3.2, 08's Theorem 7.4 and 09's equation (8) prove it again. Canonicalization: 08's Proposition 2.3, 05 §2.2, 09's Corollary 3.3. Size obstruction: all three.
Section 3	05 §4 (Theorems 4.1, 4.2), printed as the general embedding; remark 3.2 is the merge's.
Part I	08 §§4–10. Remark 4.4 (credit to <code>hol:lem:integer</code>) is the merge's; the floor theorem keeps only its effectivity.
Part II	05 §§3, 5, 6. The floor theorem keeps only its effectivity; the Schützenberger attribution is added.
Sections 14 to 16	09 §§3–5; the division corollary cites <code>odg:prop:division</code> for existence and uniqueness.
Section 17	09 §6 as the main statement; proposition 17.3 is the merge's; the ω_1^{CK} proposition is 09's Proposition 6.3 with 05's Proposition 10.1 as its uniform half.
Sections 18 and 19	09 §§7, 9; the Diophantine transfer cites <code>odg:cor:H10</code> .
Section 20	08 §11 (base; d blocks) with 05 §7 and 09 §§8.1–8.3 as the case $d = 1$; strict comparison from 05's Theorem 7.1, non-semidecidability from 09's Theorem 8.2; the no-translator theorem is 08's Theorem 11.4 and 09's Theorem 8.3 printed once; credit to <code>cas:thm-zerotest</code> .
Section 21	05 §8; credit to <code>cas:rem-moralA</code> .
Section 22	08 §12; credit to <code>cas:prop-leading</code> , <code>cas:rem-rankone</code> .
Section 23	05 §9 (coding and theorem, main route); 08 §13 (Kleene–Brouwer lemma, second route, promised and checked equality, corrected) and 09's Theorem 8.5 (second route).
Section 24	08 §14, 05 §11, 09 §10.
Sections 25 and 26	The merge, with the sources' directions (08 §16, 05 §12, 09 §11.3).

Section, theorem and equation numbers of the sources refer to their own manuscripts, which are not shipped. Where the sources overlap, the statement with the weakest hypotheses is printed. No source proof needed a hypothesis that the printed statement lacks: the sum-of-squares step of the transfer theorem needs an ordered coefficient field, which the Hahn setting supplies; 08's floor proof uses only [corollary 7.3](#) and the rank-zero integer search; the stream reductions use only $\mathbb{Q}$ -valued primitive-recursive coefficients.

A.2 Dependency map

For the towers the dependencies run as follows (08): exact ordered real coefficients; leading-polynomial integer-root bounds; finite Euler certificates; effective D-finite ring operations; fraction fields and leading data; finite-rank recursion; truncation and triangular integer parts; floor and ordinal recognition; effective insertion of finitely many labels. The next rank is never used to justify the preceding rank's zero test, and the root procedure needs only real leading coefficients. For rational-omega expressions (09): finite recursive equality; separated grid; polynomial-part recursion; omnific recognition and floor; the support barrier follows from finite lattice rank, not from the floor theorem. For 05: hereditary normalization and ordinal bounds; shielding; fraction supports and the ε_0 calculus; matrix adjugates and positive-scale evaluation. The negative results are independent of all three constructions: they use normal-form uniqueness, fixed positive support blocks, bounded machine simulation, and, only for validity, the completeness of well-founded recursive trees. The translator theorems join the tracks by observing that every spike value has a rational monomial name.

A.3 Renamed symbols

Table 2: Symbols renamed from the sources.

Source	Source symbol	Here, and why
08	$\ell(x)$ (leading exponent)	$\deg_\omega x$, the collection's symbol; 09's ℓ is support length.
08	C (base field)	k ; 05's $\mathcal{C}$ and 09's C are other objects.
08	D (code set)	N ; D is the constant ring of the shielding theorem.
08	$N(a)$ (canonicalization)	$\text{can}(a)$; N is the code set.
08	E^{chk}	$\text{Eq}_\nu^{\text{chk}}$.
08	L (Euler operator)	$\mathbb{L}$; L is the shielded coefficient field.
08	$\mathcal{A}(K, t)$ (modular class)	$\mathfrak{C}(K, t)$.
08	$\text{Tr}_{\geq g}, \text{Tr}_{>g}$	$\tau_{\geq g}, \tau_{>g}$, shared with 09.
08	$G_r = \prod (1 - t_j)^{-1}, G(t)$	$\text{Geo}_r, \text{Geo}(t)$; G_r is the exponent group.
08	$A(t), E(t), E_-(t)$	$\text{Fac}(t), \text{Ex}(t), \text{Ex}(-t)$.
08	B in $A \subseteq B$ (label sets)	A' ; $B(P)$ is the jet bound.
08	S_e, q (tree sums, midpoint map)	S_T, μ ; $\Sigma(T)$ and κ are 05's.
08	k (block index), ℓ (product index)	j, l ; k is the coefficient field.
05	K (coefficient field), $\mathbb{A}$	$k, \mathbb{R}_{\text{alg}}$.
05	$E (= \varepsilon_0)$	ε_0 ; Ex is the exponential series.
05	$\mathcal{F} = \text{Frac } \mathcal{H}, \mathcal{C}$	$\text{Frac}(\mathcal{H}), \mathcal{S}_{\varepsilon_0} = \mathcal{S}_{\varepsilon_0}(\mathcal{D}, \text{Frac}(\mathcal{H}))$; $\mathcal{F}_A$ and $\mathcal{Q}$ are the other fields.
05	F, D in the shielding theorem	L, D ; Eq_F is Eq_L .
05	$C(p)$ (canonicalizer)	$\text{can}(p)$.
05	$P + c + \eta$ (floor decomposition)	$x_{>0} + \text{ct}(x) + x_{<0}$.

Source	Source symbol	Here, and why
05	$F, \widehat{F}, P, Q$ (matrix series)	$\Phi, \widehat{\Phi}, P, Q$.
05	d (matrix dimension), $B = d + e$	$m, b = m + m'$; d is the number of blocks.
05	λ (leading Hahn exponent)	$v(\Phi)$, a valuation in the t -convention.
05	N (nilpotent matrix)	N_0 ; N is the code set.
05	Z_r	ζ_r ; Z denotes raw block names.
05	$X = \omega^\omega$ (streams), $S(a)$	$U, Z(a)$, as in 08.
05	h_e (halted within n steps)	$\bar{h}_e$; h_e is the first-halt spike of 08 and 09.
05	A_e, B (switch family)	A_e^{sw}, B .
05	$q, e(s), (\ell_s, r_s), Z(T), S_T$	$\iota, \kappa(s), (a_s, b_s), \Sigma(T), \kappa(T)$; e is a machine index.
09	$\mathcal{F}, \mathcal{F}_d, \mathcal{A}$	$\mathcal{Q}, \mathcal{Q}_d, \mathcal{I}$; $\mathcal{F}_A$ is 08's field.
09	$\lambda(x), \ell(x)$	$\deg_\omega x, \text{ord}(\text{supp } x)$.
09	$K_r, A_r, A_{r-1}^{\text{tail}}$	$\mathcal{K}_r, \mathcal{I}_r, \mathcal{I}_{r-1}^{\text{tail}}$; K is 08's leading-data field.
09	$T_r(R), C_r(R)$	$\tau_{\geq 0}R, \text{ct}(R)$.
09	D (code set; $\sum \omega^{1/n}$)	N ; H .
09	B, B_d, C_d, C	$B, B_d^\#, E_d, E$.
09	D_i (jet degree bounds), G (support group)	b_i, Γ .
09	t (terms), $S(a), Z_i$	$s, Z(a), Y_i$.

A.4 Stale repository statements

All three sources pin `a45d722`, 30 commits before the placement. At the pin the computer-algebra, computable-surreals, omnific Diophantine and holonomic-rigidity reports already contained every label cited here. The sources' reviews were targeted, and the following statements are corrected; the pins are kept as provenance.

- 08 read the computer-algebra README (lines 1–180) and describes the report correctly as containing effective rational monomial cores and zero-test and high-rank leading-term obstructions. Its novelty ledger then presents the localization of raw equality to at most one monomial as part of its contribution, and prints the two-block sign example; both were at the pin `cas:rem-moralB` and `cas:prop-leading`, and are credited here ([sections 20](#) and [22](#)).
- 05 read the omnific Diophantine report only through its line 150 and says that it “uses normal-form constant terms to organize the ring”. That is correct, but at the pin the report already contained the floor, division and Hilbert’s-tenth-problem results (`odg:thm:floor`, `odg:prop:division`, `odg:cor:H10`), which 05’s and 09’s corresponding results duplicate without citing.
- 09 writes that “similar constant-term arguments already belong to the repository’s omnific arithmetic program”, citing the catalogue; the precise reference is `odg:cor:H10` with `odg:eq:existencetransfer`. 09’s description of the computable-surreals report as “the full 51-page report” still matches that report’s README.

4. Neither 05 nor 09 cites the computer-algebra report, whose Reductions A and B anticipate their stream reductions, and no source cites `hol:lem:integer`; the credits are added.
5. At the pin the computer-algebra report said that an omnific-integer convention “is a different target again” from its $\Pi \oplus \mathbb{Z}$ floor. This was corrected in commit `1e3f14b` before the placement: the report now identifies $\Pi \oplus \mathbb{Z}$ with **Oz** (`found:eq:omnific`).
6. All three sources say that the repository has no Lean formalization of omnific integers; this remains true, and `docs/FORMALIZATION.md` maps no label of this report.
7. 08 cites the Chen–Fang–van der Hoeven zero test as a 2026 arXiv preprint; the computer-algebra report records its publication in the ISSAC 2026 proceedings [5].

A.5 Corrections

1. 08’s Proposition 13.4 stated that “equality under the validity promise has a co-c.e. test for inequality”. Its proof (a finite node in exactly one tree witnesses inequality) and its own summary table (“co-c.e. on valid inputs”) show that inequality is c.e. and promised equality co-c.e.; the statement is corrected ([proposition 23.5](#)). Wording only.
2. 05 attributes its matrix zero test to weighted-automata expositions only; the underlying rational-series theory is Schützenberger’s, which is now named ([section 13](#)).

No theorem of the three sources was found to be false. The placement dossier re-read every proof and independently rechecked 30 finite instances (among them 05’s collision example, the Hankel ranks $s + 1$ for $s \leq 12$, the dyadic coding on all sequences of length at most three over $\{0, \dots, 3\}$, 08’s certificates and integer-root example, and 09’s division example).

B What this report does not claim

Every limitation, disclaimer and priority caveat stated by a source is kept here, numbered per source, followed by those of the merge.

B.1 Source 08 (27 items)

- 08.1 The proofs are mathematical arguments, not Lean formalizations; no Lean file is supplied, no theorem is asserted to have passed the repository’s axiom audit, and classical normal-form semantics is used directly, not a claim that every semantic bridge in the repository is formalized.
- 08.2 Several combined results are proposed as original formulations; the literature search is not a proof of priority and supports only “to our knowledge”; it cannot certify absence from published literature, unpublished work or uninspected repository companions.
- 08.3 Conway normal forms, the omnific support criterion and Hahn arithmetic are classical; sparse exact monomial algebra and representation-sensitive obstructions are classical in substance and already in the repository.
- 08.4 D-finite ring closure and linear-equation zero tests are classical; the construction is not presented as the first zero test beyond rational functions, and more general transseries zero tests are acknowledged.

- 08.5 The integer-root procedure is an explicit elementary construction with standard ingredients; no broad priority claim is made for a root-bound observation.
- 08.6 Raw equality hardness is a standard halting method, localized; the Ershov hierarchy and the reduction techniques are established; the Π_1^1 result is a localized formulation, not a new discovery of the completeness of well-foundedness.
- 08.7 The differential operator at a level treats the lower field as constant; no compatibility with a global surreal derivation is assumed.
- 08.8 No algorithm is asserted for recognizing that a coefficient program is D-finite or for finding its equation, and a general routine deciding whether a raw program admits a certificate is neither supplied nor implied.
- 08.9 No analytic convergence is claimed, and no real closedness of the tower fields.
- 08.10 The modular generalization does not justify inserting arbitrary differential-equation solvers; a zero test, a solution convention and terminating coefficient algorithms remain hypotheses.
- 08.11 The triangular form is not a claim that every coefficient has a finite Conway normal form; a truncation is returned as an expression, not a finite list.
- 08.12 Ordinal recognition does not decide nonnegativity of arbitrary recurrences.
- 08.13 ε_0 is a design choice, not a ceiling for effective omnific notation; “least ordinal upper bound” is not “surreal supremum”.
- 08.14 The construction is not a notation system for every omnific integer: it needs an exact coefficient field, finitely many scales per name and a specified certificate class, and it is not closed under all analytic functions or cuts.
- 08.15 Support bounds concern rank, not birthdays; the ε_0 trace concerns the chosen label syntax.
- 08.16 The d -c.e. classification concerns zero-constant raw d -block names with primitive-recursive rational coefficients and finitely many blocks; the Π_1^1 result concerns a different raw syntax; moving any conclusion to another representation needs a new proof.
- 08.17 The count-promise algorithm does not certify the count and can diverge or be wrong on a false count.
- 08.18 No uniform complexity bound; jets can have length $M + 1$.
- 08.19 A “not completed” status is not a proof, and failing to import a raw stream says nothing about its value.
- 08.20 The prototype lacks arbitrary coefficient backends, higher-rank operators, the full R_2 recognition and floor, and all ε_0 -labelled embeddings; these are mathematical algorithms, not software capabilities.
- 08.21 The code is not formally verified, is not a hardened parser for untrusted input, and is not a security boundary (callers can mutate Python objects); it emits no proof-assistant object or complete proof trace.
- 08.22 No finite test establishes a halting reduction, a completeness theorem, a support theorem for arbitrary trees, or correctness on every input.

- 08.23 During development a test bug (reversed neighbour selection in the dyadic audit) was found and fixed; the record reports only the final run, not that no intermediate test failed; the lemma itself did not rely on the erroneous code.
- 08.24 The repository inspection was targeted (the computable-surreals article lines 1–230 and the computer-algebra README lines 1–180, with the catalogue); the repository’s formalization status is not transferred; nothing was modified.
- 08.25 The ordinal-notation overview is motivation, not a technical source.
- 08.26 The further questions are not asserted results; the finite mind-change classification is not asserted to extend to ordinal priorities.
- 08.27 The PDF checks were document-quality checks, not proof of correctness.

B.2 Source 05 (26 items)

- 05.1 An AI-assisted research manuscript, not refereed; no theorem has been independently refereed or newly formalized in Lean, and no repository-wide Lean build was run.
- 05.2 The formulations appear new after a limited search; they are contribution candidates, not a literature-wide priority claim; some narrowly quoted queries returned nothing, which is weak evidence.
- 05.3 The audit was targeted (catalogue, the computable-surreals README and article lines 1–295, the omnific Diophantine article lines 1–150); nothing is claimed absent from uninspected parts, and no repository theorem is assumed formally verified because a related module exists; no repository files were changed.
- 05.4 Hereditary normalization is a natural effective use of classical normal-form uniqueness; the one-variable matrix zero test is classical and close to weighted-automata equivalence; undecidability of zero testing for unrestricted data is classical; these are infrastructure, not new discoveries.
- 05.5 Fixed-support hardness is a standard-style reduction, not a new halting theorem; the dyadic validity result is a restricted realization of a classical obstruction; effective boundedness is an elementary specialization.
- 05.6 No decidability of factorization, of elementary theories or of general Diophantine solvability is inferred.
- 05.7 The hereditary normalizer has no polynomial bit-complexity bound.
- 05.8 Λ exists, but no uniform algorithm produces it; it is presentation data.
- 05.9 The transfer theorem is for ordered coefficient fields; over arbitrary fields there is only a truth-table reduction; the abstract existence of L does not make its raw-name equality decidable.
- 05.10 Shielding does not embed L as a subfield of $\mathbf{Oz}$.
- 05.11 $\mathcal{S}_{\varepsilon_0}$ is not closed under Ω on arbitrary inputs, roots or ordinal sums, and is not all of $\mathbf{Oz}$; fractions need not be reduced, and no multivariate gcd is hidden.
- 05.12 $\text{Frac}(\mathcal{H})$ is not real closed.

- 05.13 Not every rational multivariate series has a separated-resolvent form $(1/(1 - z_1 - z_2))$.
- 05.14 The front bound is finite, not polynomial, when r is part of the input; univariate polynomial-time automata bounds are not quoted for the multiscale case; the variable-rank bit complexity is not settled.
- 05.15 With colliding scales the tensor product need not give the Hadamard product of collected coefficients.
- 05.16 The Π_1^1 theorem classifies validity, not promised equality.
- 05.17 The boundedness proposition is not a uniform algorithm, and a bare symbol for an ordinal is not a computable copy of its well-order.
- 05.18 $\varepsilon_0 - 1$ is also an upper bound: no surreal supremum is claimed.
- 05.19 Stream-to-compressed conversion is partial and certificate-producing; a sound implementation reports “not established” rather than a false equality.
- 05.20 The finite tests are regression evidence, not proofs; no test computes the halting set or decides well-foundedness; the package does not implement arbitrary omnific arithmetic or validate arbitrary well-orders.
- 05.21 A rerun rewrites the verification record, whose checksum then changes.
- 05.22 A type-level decidability assertion in Lean is not an executable verified normalizer, and a Python test is not a substitute for a proof of semantic preservation.
- 05.23 Lurie’s paper was consulted through its abstract and record, not reanalysed; L’Innocente–Mantova is context, not an algorithm for notation equality; Arvind et al. are cited for context, not as the same algorithm.
- 05.24 The ordinal-notation overview is motivational only.
- 05.25 The extensions of its final section are not completed results.
- 05.26 Extensions to richer exponent languages must state their extra decision assumptions.

B.3 Source 09 (24 items)

- 09.1 An AI-assisted research draft, not independently refereed or formalized in Lean; no repository build was run and no repository files were changed.
- 09.2 The specific effective theorems and representation comparisons are proposed contributions after a targeted check, not certified priority claims.
- 09.3 The article concerns one language, not a universal decision claim; “computable” refers to a stated representation, and the language does not capture every reasonable computable surreal (Hamkins–Turetsky, Lurie).
- 09.4 Ω is Conway’s map, not the surreal exponential; no decision result for unrestricted exp and log is claimed.
- 09.5 Constants are exact rationals, not arbitrary computable reals; the real trace is $\mathbb{Q}$, and $\mathcal{Q}$ is not real closed.
- 09.6 $\mathcal{Q}$ is not a completion of $\mathbf{Oz}$ and not all countably or effectively described surreals.

- 09.7 The support bound is necessary, not sufficient.
- 09.8 No polynomial-time or other complexity bound; expression swell is allowed.
- 09.9 Ground equality is decidable, existential Diophantine solvability is not; the transfer is included to prevent a scope error, not as a new result.
- 09.10 The ω_1^{CK} cofinality statement needs uniform presentations; the proposition is a classical observation and does not apply directly to $\mathcal{I}$.
- 09.11 Abstract truncation closure is Camacho's; geometric omnific examples are classical; $\sum \omega^{1/n}$ is not a new series.
- 09.12 Classical ingredients are not relabelled: normal forms, **Oz** and its constant-term retraction, Hahn geometric inversion, Cantor normal forms, computable well-orders, the halting and tree completeness inputs, Cayley–Hamilton tests, clearing denominators.
- 09.13 The separated rational basis fails with irrational coefficients.
- 09.14 Grid transplantation does not commute with Ω .
- 09.15 No Euclidean termination is claimed.
- 09.16 The recurrence test is not the Skolem problem.
- 09.17 The certified intermediate languages use elementary algebra and are not new general theorems about automata or recurrences.
- 09.18 The prototype is not a CAS, a parser or production software, has no halting oracle, rejects floating-point input, and does not support directly constructed malformed objects.
- 09.19 The finite tests do not prove arbitrary-depth termination, support well-ordering, ordinal bounds or undecidability, do not verify the Π_1^1 reduction, and do not enumerate the support of **E**.
- 09.20 A classical formalized isomorphism, especially one using choice, does not implement a computable normalizer.
- 09.21 The repository review read the catalogue, the computable-surreals README and the normal-form bridge note, not the full computability article or every Lean declaration.
- 09.22 The Π_1^1 statement concerns the flexible support language, not fixed-stream equality or validity of rational–omega terms.
- 09.23 The ordinal-notation overview is orientation, not a source of proofs.
- 09.24 The research directions are not claims.

B.4 The merge (6 items)

- M.1 The incomparability of the three omnific rings ([proposition 17.3](#)) is not established for the pair $(\mathcal{R}_\varepsilon, \mathcal{I})$ when $k = \mathbb{Q}$; whether $\sum n! \omega^{\omega^{-n}}$ has a rational–omega term or lies in $\text{Frac}(\mathcal{H})$ is open ([question 26.8](#)).
- M.2 The characterizations of [remark 3.2](#) are the triangular and grid-ring theorems; the shielding theorem alone gives only the embedding.

- M.3 Only 05's coding gives decidable supports; the midpoint codings are not claimed to.
- M.4 Results printed by citation (the floor, division, the Diophantine transfer, the stream reductions, the two-scale sign obstruction, the rational monomial core, integer evaluation) are proved in the cited reports; this report adds only the stated effective or omnific refinements.
- M.5 No question named by an earlier report is answered.
- M.6 The reruns for this report used Python 3.14.4 and checked finite instances only; the dossier's 30 independent checks are likewise finite.

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